⚡ OFFICIAL HARDWARE COMPATIBILITY & DEVICE INTEGRATION MANUAL

# GEMS Supported Hardware & Devices Reference

Exhaustive Technical Specification for All 168 Native Hardware Profiles (176 Registered Templates): Real-Time Telemetry Read Out, Writable Control Setpoints, Communication Protocols, and Closed-Loop Automation Capabilities.

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| SOFTWARE VERSION<br>**v9.35.0-release (Official Tag)** | NATIVE TEMPLATES<br>**168 Hardware Profiles (176 Total)** | HARDWARE CATEGORIES<br>**5 Native Device Classes** |
| CONTROL LOOP CYCLE<br>**1 Hz Sub-Second Closed-Loop** | TARGET ARCHITECTURE<br>**Raspberry Pi (Debian/Linux)** | EXECUTION MODEL<br>**100% Local-First / Zero Cloud** |

## Table of Contents & Device Count by Category


| | | | |
|---|---|---|---|
| 1. Solar & Hybrid Inverters | 37 Devices | 2. Battery Energy Storage (BESS) | 18 Devices |
| 3. EV Charging Stations & Wallboxes | 61 Devices | 4. Grid Smart Meters & Analyzers | 40 Devices |
| 5. Smart Relays & Load Actuators | 12 Devices | Total Verified Native Profiles | 168 Profiles |


⭐ **REFERENCE HARDWARE PREFERENCE DIRECTIVE**

Per system architecture standards, **RAEDIAN** is designated as the primary reference template for EV Charging Stations (NEO, NEX, and Gemini dual-socket series), and **Solis** is designated as the primary reference template for Hybrid Solar Inverters and Battery Energy Storage Systems (including S5/S6 Hybrid and Solis Retrofit series). Both vendors feature complete sub-second register control and active curtailment setpoints.

# 1. Executive Overview & Unified Hardware Abstraction Layer

GEMS (Grid Energy Management System) is an edge Energy Management System engineered specifically to execute on local Raspberry Pi hardware without reliance on external cloud servers. The entire system is **local-first**: all automated power balancing, solar surplus tracking, Flanders peak shaving, and main fuse protection algorithms run directly inside the local Go binary (`gems` daemon), querying connected hardware and recalculating setpoints every single second.

## Commercial Support Tiers & Unified Hardware Interfaces

Every hardware driver in GEMS is classified into a commercial support tier (**Verified** for field-validated Tier-1 platforms displayed by default, **Experimental** accessible via the Installer Mode toggle, and **Monitor Only** for closed-protocol read-only telemetry) and implements standardized capability interfaces:

### `DevicePoller` (Universal Base)

Implemented across all 168 hardware profiles. Standardizes `Connect()`, `Poll()`, `Status()`, and `Close()`. Standardizes real-time power (W), battery power (W), grid flow (W), total energy (kWh), and battery SOC (%).

### `ChargeController` (EV Chargers)

Provides `SetChargeCurrent(amps float64)` (dynamic throttling from 6A to 32A in 1A steps) and `SetPhaseMode(phases int)` (automated switching between 1-phase and 3-phase charging modes).

### `BatteryController` (Inverters & BESS)

Provides `ChargeBattery(powerW float64)` and `DischargeBattery(powerW float64)` for automated Day-Ahead EPEX spot price arbitrage and Flemish capacity tariff peak shaving.

### `InverterController` (Solar Generation)

Provides `SetActivePowerLimit(powerW float64)` for automated solar curtailment during negative electricity spot market hours and Dutch zero-export regulatory compliance.

### `RelayController` (Smart Switches)

Provides `SetState(on bool)` for automated potential-free contactor actuation, powering domestic hot water (DHW) immersion heaters and triggering SG-Ready heat pump boost states.

### `PhasePoller` & `PhaseVoltagePoller`

Returns sub-second per-phase currents ($I_{L1}$, $I_{L2}$, $I_{L3}$) and voltages ($V_{L1}$, $V_{L2}$, $V_{L3}$) for dynamic phase balancing and main grid breaker protection.

### `MeterPeakPoller` (Belgian Capacity Tariff)

Implemented by P1 smart meters. Reads the official 15-minute rolling average peak demand register (W) directly from Fluvius DSMR meters to enforce capacity tariff ceilings.

### `MultiConnectorPoller` (Dual-Socket EVSE)

Implemented by dual-socket chargers (e.g. RAEDIAN Gemini). Manages independent telemetry, sessions, and charging currents across multiple physical charging ports simultaneously.

# 2. What GEMS Actually Does With Devices (Strategy Engine)

GEMS integrates disparate hardware from dozens of manufacturers into a cohesive, intelligent energy ecosystem. Below are the 9 automated closed-loop strategies executed every second by the local GEMS control engine:

### 1. 🌞 Eco Mode (100% Solar Self-Consumption)

Continuously computes instantaneous solar surplus ($P_{surplus} = P_{solar} - P_{house}$). Surpluses are routed sequentially: first charging the home battery, then modulating EV charge currents (6A–32A), and finally triggering hot water immersion heaters or heat pump SG-Ready boost states.

### 2. 🇧🇪 Flanders Capacity Tariff & Dynamic Ceiling

Monitors household demand every second and computes projected 15-minute rolling peak imports. When demand approaches the monthly ceiling, GEMS throttles EV charging to 6A or 0A, disengages non-critical relays, and commands battery discharge. Features an Adaptive Ceiling: if an unavoidable peak occurs (e.g. 4.8 kW from cooking), GEMS raises its operational ceiling to that level for the remainder of the month so the car charges faster at zero extra tariff cost.

### 3. 🇳🇱 Netherlands Zero-Export & Curtailment

Protects Dutch installations against grid export penalties and salderingsregeling phase-outs. GEMS commands solar inverters to reduce generation (`SetActivePowerLimit`) whenever export limits are exceeded and storage batteries are saturated.

### 4. 💶 Negative Spot Protection & Flemish GSC Shield

Protects homeowners with dynamic EPEX Day-Ahead energy tariffs from paying export penalties during negative electricity pricing. In Flanders, GEMS applies an automated GSC Subsidy Shield (e.g. €250/MWh): solar export remains active down to −€0.250/kWh, curtailing only if the market penalty strictly exceeds the value of the green certificate.

### 5. 🔋 EPEX Day-Ahead Dynamic Battery Arbitrage

Integrates 24-hour EPEX Spot Day-Ahead prices (updated daily at 14:00 CET) with Open-Meteo solar forecasts. Automatically commands the home battery to force-charge during negative/cheapest off-peak hours and discharge during evening peak price windows.

### 6. 🚗 Smart EV Charging (Cheapest Hours Boost)

Scans the 24-hour EPEX tariff curve to identify the $N$ cheapest hours of the day. Automatically ramps EV charging to maximum power (16A/32A) during those hours, throttling down to 0A during expensive price peaks.

### 7. ⚡ Dynamic Phase Load Balancing & Fuse Protection

Continuously inspects individual phase currents ($I_{L1}$, $I_{L2}$, $I_{L3}$) against the main grid connection breaker limit (e.g. 25A, 32A, 40A, 63A). If any individual phase exceeds 90% of the breaker rating, GEMS instantly throttles the EV charger within milliseconds to prevent tripping the main fuse.

### 8. 🔌 Dynamic Phase Switching (1-Phase / 3-Phase)

EV charging requires a minimum current of 6A (1.4 kW on 1-phase; 4.1 kW on 3-phase). For supported chargers (e.g. RAEDIAN, go-e, Alfen, Mennekes, Peblar, Huawei), GEMS commands 1-phase mode during modest solar surplus (1.4 kW–4.1 kW) and automatically switches to 3-phase mode when surplus exceeds 4.2 kW.

### 9. 🚰 Thermal Storage & SG-Ready Heat Pump Boost

Actuates contactors and smart plugs for domestic hot water (DHW) boilers or triggers SG-Ready Heat Pump Operating State 3 (Boost Mode) during solar excess or negative price windows. Enforces strict minimum run-time and off-time hysteresis to safeguard heat pump compressors against short-cycling.

### 🔒 Local Fail-Safe Watchdogs

All writable setpoints (EV current, inverter power limits, battery discharge) feature built-in hardware watchdog timeouts. If communication is lost, devices automatically revert to safe autonomous default modes.

# 3. Hardware Summary Matrix by Category

The tables below provide a quick-reference overview of all 168 native hardware profiles supported in GEMS across each category:

## ☀️ 1. Solar & Hybrid Inverters (37 Devices)

Inverter templates interface via Modbus TCP (SunSpec or vendor-specific registers) or local HTTP REST APIs. They continuously report photovoltaic (PV) generation, battery charge/discharge wattage, state of charge (SOC %), and grid flow. Advanced units support real-time active power curtailment for negative spot price protection and battery force-charge/discharge commands for day-ahead spot arbitrage.

| DEVICE & VENDOR | TEMPLATE ID | PROTOCOL & PORT | 🔍 READ OUT | 🎛️ CONTROLLED | ⚡ AUTOMATION ROLE |
|---|---|---|---|---|---|
| **Afore Hybrid Inverter**<br>Afore | `afore_inverter` | **MODBUS**<br>Modbus TCP via Ethernet / Wi-Fi dongle (Default Port: 502 | Real-time solar PV generation (W), Battery charge/discharge power (W), Grid import/export power (W), Cumulative solar production (kWh), Battery State of Charge (SOC %), 3-Phase inverter output currents (L1, L2, L3 in Amps). | EMS Energy Mode & Forced Charge/Discharge Power Setpoints (Holding `3000`/`3002`), plus Active Power Derating Limit (`0–100%`). | Closed-loop hybrid management for Afore AF-S and AF-T inverters: dynamically charges/discharges storage for Flanders capacity peak shaving and EPEX spot arbitrage while curtailing PV export during negative price windows. |
| **Alpha ESS Storion SMILE**<br>Alpha ESS | `alpha_ess_inverter` | **NETWORK TCP**<br>Modbus TCP via local network dongle (Default Port: 502 or 8502 | Solar PV active power (W), Battery charging/discharging power (W), Net grid import/export (W), Cumulative energy (kWh), Battery SOC (%), 3-Phase currents (L1, L2, L3 in Amps). | Dispatch Mode Control (`0x084F`), Active Power Offset Setpoint (`0x0850`–`0x0851` with 32000 W bias), Dispatch SOC Target, and Grid Feed-In Power Limit. | Full closed-loop Alpha ESS Storion SMILE control: dispatches exact wattage charge/discharge setpoints for EPEX Day-Ahead arbitrage and Flemish capacity ceiling protection, plus active solar curtailment. |
| **Anker SOLIX X1 Hybrid Inverter**<br>Anker | `anker_inverter` | **MODBUS**<br>Modbus TCP (Default Port: 502 | Solar PV power (W), Battery flow (W), Grid exchange power (W), Total lifetime energy (kWh), Battery SOC (%), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts). | Anker SOLIX X1 EMS Operating Mode and Forced Charge/Discharge Power Setpoints (W), plus Active Power Export Curtailment. | Closed-loop residential 3-phase hybrid control for Anker SOLIX X1: executes automated battery peak shaving, off-peak spot charging, and negative-price solar curtailment. |
| **Deye / Sunsynk Hybrid Inverter**<br>Deye | `deye_inverter` | **MODBUS**<br>Modbus TCP via Ethernet LAN or Solarman/Deye Wi-Fi logger (Default Port: 502 or 8899 | Solar PV production (W), Battery charge/discharge power (W), Grid meter power (W), Lifetime energy (kWh), Battery SOC (%), 3-Phase inverter currents (L1, L2, L3 in Amps). | Time-of-Use Grid Charge & Discharge Power Setpoints, Minimum/Target SOC Windows, and Dynamic Max Solar Sell Power Limit (Holding `143` / `340`, LV & HV auto-scaled). | Full closed-loop control for Deye and Sunsynk LV/HV hybrid inverters: executes sub-second Flemish capacity peak shaving, Day-Ahead EPEX battery arbitrage, and zero-export / negative-price curtailment. |
| **Enerlution Hybrid Inverter**<br>Enerlution | `enerlution_inverter` | **MODBUS**<br>Modbus TCP (Default Port: 502 | Solar PV power (W), Battery power (W), Grid power (W), Cumulative energy (kWh), Battery SOC (%), 3-Phase currents (L1, L2, L3 in Amps). | Active Power Curtailment Limit (W or % via Modbus registers 40400/40441). | Full active curtailment integration: automatically reduces inverter power output during negative EPEX Day-Ahead spot price hours (Netherlands Mode / Zero-Export) while respecting Flemish GSC subsidy offsets; manages solar surplus routing into battery and EV. |
| **Enphase IQ Gateway / Envoy**<br>Enphase | `enphase_envoy` | **REST / HTTP**<br>Local HTTP REST API via token/digest authentication (Default Port: 80 / 443 | Microinverter aggregated solar PV generation (W), IQ Battery charge/discharge wattage (W), Net grid import/export (W), Total lifetime generation (kWh), Battery SOC (%), Per-phase currents and voltages. | Local Enphase IQ Gateway HTTPS Bearer JWT dynamic power-export limiting and IQ Battery storage mode dispatch. | Local-first integration with Enphase IQ Gateway Metered and IQ Batteries: streams production/consumption/storage metrics and coordinates negative-price export limiting. |

| DEVICE & VENDOR | TEMPLATE ID | PROTOCOL & PORT | 🔍 READ OUT | 🎛️ CONTROLLED | ⚡ AUTOMATION ROLE |
|---|---|---|---|---|---|
| **FoxESS H1 / H3 / H3-Pro Hybrid Inverter**<br>FoxESS | `foxess_inverter` | **MODBUS**<br>Modbus TCP via LAN or RS485 Ethernet converter (Default Port: 502 | Solar PV generation (W), Battery power (W), Grid active power (W), Total energy (kWh), Battery SOC (%), 3-Phase currents (L1, L2, L3 in Amps). | FoxESS Work Mode (`41000` / `44000`), Remote EMS Power Setpoints (W), Min SOC Floor, and Active Export Power Limit. | Full closed-loop control across FoxESS H1, H3, and H3-Pro hybrid series: dispatches forced battery charging/discharging for spot arbitrage and capacity peak shaving alongside active solar curtailment. |
| **Fronius Gen24 Inverter**<br>Fronius | `fronius_inverter` | **MODBUS**<br>Modbus TCP via SunSpec protocol (Default Port: 502 | Solar PV generation (W), Battery charge/discharge wattage (W), Grid power flow (W), Cumulative energy (kWh), BYD / Fronius Battery SOC (%). | SunSpec Model 123 Active Power Limit (`WMaxLimPct` single-register FC06) and SunSpec Model 124 Storage Charge/Discharge Control (`StorCtl_Mod`, `OutWRte`, `InWRte`). | Full closed-loop Fronius Symo / Primo / Gen24 Plus control: regulates BYD Battery-Box HVS/HVM charge/discharge rates for spot arbitrage and peak shaving, and curtails PV output during negative market prices. |
| **GoodWe Inverter**<br>GoodWe | `goodwe_inverter` | **MODBUS**<br>Modbus TCP / UDP via LAN or Wi-Fi+LAN kit (Default Port: 502 or 8899 | Solar PV power (W), Battery charge/discharge power (W), Net grid flow (W), Lifetime energy (kWh), Battery SOC (%). | GoodWe ET/EH/BT/BH EMS Power Mode (`47511` modes `1`/`2`/`8`/`9`), EMS Power Setpoint (`47512` W), and Grid Export Power Limit Enable & Value (`47509`–`47510`). | Verified closed-loop hybrid control for GoodWe ET/EH/BT/BH inverters: executes automated battery peak shaving, Day-Ahead spot arbitrage, and dynamic PV export curtailment. |
| **Growatt Inverter**<br>Growatt | `growatt_inverter` | **MODBUS**<br>Modbus TCP via ShineLAN-X Ethernet dongle (Default Port: 502 | Solar PV active power (W), Battery power (W), Grid feed-in/consumption (W), Cumulative kWh, Battery SOC (%). | Growatt SPH / MOD TL3-XH Priority Mode (`1044` / `3049`), Charge/Discharge Rate & Stop SOC (`1070`–`1092`), and Active Power Rate / Export Limit (`3`, `122`–`123`). | Verified closed-loop control for Growatt SPH and MOD hybrid/string inverters: manages forced battery charging, peak discharge, and negative spot price PV curtailment. |
| **Huawei Hybrid Inverter**<br>Huawei | `huawei_inverter` | **MODBUS**<br>Modbus TCP via SDongleA-05 (Ethernet/WLAN) or SmartLogger3000 (Default Port: 502 | Active solar PV generation (W, reg 32080), LUNA2000 Battery charge/discharge power (W, reg 37001), Grid import/export power via DTSU666-H (W, reg 37113), Cumulative generation (kWh, reg 32106), Battery SOC (%, reg 37004). | Forced Battery Charge / Discharge Power Setpoint (W, reg 40118/40120), Active Power Curtailment Limit (W or % of rated power, reg 40118/40125). | Full bidirectional battery and curtailment control: commands LUNA2000 to discharge during household peak spikes (Flanders capacity peak shaving); commands grid force-charging during negative/cheap EPEX spot hours; curtails active PV output during negative export prices or zero-export mandates. |
| **LG ESS Home (8 / 10 / 15)**<br>LG | `lg_ess` | **REST / HTTP**<br>Local HTTP REST API (Default Port: 80 / 443 | Solar PV production (W), Battery power flow (W), Grid import/export (W), Cumulative energy (kWh), Battery SOC (%). | Local LG ESS Home `/v1/user/setting/batt` forced-charge / self-consumption control and PCS power curtailment. | Local zero-cloud HTTPS integration for LG ESS Home 8 / 10 / 15 systems with automated off-peak battery charging and grid flow balancing. |
| **Marstek Venus / Jupiter Storage**<br>Marstek | `marstek_storage` | **MODBUS**<br>Modbus TCP (Default Port: 502 | Solar PV power (W), Battery charge/discharge power (W), Grid exchange power (W), Total energy (kWh), Battery SOC (%). | Marstek Venus C / E / Jupiter Passive Mode Enable (`42000` = `0x55AA`), Direction Control (`42010`), and Exact Charge/Discharge Watt Setpoints (`42020`–`42021`). | Full closed-loop AC-coupled storage control for Marstek Venus C/E and Jupiter batteries: absorbs excess rooftop solar, shaves domestic capacity peaks, and executes Day-Ahead spot arbitrage. |
| **SAJ eSolar (H1 / H2 / R5 / R6 Hybrid Inverter)**<br>SAJ | `saj_hybrid` | **MODBUS**<br>Modbus TCP (Default Port: 502 | Solar PV generation (W), Battery power (W), Grid active power (W), Total energy (kWh), Battery SOC (%), 3-Phase currents (L1, L2, L3 in Amps). | SAJ H1 / H2 / R5 / R6 AppMode / Passive Charge & Discharge Wattage Setpoints and Active Power Derating Limit (`0.1%`). | Closed-loop hybrid management for SAJ H1/H2 systems: coordinates 3-phase PV curtailment, forced off-peak battery charging, and peak demand shaving. |
| **SMA Sunny Boy / Tripower / Storage Inverter**<br>SMA | `sma_inverter` | **MODBUS**<br>Modbus TCP via Speedwire / SunSpec (Default Port: 502 | Solar PV generation (W, reg 30775), Battery charge/discharge power (W, reg 31393), Grid active power (W, reg 30865), Total energy (kWh, reg 30513), Battery SOC (%, reg 30845), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts). | SMA Active Power Limit (`WModCfg` / `WMaxLimPct`) and Bidirectional Battery Control across both Sunny Island / Sunny Boy Storage (`40151=802`, `40149`) and Sunny Tripower Smart Energy (`40236`, `40793`–`40801`). | Complete closed-loop solar curtailment and battery arbitrage across SMA Sunny Boy, Sunny Tripower, Sunny Tripower Smart Energy (STP-SE), and Sunny Boy Storage. |

| DEVICE & VENDOR | TEMPLATE ID | PROTOCOL & PORT | 🔍 READ OUT | 🎛️ CONTROLLED | ⚡ AUTOMATION ROLE |
|---|---|---|---|---|---|
| **Senergy (SE Series / Hybrid Inverter)**<br>Senergy | `senergy_inverter` | **MODBUS**<br>Modbus TCP (Default Port: 502 | Solar PV active power (W), Battery charge/discharge power (W), Grid exchange (W), Cumulative energy (kWh), Battery SOC (%). | Senergy SE Hybrid Passive Mode Charge/Discharge Power Setpoints (W) and Active Power Output Limit (%). | Closed-loop control for Senergy hybrid inverters: automates battery peak shaving, off-peak charging, and negative-price PV curtailment. |
| **Sigenergy SigenStor (Hybrid Inverter & Storage)**<br>Sigenergy | `sigenergy_hybrid` | **MODBUS**<br>Modbus TCP via Sigen Gateway / Inverter (Default Port: 502 | Solar PV power (W), Sigen Battery flow (W), Grid exchange power (W), Total energy (kWh), Sigen Battery SOC (%), 3-Phase currents (L1, L2, L3 in Amps). | SigenStor Remote EMS Enable (`40029`), Operating Mode (`40031`), Max Charge/Discharge Power Setpoints (`40032`–`40034` U32 W), and PV Curtailment Limit (`40038`). | High-precision closed-loop EMS control for Sigenergy SigenStor 5-in-1 energy systems: executes sub-second capacity peak shaving, EPEX spot arbitrage, and PV export curtailment. |
| **SofarSolar (HYD / G3 / AMASS Hybrid Inverter)**<br>SofarSolar | `sofarsolar_hybrid` | **MODBUS**<br>Modbus TCP via Ethernet/Wi-Fi stick (Default Port: 8899 or 502 | Solar PV power (W), Battery charge/discharge power (W), Grid power flow (W), Lifetime energy (kWh), Battery SOC (%), 3-Phase currents (L1, L2, L3 in Amps). | SofarSolar HYD 5K–20KTL-3PH Passive Storage Setpoints (`0x1187` mode, `0x1189`–`0x118D` signed S32 W) and Active Power / Anti-Reflux Export Limit. | Verified closed-loop control for SofarSolar HYD 3PH G3 and ME3000SP hybrid systems: regulates battery charge/discharge wattage and solar curtailment. |
| **SolaX Inverter**<br>SolaX | `solax_inverter` | **MODBUS**<br>Modbus TCP via Pocket LAN or Pocket Wi-Fi (Default Port: 502 | Solar PV active power (W), Battery power (W), Grid active power (W), Total energy (kWh), Battery SOC (%). | SolaX X1/X3-Hybrid G4 Remote Control Power Mode (`0x007C`–`0x0080`, exact active power target W + watchdog) with `0x001F` Manual Mode fallback, plus Export Control User Limit (`0x0042`). | Verified sub-second closed-loop control for SolaX X1/X3-Hybrid and Fit systems: executes dynamic solar self-consumption, spot-price arbitrage, and PV curtailment. |
| **SolarEdge 3-Phase Inverter**<br>SolarEdge | `solaredge_3phase` | **MODBUS**<br>Modbus TCP via SunSpec protocol (Default Port: 1502 or 502 | Solar PV generation (W), Battery charge/discharge power (W), Grid meter power (W), Total energy (kWh), Connected Energy Bank SOC (%). | SolarEdge StorEdge Control Mode `4` (Remote), Charge/Discharge Limit (`0xE00E`/`0xE010` float32 W), Mode Command (`0xE00D`), and Advanced Power Control (`0xF001` / SunSpec M123). | Full closed-loop 3-phase SolarEdge Home Hub & StorEdge control: modulates home battery charge/discharge wattage and curtails PV generation during negative spot prices. |
| **SolarEdge Inverter**<br>SolarEdge | `solaredge_inverter` | **MODBUS**<br>Modbus TCP via SunSpec protocol (Default Port: 1502 or 502 | Solar PV generation (W), Battery charge/discharge power (W), Grid active power (W), Total energy (kWh), Battery SOC (%). | SolarEdge StorEdge Control Mode `4` (Remote), Charge/Discharge Limit (`0xE00E`/`0xE010` float32 W), Mode Command (`0xE00D`), and Advanced Power Control (`0xF001` / SunSpec M123). | Full closed-loop 1-phase SolarEdge HD-Wave & Home Hub control: regulates StorEdge / Home Battery charge and discharge setpoints and curtails solar production when required. |
| **Solis Hybrid & Retrofit Inverters (Primary Reference)**<br>Solis<br>⭐ Reference | `solis_hybrid` | **MODBUS**<br>Modbus TCP via direct internal Ethernet / Ginlong Wi-Fi/LAN stick (Default Port: 502 or 8899 | Solar PV Active Generation Power (W, reg 33079), Battery Charge/Discharge Power (W, reg 33149), Battery SOC (%, reg 33139), Grid Meter Active Power (W, reg 33130), Total Cumulative Energy Yield (kWh, reg 33029), 3-Phase AC Currents (0.1A, regs 33073-33075), 3-Phase Voltages (0.1V, regs 33076-33078). | Active Power Curtailment Mode and Limit (W / % via holding registers 3051/3052), Forced Battery Charging & Discharging Setpoint (W via registers 43110/43135/43136). | Designated primary reference platform for hybrid solar and battery energy storage. Executes sub-second closed-loop active power curtailment during negative spot price hours (Netherlands Mode / Zero-Export), manages automatic solar surplus sinking into battery storage, dynamically throttles charging during Flanders capacity tariff peak shaving, and orchestrates spot market arbitrage (EPEX Day-Ahead price force charging and peak discharging). Solis Retrofit models operate seamlessly as AC-coupled storage units with dedicated battery control and zero solar PV string dependency. |
| **Sungrow Inverter**<br>Sungrow | `sungrow_inverter` | **MODBUS**<br>Modbus TCP via direct internal LAN RJ45 port (Default Port: 502 | Solar PV generation (W, reg 5016), Battery charge/discharge power (W, reg 13021), Grid exchange power (W, reg 13009), Cumulative generation (kWh, reg 5003), SBR Battery SOC (%, reg 13022). | Sungrow SH-RS / SH-RT / SH-T EMS Mode Selection (`13049`), Charge/Discharge Command (`13050` `0xAA`/`0xBB`/`0xCC`), Power Setpoint (`13051` W), and Active Power Limitation (`5006`–`5007`). | Verified high-speed closed-loop control for Sungrow SH hybrid and SG string inverters: executes sub-second peak shaving, EPEX battery arbitrage, and PV curtailment. |

| DEVICE & VENDOR | TEMPLATE ID | PROTOCOL & PORT | 🔍 READ OUT | 🎛️ CONTROLLED | ⚡ AUTOMATION ROLE |
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| **Victron Inverter/GX**<br>Victron | `victron_inverter` | **MODBUS**<br>Modbus TCP via Venus OS (Default Port: 502 | Solar PV DC/AC power (W, reg 811), Battery charge/discharge power (W, reg 842), Net AC grid import/export (W, reg 820), Total energy (kWh), Battery SOC (%, reg 843). | Victron Venus OS / Cerbo GX ESS Grid Power Setpoint (`2700` S16 / `2703` S32 W), ESS Discharge Disable, and Signed Max PV Feed-In Power Limit (`2706` in 100 W steps). | Deep closed-loop integration with Victron MultiPlus-II / Quattro GX ESS installations: steers AC grid setpoint for battery arbitrage/peak shaving and curtails excess feed-in. |
| **ABB / Fimer Solar & Hybrid Inverter**<br>ABB / Fimer | `abb_fimer_inverter` | **MODBUS**<br>Modbus SunSpec over TCP/RTU (Default Port: 502 | Solar PV Active Power (W, SunSpec standard register 40083), Battery State of Charge (SOC %), Battery Charge / Discharge Power (W), Grid Feed-in Power (W), Cumulative Generation Energy (kWh). | SunSpec Model 123 Active Power Curtailment (`WMaxLimPct`, `WMaxLim_Ena`) and SunSpec Model 124 Storage Charge/Discharge Rate Control (`StorCtl_Mod`, `OutWRte`, `InWRte`). | Full closed-loop SunSpec control for ABB / Fimer PV and REACT 2 hybrid installations: executes negative spot price solar curtailment, zero-export limiting, and automated battery peak shaving and Day-Ahead spot arbitrage. |
| **Autarco Solar Inverter**<br>Autarco | `autarco_inverter` | **NETWORK TCP**<br>Modbus TCP via local network dongle/gateway (Default Port: 502 | Solar PV Active Power (W), Total Daily & Lifetime Generation (kWh), Grid AC Voltage (V), AC Frequency (Hz). | Remote Control Battery Charge/Discharge Power Setpoints (`43135`–`43136`) and Active Power Curtailment Limit (`43052` / `0–100%`). | Full closed-loop hybrid inverter and PV curtailment control for Autarco S2.x / MH / LX systems: coordinates solar self-consumption, spot-price battery arbitrage, and Flemish capacity peak shaving. |
| **Chisage ESS Hybrid Inverter**<br>Chisage ESS | `chisage_ess` | **MODBUS**<br>Modbus TCP / RTU via RS485 Ethernet Gateway (Default Port: 502 | Solar PV Active Power (W), Battery Charge / Discharge Power (W), Battery SOC (%), Grid Import / Export Power (W), Cumulative Inverter Energy (kWh). | Monitor-Only (Read-Only Telemetry — closed register control map). | Integrates commercial, industrial, and residential Chisage ESS hybrid solar and battery telemetry into GEMS site load balancing. |
| **Delta Electronics Solar Inverter**<br>Delta Electronics | `delta_inverter` | **MODBUS**<br>Modbus SunSpec over TCP (Default Port: 502 | Solar PV Generation Power (W), DC String Voltages and Currents, Cumulative Generation (kWh), Grid AC Power (W). | SunSpec Model 123 Active Power Curtailment (`WMaxLimPct`, `WMaxLim_Ena`) and SunSpec Model 124 Basic Storage Control. | Standardized SunSpec closed-loop telemetry and active power curtailment for Delta residential and commercial solar/hybrid inverters. |
| **HYXiPOWER Hybrid Inverter**<br>HYXiPOWER | `hyxipower_inverter` | **MODBUS**<br>Modbus TCP via Wi-Fi/Ethernet dongle (Default Port: 502 | Solar PV Generation (W), Battery Charge / Discharge Power (W), Battery SOC (%), Grid Exchange (W), Cumulative Energy (kWh). | Autonomous / Read-Only telemetry poller. | Streams PV production and battery storage metrics to the GEMS flow chart and telemetry pipeline. |
| **KSTAR BluE Hybrid Inverter**<br>KSTAR | `kstar_inverter` | **MODBUS**<br>Modbus RTU / TCP via RS485 Ethernet converter (Default Port: 502 | Solar PV Active Power (W), Battery Charge / Discharge Power (W), Battery SOC (%), Grid Power Flow (W), Phase Currents (L1, L2, L3 in Amps). | Autonomous / Read-Only telemetry poller. | Feeds hybrid solar and battery metrics into the GEMS load balancer and multi-tariff cost optimization engine. |
| **Solinteg Integ M Hybrid Inverter**<br>Solinteg | `solinteg_inverter` | **MODBUS**<br>Modbus RTU / TCP via LAN/Wi-Fi stick (Default Port: 502 | Solar PV Active Power (W), Battery Charge / Discharge Power (W), Battery SOC (%), Grid Net Exchange Power (W), Cumulative Generation (kWh). | Solinteg Integ M series Remote EMS Mode, Charge/Discharge Power Setpoints, and Active Power Limit. | Verified closed-loop 3-phase high-voltage hybrid control for Solinteg Integ M (4–50 kW) commercial and residential inverters. |
| **Solplanet / AISWEI Inverter**<br>Solplanet (AISWEI) | `solplanet_inverter` | **MODBUS**<br>Modbus RTU / TCP via Ethernet/Wi-Fi Ai-Dongle (Default Port: 502 | Solar PV Active Power (W), Total Yield Energy (kWh), AC Voltage and Current per phase, Grid Import/Export (W). | AISWEI / Solplanet Battery Working Mode (`1103` wire / `41104` doc), Signed Charge/Discharge Power Setpoint (`1152` wire S16 W), and Active Power Control (`4000` / `5402` `0.01%`). | Verified closed-loop control for Solplanet ASW H-T2/T3 and KACO blueplanet hybrid NH3 inverters: dispatches exact Watt storage setpoints and dynamic PV curtailment. |
| **APsystems EZ1-M / EZ1-D Local Microinverter**<br>APsystems | `apsystems_ez1` | **REST / HTTP**<br>Local HTTP JSON API (Default Port: 8050 | Channel 1 & Channel 2 PV Active Power (W), Lifetime Cumulative Generation Energy (kWh). | Direct Local Max AC Output Power Setpoint (`/setMaxPower?p=30..800` W). | 100% cloud-free local microinverter telemetry and stepless active power curtailment during negative EPEX spot prices or zero-export operation. |

| DEVICE & VENDOR | TEMPLATE ID | PROTOCOL & PORT | 🔍 READ OUT | 🎛️ CONTROLLED | ⚡ AUTOMATION ROLE |
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| **FENECON Home / OpenEMS Edge (FEMS)**<br>FENECON | `fenecon_fems` | REST / HTTP<br>Local HTTP REST API (Default Port: 8084 | Solar PV Production Power (W), Cumulative PV Energy (kWh), Battery Charge/Discharge Power (W), Battery SOC (%), Grid Active Power (W). | Exact ESS Charge / Discharge Active Power Setpoint (`ess0/SetActivePowerEquals` W) and PV Active Power Curtailment (`pvInverter0/SetActivePowerLimit` W). | Full closed-loop local integration for FENECON Home & OpenEMS systems: coordinates EPEX Day-Ahead battery arbitrage, Flemish capacity ceiling peak shaving, and negative-price curtailment. |
| **Hoymiles DTU-Pro / OpenDTU Microinverter**<br>Hoymiles | `hoymiles_dtu` | REST / HTTP<br>Modbus TCP (Default Port: 502 | Aggregated & Per-Port Solar PV Active Power (W), Cumulative Solar Yield (kWh). | Active Power Output Limit (`0xC006` % on DTU-Pro Modbus; volatile RAM `limit_type=1` % on OpenDTU without EEPROM wear). | Zero-cloud microinverter monitoring and dynamic PV curtailment during negative spot market hours. |
| **KOSTAL Plenticore Plus / G2 / G3 Hybrid**<br>KOSTAL | `kostal_plenticore` | MODBUS<br>Modbus TCP (Default Port: 1502 | Total DC PV Power (W), Cumulative Yield (kWh), Battery Charge/Discharge Power (W), Battery SOC (%), Grid Sensor Power (W), 3-Phase Voltages & Currents. | External Battery Management Signed Power Setpoint (Holding `1034` Float32 W) and SunSpec Model 123 Active Power Curtailment (`WMaxLimPct`). | Full closed-loop control for KOSTAL Plenticore hybrid installations: executes battery peak shaving, Day-Ahead spot arbitrage, and negative-price solar curtailment. |
| **KACO blueplanet hybrid NH3 / gridsave TL3**<br>KACO | `kaco_inverter` | MODBUS<br>Modbus TCP / RTU (Default Port: 502 | Solar PV Active Power (W), Cumulative Energy (kWh), Battery Charge/Discharge Power (W), Battery SOC (%), Live BMS Charge/Discharge Power Limits (`40202`–`40203` W), 3-Phase Currents & Voltages. | SunSpec Model 123 Active Power Curtailment (`WMaxLimPct`, `WMaxLim_Ena`) and Model 124 Storage Control (`WChaMax`, `WChaGra`, `WDisChaGra`, `StorCtl_Mod`, `WChaMaxRvrtTms=300s` Watchdog) plus NH3 EMS fallback (`40210`–`40212`). | Full closed-loop control for KACO blueplanet hybrid NH3 and commercial gridsave TL3 installations: executes battery peak shaving, EPEX Day-Ahead spot arbitrage, and negative-price solar curtailment. |

## 🔋 2. Dedicated Battery Energy Storage Systems (BESS) (18 Devices)

Dedicated BESS integrations interface directly with stationary battery controllers and BMS gateways over Modbus TCP or local LAN APIs. They provide deep pack telemetry (SOC %, voltage, cell temperature, power flow) and support dynamic power limits, scheduled force-charging during cheap/negative hours, and autonomous peak shaving for commercial and residential tariffs.

| DEVICE & VENDOR | TEMPLATE ID | PROTOCOL & PORT | 🔍 READ OUT | 🎛️ CONTROLLED | ⚡ AUTOMATION ROLE |
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| **Siemens Junelight Smart Battery**<br>Siemens | `siemens_junelight` | MODBUS<br>Modbus TCP (Default Port: 502 | Battery charge/discharge power (W), Battery State of Charge (SOC %), Net grid power flow (W), Internal battery status. | Siemens Junelight Smart Battery EMS Operating Mode and Forced Charge/Discharge Power Setpoints (W). | Verified integration with Siemens Junelight residential storage for solar self-consumption, peak shaving, and off-peak charging. |
| **Solarwatt (EnergyManager / MyReserve Matrix)**<br>Solarwatt | `solarwatt_storage` | REST / HTTP<br>Local HTTP REST API (Default Port: 80 | Battery power flow (W), Battery SOC (%), Solar generation (W), Grid exchange (W), Cumulative energy (kWh). | Solarwatt EnergyManager / Reserve local `/rest/items` battery charge/discharge control and telemetry ingestion. | Verified zero-cloud local LAN integration for Solarwatt EnergyManager Pro and MyReserve / Battery flex storage. |
| **sonnenBatterie (eco 6 / 8 / 10 / hybrid)**<br>Sonnen | `sonnen_batterie` | REST / HTTP<br>Local HTTP REST API v1/v2 (Default Port: 80 | Battery charge/discharge wattage (W), Battery SOC (%), Solar PV production (W), Grid feed-in / consumption (W). | sonnenBatterie Local API v2 `PUT /api/v2/configurations` (`EM_OperatingMode=1` Manual / `2` Self-Consumption) and `POST /api/v2/setpoint/charge\|discharge/{W}`. | Complete closed-loop control for sonnenBatterie eco 8 / 10 / performance: switches cleanly into Manual API mode during forced arbitrage or peak shaving and restores autonomous self-consumption afterward. |
| **Duracell Energy Residential Storage**<br>Duracell Energy | `duracell_energy` | MODBUS<br>Modbus RTU / TCP via Gateway (Default Port: 502 | Battery Charge / Discharge Power (W), Battery State of Charge (SOC %), Grid Power Flow (W), Inverter AC Power (W). | Monitor-Only (Read-Only Telemetry). | Monitors Duracell residential battery SOC and power exchange for site load balancing and EV solar surplus routing. |

| DEVICE & VENDOR | TEMPLATE ID | PROTOCOL & PORT | 🔍 READ OUT | 🎛️ CONTROLLED | ⚡ AUTOMATION ROLE |
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| **Dyness Battery Storage System**<br>Dyness | `dyness_battery` | **MODBUS**<br>Modbus RTU / TCP via Gateway / Inverter BMS Port (Default Port: 502 | Battery Charge / Discharge Power (W), Battery State of Charge (SOC %), Total Pack Voltage (V), Current (A), Cell Temperatures (°C). | Dyness Tower / Powerbox / Junior Box Modbus EMS Operating Mode and Charge/Discharge Power Limits (W). | Verified BMS & PCS storage integration for Dyness HV/LV battery stacks with closed-loop charge/discharge dispatch. |
| **Elecnova Commercial & Industrial BESS**<br>Elecnova | `elecnova_bess` | **MODBUS**<br>Modbus TCP via Industrial Gateway (Default Port: 502 | Battery Active Power (W), Battery State of Charge (SOC %), Grid Exchange Power (W), Cumulative In / Out Energy (kWh). | Monitor-Only (Read-Only Telemetry). | Monitors commercial BESS power flow, state of charge, and cumulative energy on the live dashboard. |
| **Goneo Residential Storage System**<br>Goneo | `goneo_storage` | **MODBUS**<br>Modbus TCP / RTU via Wi-Fi/Ethernet Module (Default Port: 502 | Battery Charge / Discharge Power (W), Battery SOC (%), Grid Power Flow (W), Total Energy (kWh). | Monitor-Only (Read-Only Telemetry). | Monitors residential battery storage SOC and power exchange for coordinated EV and relay load balancing. |
| **Mastervolt MLI Ultra / Combi CZone**<br>Mastervolt | `mastervolt_storage` | **MODBUS**<br>Modbus RTU / TCP via CZone Bridge / MasterBus gateway (Default Port: 502 | Battery Power (W), Battery SOC (%), Inverter AC Output Power (W), DC Bus Voltage (V), Battery Current (A). | Monitor-Only (Read-Only Telemetry). | Provides off-grid, marine, and residential Mastervolt / CZone battery system telemetry monitoring. |
| **Power Sonic C&I Energy Storage**<br>Power Sonic | `powersonic_bess` | **MODBUS**<br>Modbus TCP via BMS/PCS Controller (Default Port: 502 | Battery Active Power (W), Battery State of Charge (SOC %), Grid Power Flow (W), Fault Status Registers. | Monitor-Only (Read-Only Telemetry). | Monitors commercial BESS power flow and state of charge on the live GEMS dashboard. |
| **Poweroad Residential & C&I Storage**<br>Poweroad | `poweroad_storage` | **MODBUS**<br>Modbus RTU / TCP via Gateway (Default Port: 502 | Battery Power (W), Battery State of Charge (SOC %), Grid Flow (W), Cumulative In / Out Energy (kWh). | Monitor-Only (Read-Only Telemetry). | Monitors residential and light-commercial Poweroad battery storage telemetry. |
| **Pylontech Battery Storage (US / Force)**<br>Pylontech | `pylontech_storage` | **MODBUS**<br>Modbus RTU / TCP via Inverter Gateway or SC0500/SC1000 controller (Default Port: 502 | Battery Charge / Discharge Power (W), State of Charge (SOC %), Pack Voltage (V), Pack Current (A), SOH (%). | Pylontech US2000/US3000/US5000 & Force-H1/H2 BMS/EMS Charge and Discharge Power Setpoints via Modbus TCP/RTU. | Verified closed-loop telemetry and dispatch for Pylontech low-voltage and high-voltage stationary storage racks. |
| **Socomec SUNSYS C-System / HES BESS**<br>Socomec | `socomec_bess` | **MODBUS**<br>Modbus TCP via Diris Digiware / BESS Controller (Default Port: 502 | Battery Active Power (W), Battery SOC (%), Live BMS Max Charge/Discharge Limits (`0xC008`–`0xC009` W), Grid Active Power Flow (W), Cumulative Energy (kWh). | Socomec SUNSYS PCS Remote Active Power Control (`0xC100` Mode, `0xC101`–`0xC102` S32 0.1 kW Setpoint, `0xC103=300s` Hardware Watchdog) with SunSpec Model 124 Storage Control fallback. | Verified closed-loop C&I battery control for Socomec SUNSYS HES L/XL and PCS+: executes industrial peak shaving, capacity ceiling protection, and EPEX spot market arbitrage. |
| **Sunwoda AtSmart Energy Storage**<br>Sunwoda Energy | `sunwoda_energy` | **MODBUS**<br>Modbus TCP via BMS Communication Hub (Default Port: 502 | Battery Charge / Discharge Power (W), Battery State of Charge (SOC %), Grid Power Flow (W), Temperature (°C). | Monitor-Only (Read-Only Telemetry). | Monitors Sunwoda battery state of charge and power exchange on the live GEMS dashboard. |
| **Sessy Smart Battery (AC Home Storage)**<br>Sessy | `sessy_battery` | **REST / HTTP**<br>Local HTTP REST API with Basic Auth (`/api/v1/power/status` | Battery Charge/Discharge Power (W), State of Charge (SOC %), Exported Energy (kWh), Per-Phase RMS Voltages (V) & Currents (A). | Dynamic Strategy Selection (`POWER_STRATEGY_API` vs `POWER_STRATEGY_NOM`) and Signed Watt Setpoint (`-W` Charge / `+W` Discharge). | Zero-cloud local control of Dutch-engineered Sessy retrofit batteries for solar self-consumption, Flanders capacity peak shaving, and EPEX spot arbitrage. |
| **Tesla Powerwall 2 / 3 & Backup Gateway (Local HTTPS)**<br>Tesla Energy | `tesla_powerwall` | **REST / HTTP**<br>Local HTTPS Gateway API (`POST /api/login/Basic` | Battery Power (W), Usable State of Energy (SOC %), Site Grid Power (W), Solar Power (W), Cumulative Energy (kWh), 3-Phase Site Currents & Voltage. | Monitor-Only (Local Gateway v23+ firmware enforces read-only local telemetry). | High-speed local HTTPS ingestion of stationary Powerwall site, solar, and battery flows to drive EV charging surplus regulation and capacity peak protection. |

| DEVICE & VENDOR | TEMPLATE ID | PROTOCOL & PORT | 🔍 READ OUT | 🎛️ CONTROLLED | ⚡ AUTOMATION ROLE |
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| **VARTA pulse / neo / element Home Storage**<br>VARTA | `varta_storage` | MODBUS<br>Modbus TCP (Default Port: 502 | Battery Charge/Discharge Power (W), Battery SOC (%), Cumulative Charged Energy (kWh), Grid Sensor Power (W), Attached PV Sensor Power (W). | VARTA EMS Active Power Control (`2065` Command: Charge/Discharge/Auto, `2066` Power W; fallback `1074`). | Closed-loop local Modbus TCP control for VARTA AC-coupled storage systems: automates off-peak spot charging and household peak shaving. |
| **Zendure SolarFlow AC / Hyper 2000**<br>Zendure | `zendure_solarflow` | REST / HTTP<br>Local HTTP JSON API (`GET /properties/report` | Solar Input Power (W), Battery Pack Charge & Discharge Power (W), State of Charge (`electricLevel` %). | Smart AC Mode (`acMode` Charge/Discharge) and Stepless `inputLimit` / `outputLimit` Watt Setpoints (`0–2400 W`). | 100% local zero-cloud control for plug-in and AC-coupled Zendure storage: absorbs surplus PV and discharges to cover household evening peaks. |
| **Sinexcel PWS Bi-Directional PCS / Commercial BESS**<br>Sinexcel | `sinexcel_bess` | MODBUS<br>Modbus TCP / RTU (Default Port: 502 | Battery Active Power (`106` 0.1 kW), Battery SOC (`109` 0.1%), Cumulative Discharged Energy (`110`–`111` 0.1 kWh), Live BMS Max Charge/Discharge Power Limits (`112`–`113` 0.1 kW), 3-Phase Voltages & Currents. | Remote PCS Start/Standby (`600`), Constant-Power PQ Mode (`601`), Signed Active Power Setpoint (`602` S16 0.1 kW), and 60-Second Fail-Safe Watchdog (`610`). | Closed-loop industrial and C&I battery control for Sinexcel PWS storage converters: automates high-power peak shaving, grid transformer capacity protection, and Day-Ahead spot arbitrage. |

## ⚡ 3. EV Charging Stations & Wallboxes (61 Devices)

EVSE integrations communicate via native local OCPP 1.6-J over WebSockets, high-speed Modbus TCP/RTU, or local REST APIs. They receive dynamic charge current setpoints (throttling 6A to 32A) calculated every second by the GEMS control loop. Supported multi-contactor units also feature opt-in 1-phase to 3-phase switching with a mandatory 20-second anti-welding cooldown.

| DEVICE & VENDOR | TEMPLATE ID | PROTOCOL & PORT | 🔍 READ OUT | 🎛️ CONTROLLED | ⚡ AUTOMATION ROLE |
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| **ABB Charger**<br>ABB | `abb_charger` | MODBUS<br>Modbus TCP via Ethernet / Wi-Fi (Default Port: 502 | Active charging power (W, reg 0x401C), Cumulative session energy (kWh, reg 0x401E). | Dynamic Charging Current Throttling (Amps, reg 0x4100, 6A to 32A in 1mA steps). | Sub-second current throttling over Modbus TCP: dynamically modulates charging current based on solar surplus (Eco mode) and throttles down immediately during capacity peaks (Flanders mode); coexists seamlessly with third-party billing/reimbursement backends. |
| **ABL eM4 Single/Twin (SBCx)**<br>ABL | `abl_em4_charger` | MODBUS<br>Modbus TCP (Default Port: 502 | Active charging power (W, reg 0x300D), Total energy (kWh, reg 0x300F). | Dynamic Charging Current Throttling (Amps, reg 0x3001, 6A to 32A). | Direct Modbus control: throttles single or dual socket charging currents to match rooftop solar surplus and Flanders capacity limits without disconnecting corporate billing. |
| **ABL eMH1 / eMH2 / eMH3**<br>ABL | `abl_emh_charger` | MODBUS<br>Modbus RTU over TCP Bridge (Default Port: 502 | Real-time charging power (W), Session energy (kWh). | Charging Current Throttling (Amps, 6A to 32A). | Integrates legacy and modern ABL eMH chargers; provides automated solar-following charging and peak shaving. |
| **AUTEL AC MaxiCharger**<br>AUTEL | `autel_charger` | OCPP 1.6-J<br>Native OCPP 1.6-J WebSocket Server (Port: 8887 | Real-time charging power (W), Session energy (kWh), Vehicle connection state, 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts). | Dynamic Current Throttling via SetChargingProfile (6A to 32A), Phase Switching (1P / 3P). | Autonomous residential charging over native OCPP: adjusts charging speed to solar excess and halts charging during peak tariff periods. |

| DEVICE & VENDOR | TEMPLATE ID | PROTOCOL & PORT | 🔍 READ OUT | 🎛️ CONTROLLED | ⚡ AUTOMATION ROLE |
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| **Alfen Charger**<br>Alfen | `alfen_charger` | MODBUS<br>Modbus TCP / Active Load Balancing license (Default Port: 502 | Active charging power (W), Total energy (kWh), Per-phase charging currents (Amps). | Modbus Slave Max Current (`1210` IEEE-754 Float32 Amps with station watchdog) and Contactor Phase Switching (`1215`: `1` = 1-Phase, `3` = 3-Phase). | Preferred Benelux Modbus TCP integration for Alfen Eve Single & Double Pro-line: sub-second solar surplus tracking, main fuse protection, and automated 1P/3P switching. |
| **Alpitronic Hypercharger**<br>Alpitronic | `alpitronic_charger` | OCPP 1.6-J<br>Modbus TCP / OCPP 1.6-J (Default Port: 502 or 8887 | Active DC charging power (kW/W), Cumulative delivered energy (kWh), Connector status. | Dynamic Charging Power Limit (kW / Amps setpoints). | Dynamically limits DC fast charging output to stay strictly below site grid connection capacity and Flemish peak limits during high-demand periods. |
| **BMW i Wallbox (Connect/Pro)**<br>BMW | `bmw_charger` | MODBUS<br>Modbus TCP via LAN (Default Port: 502 | Charging power (W), Session energy (kWh). | Dynamic Current Setpoint (Amps, 6A to 32A). | Local LAN current regulation; coordinates charging with home solar availability and off-peak EPEX price hours. |
| **Bender Charger**<br>Bender | `bender_charger` | MODBUS<br>Modbus TCP (Default Port: 502 | Active charging power (W), Session energy (kWh), Pilot state. | Bender CC612 / CC613 HEMSw Limit (`1001` 0.1 A on FW >= 4.70, fallback `1000` 1 A) and Dynamic 1-Phase / 3-Phase Switching (`1002`). | Direct local Modbus TCP control for Bender CC612/CC613 controllers (EBEE, Compleo, Mennekes OEMs) with sub-amp throttling and 1P/3P solar switching. |
| **Blitzpower / Phoenix CharX**<br>Blitzpower | `blitzpower_charger` | MODBUS<br>Modbus TCP (Default Port: 502 | Charging power (W), Cumulative energy (kWh). | Charging Current Throttling (Amps, 6A to 32A). | Fast Modbus register regulation: modulates current setpoints for solar surplus charging and peak shaving. |
| **Delta AC MAX / SLIM / Ultra Fast Charger**<br>Delta | `delta_charger` | MODBUS<br>Modbus TCP (Default Port: 502 | Active charging power (W), Total energy (kWh). | Charging Current Setpoint (Amps, 6A to 32A). | Modulates Delta wallbox current setpoint via Modbus TCP; optimizes charging speed to absorb solar production without importing expensive grid power. |
| **EN+ AC EV Charger**<br>EN+ | `enplus_charger` | OCPP 1.6-J<br>Native OCPP 1.6-J WebSocket Server (Port: 8887) | Charging power (W), Energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages. | Charging Current Throttling (Amps, 6A to 32A), Phase Switching (1P / 3P). | Autonomous solar tracking and peak shaving via OCPP SetChargingProfile commands. |
| **ETEK EKEPC2-C/S Charge Controller**<br>ETEK | `etek_charger` | MODBUS<br>Modbus TCP via Gateway (Default Port: 502 | Charging power (W), Session energy (kWh). | Dynamic Current Throttling (Amps, 6A to 32A). | Direct register writes to adjust charge rate based on real-time household power balance. |
| **EVBox Elvi / Livo / BusinessLine**<br>EVBox | `evbox_charger` | OCPP 1.6-J<br>Native OCPP 1.6-J WebSocket Server (Port: 8887 | Real-time charging power (W), Total delivered energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts), Connector status. | Dynamic Current Throttling via SetChargingProfile (6A to 32A), Phase Switching (1P / 3P). | Full native OCPP support for EVBox: dynamically throttles charging current in 1A increments to absorb rooftop solar and prevent capacity tariff penalties. |

| DEVICE & VENDOR | TEMPLATE ID | PROTOCOL & PORT | 🔍 READ OUT | 🎛️ CONTROLLED | ⚡ AUTOMATION ROLE |
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| **Easee Charger**<br>Easee | `easee_charger` | REST / HTTP<br>Easee Cloud REST API (Cloud credentials authentication) | Active charging power (W), Total session energy (kWh), Charger status. | Easee API dynamic circuit/charger current (6–32 A), session `pause_charging` / `resume_charging`, and automatic `phaseMode` 1-Phase / 3-Phase switching. | Verified smart charging control for Easee Home, Charge, and Charge Lite/Up: modulates charging amps, pauses/resumes cleanly, and switches between 1-phase and 3-phase solar modes. |
| **Enovates SingleWallBox**<br>Enovates | `enovates_charger` | MODBUS<br>Modbus TCP via LAN (Default Port: 502 | Charging power (W), Total energy (kWh). | Dynamic Current Setpoint (Amps, 6A to 32A). | Direct local Modbus control; operates in harmony with external CPO billing platforms while GEMS manages local load balancing and peak shaving. |
| **Ensto Chago Wallbox / Pro**<br>Ensto | `ensto_charger` | MODBUS<br>Modbus TCP (Default Port: 502 | Active charging power (W), Total energy (kWh). | Charging Current Throttling (Amps, 6A to 32A). | High-reliability Scandinavian commercial/home charger; throttles charge current to respect domestic supply limits and solar surplus. |
| **Etrel INCH / sonnenCharger**<br>Etrel | `etrel_charger` | MODBUS<br>Modbus TCP via Ethernet (Default Port: 502 | Active power (W), Session energy (kWh). | Dynamic Current Setpoint (Amps, 6A to 32A). | Local Modbus TCP regulation; optimizes charging to match sonnen or third-party battery states and rooftop solar. |
| **Fronius Wattpilot (Home / Go)**<br>Fronius | `fronius_wattpilot` | OCPP 1.6-J<br>Native OCPP 1.6-J WebSocket Server (Port: 8887) | Active power (W), Energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts). | Dynamic Current Throttling (Amps, 6A to 32A), Phase Switching (1P / 3P). | Operates Wattpilot in zero-cloud mode directly connected to GEMS; supports dynamic 1-phase / 3-phase switching and solar surplus tracking. |
| **GoodWe HCA Series EV Wallbox (Gen 2)**<br>GoodWe | `goodwe_charger` | MODBUS<br>Modbus TCP (Default Port: 502 | Charging power (W), Session energy (kWh). | Charging Current Setpoint (Amps, 6A to 32A). | Coordinates GoodWe wallboxes with GoodWe hybrid inverters; prioritizes battery charging before routing surplus into the EV. |
| **Hager witty (one / plus / pro / park 2)**<br>Hager | `hager_charger` | OCPP 1.6-J<br>Native OCPP 1.6-J WebSocket Server (Port: 8887) | Active power (W), Session energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages. | Dynamic Current Throttling (Amps, 6A to 32A), Phase Switching (1P / 3P). | Zero-cloud residential management of Hager witty wallboxes; solar excess following and capacity tariff peak shaving. |
| **Huawei FusionCharge (SCharger-7KS / 22KT)**<br>Huawei | `huawei_scharger` | OCPP 1.6-J<br>Native OCPP 1.6-J WebSocket Server (Port: 8887 | Active charging power (W), Session energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts), Plug state. | Dynamic Current Throttling via SetChargingProfile (6A to 32A), Phase Switching (1P / 3P). | Direct local control of Huawei FusionCharge without requiring FusionSolar cloud: coordinates perfectly with Huawei SUN2000 inverters and LUNA2000 batteries to ensure the car only charges from 100% pure solar. |
| **KEBA KeContact P30/P40**<br>KEBA | `keba_charger` | MODBUS<br>Modbus TCP via LAN (Default Port: 502 | Active charging power (W, reg 1020), Total energy (kWh, reg 1036). | KEBA KeContact P30 x/c-series & P40 Charging Current in milliamps (`5004`), Enable (`5014`), and KeContact S10 Phase Switching (`5050=3`, `5052=0/1`). | High-precision 1 mA Modbus TCP control and 1P/3P phase switching for KEBA KeContact P30 and P40 installations. |

| DEVICE & VENDOR | TEMPLATE ID | PROTOCOL & PORT | 🔍 READ OUT | 🎛️ CONTROLLED | ⚡ AUTOMATION ROLE |
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| **Mennekes AMTRON (4You / 4Business / ACU / AMEDIO)**<br>Mennekes | `mennekes_ocpp` | **OCPP 1.6-J**<br>Native OCPP 1.6-J WebSocket Server (Port: 8887) | Active power (W), Total energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages. | Dynamic Current Throttling via SetChargingProfile (6A to 32A), Phase Switching (1P / 3P). | Universal OCPP control for modern Mennekes product lines; enables solar-only charging and Flemish capacity peak prevention. |
| **Mennekes AMTRON (Xtra / Premium / HCC3 Modbus)**<br>Mennekes | `mennekes_modbus` | **MODBUS**<br>Modbus TCP via LAN (Default Port: 502 | Active charging power (W), Cumulative energy (kWh). | Dynamic Charging Current Throttling (Amps, 6A to 32A). | Direct local Modbus control for Mennekes premium hardware; eliminates cloud lag for instant peak shaving. |
| **Mercedes-Benz Wallbox**<br>Mercedes-Benz | `mercedes_wallbox` | **OCPP 1.6-J**<br>Native OCPP 1.6-J WebSocket Server (Port: 8887) | Active power (W), Energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages. | Charging Current Throttling (Amps, 6A to 32A), Phase Switching (1P / 3P). | Autonomous home charging for Mercedes-Benz Wallbox; integrates into GEMS dynamic load balancing. |
| **NRGkick (Gen 2)**<br>NRGkick | `nrgkick_gen2` | **MODBUS**<br>Modbus TCP via local Wi-Fi (Default Port: 502 | Real-time charging power (W), Energy delivered (kWh). | Charging Current Throttling (Amps, 6A to 32A), Phase Switching (1P / 3P). | Local Modbus TCP regulation for mobile/wallbox NRGkick; automatically modulates current setpoints based on solar excess. |
| **OCPP Charger**<br>Generic | `ocpp_charger` | **OCPP 1.6-J**<br>Native OCPP 1.6-J WebSocket Server (`ws://<EMS-IP>:8887/<ChargePointID>` or `ws://<EMS-IP>:8080/api/ocpp/<ChargePointID>`) | Active power (W), Delivered energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts), Plug and transaction status. | Standardized OCPP 1.6-J SetChargingProfile (Current Setpoints: 0A pause and 6A to 32A). | Vendor-agnostic central system: supports hundreds of compliant commercial and residential wallboxes worldwide; coordinates dynamic charging profiles according to GEMS site strategy. |
| **Peblar EV Charger (Home / Business / Plus)**<br>Peblar | `peblar_charger` | **MODBUS**<br>Modbus TCP (Default Port: 502 | Real-time charging power (W), Cumulative session energy (kWh). | Dynamic Current Setpoint (Amps, 6A to 32A in 0.1A steps), Phase Switching (1P / 3P). | Dutch engineering wallbox: allows GEMS to command sub-second current adjustments over Modbus TCP while Pebble/Peblar cloud handles business split-billing. |
| **Phoenix Contact Charx Charger**<br>Phoenix Contact | `phoenix_charger` | **MODBUS**<br>Modbus TCP (Default Port: 502 | Charging power (W), Total session energy (kWh). | Phoenix Contact CHARX SEC-3000 / EV-CC Modbus Charging Current Limit and Charge Enable Coil/Register. | Industrial-grade local Modbus TCP load balancing and solar surplus charging for Phoenix Contact CHARX controllers. |
| **Phoenix Contact EV-CC / CHARX (Modbus TCP)**<br>Phoenix Contact | `phoenix_contact_evcc` | **MODBUS**<br>Modbus TCP (Default Port: 502 | Active charging power (W), Total energy (kWh). | Dynamic Current Throttling (Amps, 6A to 32A), Phase Switching (1P / 3P). | Direct register writes to adjust charge rates for peak shaving and solar surplus. |
| **External CPO Wallbox (OCPP 1.6-J)**<br>Generic OCPP | `generic_cpo_ocpp` | **OCPP 1.6-J**<br>Native OCPP 1.6-J WebSocket Server (`ws://<EMS-IP>:8080/api/ocpp/<ChargePointID>`) | Active power (W), Delivered energy (kWh), 3-Phase currents and voltages. | Dynamic Current Throttling (Amps, 6A to 32A). | Local smart charging and straight-through OCPP proxy load arbitration with external CPO billing platforms based on solar excess and fuse limits. |

| DEVICE & VENDOR | TEMPLATE ID | PROTOCOL & PORT | 🔍 READ OUT | 🎛️ CONTROLLED | ⚡ AUTOMATION ROLE |
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| **Porsche Wallbox / Mobile Charger Connect**<br>Porsche | `porsche_wallbox` | OCPP 1.6-J<br>Native OCPP 1.6-J WebSocket Server (Port: 8887) | Active charging power (W), Delivered energy (kWh), 3-Phase currents and voltages. | Dynamic Current Throttling (Amps, 6A to 32A), Phase Switching (1P / 3P). | Connects Porsche chargers directly to GEMS without subscription; enforces Flanders peak limits. |
| **RAEDIAN Gemini AC Wallbox**<br>RAEDIAN<br>⭐ Reference | `raedian_gemini` | MODBUS<br>Modbus TCP via LAN (Default Port: 502 | Combined active power (W), Individual Socket 1 & Socket 2 charging power (W), Per-socket energy (kWh), Individual 3-phase currents for Socket 1 (L1, L2, L3) and Socket 2 (L1, L2, L3 in Amps), Per-socket plug status. | Continuous Dynamic Current Modulation (0A to 64A total range, dynamically split across Socket 1 & Socket 2), Individual Socket Mode (Eco / Fast / Off). | Advanced dual-socket load management: balances charging speeds between two vehicles simultaneously; dynamically adjusts combined current from 0A to 64A; protects building electrical service fuses while maximizing solar self-consumption. |
| **RAEDIAN NEO / NEX AC Wallbox**<br>RAEDIAN<br>⭐ Reference | `raedian_charger` | MODBUS<br>Modbus TCP via LAN or Wi-Fi (Default Port: 502 | Real-time active charging power (W), Session energy (kWh), 3-Phase charging currents (L1, L2, L3 in Amps), 3-Phase charging voltages (L1, L2, L3 in Volts), Socket status & error codes. | Continuous Dynamic Current Throttling (Amps, 6A to 32A in 0.1A increments), Automated Session Pause & Resume. | Preferred Modbus TCP flagship integration: single-read batching polls all 3-phase currents and voltages in a single request; sub-second current throttling responds instantly to solar surplus and Flanders capacity tariff peaks. |
| **SMA EV Charger (7.4 / 22 / eCharger)**<br>SMA | `sma_evcharger` | REST / HTTP<br>Local HTTP REST API (Default Port: 80 / 443) | Active power (W), Cumulative energy (kWh), 3-Phase currents (L1, L2, L3 in Amps). | Monitor-Only (Read-Only Telemetry — SMA EV Charger requires SEMP/SHM2 for active current control). | Monitors SMA EV Charger real-time power draw and session energy for household load accounting. |
| **Schneider Electric EVlink (Wallbox / Pro AC)**<br>Schneider Electric | `schneider_evlink` | OCPP 1.6-J<br>Native OCPP 1.6-J WebSocket Server (Port: 8887) | Active charging power (W), Energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages. | Dynamic Current Throttling via SetChargingProfile (6A to 32A), Phase Switching (1P / 3P). | Native OCPP setpoint regulation; provides smooth solar surplus charging and capacity peak avoidance. |
| **Siemens VersiCharge GEN3 Wallbox**<br>Siemens | `siemens_versicharge` | OCPP 1.6-J<br>Native OCPP 1.6-J WebSocket Server (Port: 8887 | Real-time power (W), Cumulative energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages. | Dynamic Current Throttling (Amps, 6A to 32A), Phase Switching (1P / 3P). | Connects Siemens VersiCharge GEN3 to GEMS over OCPP; executes Flanders peak shaving and solar surplus matching. |
| **Sigenergy SigenStor EVAC / EVDC Charger**<br>Sigenergy | `sigenergy_evac` | MODBUS<br>Modbus TCP via Sigen Gateway (Default Port: 502 | Charging power (W), Session energy (kWh), 3-Phase currents and voltages. | Dynamic Current Throttling (Amps, 6A to 32A), Phase Switching (1P / 3P). | Direct Modbus control over Sigen EV charging; coordinates with SigenStor battery to absorb excess solar. |

| DEVICE & VENDOR | TEMPLATE ID | PROTOCOL & PORT | 🔍 READ OUT | 🎛️ CONTROLLED | ⚡ AUTOMATION ROLE |
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| **SmartEVSE v2 / v3 (HTTP / Modbus)**<br>SmartEVSE | `smartevse_charger` | REST / HTTP<br>Local HTTP REST API / Modbus TCP (Default Port: 80 | Active power (W), Total energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts), Pilot status. | Dynamic Current Throttling (Amps, 6A to 32A via /add?current=X), Phase Switching (1P / 3P). | Local REST setpoint modulation; allows open-source charging hardware to run 100% solar surplus charging without cloud accounts. |
| **SolarEdge EV Charger**<br>SolarEdge | `solaredge_evse` | MODBUS<br>Modbus TCP via SolarEdge inverter or standalone module (Default Port: 502 / 1502 | Active charging power (W), Cumulative energy (kWh). | Dynamic Current Setpoint (Amps, 6A to 32A). | Synchronizes charging rate with SolarEdge inverter production; protects household main fuses. |
| **Veton / Phoenix CharX**<br>Veton | `veton_charger` | MODBUS<br>Modbus TCP via internal Phoenix Contact CHARX controller (Default Port: 502 | Active charging power (W), Session energy (kWh). | Dynamic Charging Current Throttling (Amps, 6A to 32A). | Direct Modbus TCP register writes; provides seamless solar surplus charging and peak shaving for premium architectural Veton chargers. |
| **Volkswagen / Elli / Skoda / SEAT (ID. Charger Connect / Pro)**<br>Volkswagen / Elli | `elli_charger` | OCPP 1.6-J<br>Native OCPP 1.6-J WebSocket Server (Port: 8887 | Active charging power (W), Total session energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts), Plug status. | Dynamic Current Throttling via SetChargingProfile (6A to 32A), Phase Switching (1P / 3P). | Direct native OCPP server integration: allows VW Group ID. Chargers to charge purely on excess solar power and avoid Flanders capacity tariff peaks without requiring expensive manufacturer cloud subscriptions. |
| **Wallbox Pulsar/Commander**<br>Wallbox | `wallbox_charger` | REST / HTTP<br>Cloud REST API (Wallbox Cloud Credentials Authentication) | Active charging power (W), Total energy (kWh), Charger status. | Wallbox API `PUT /v2/charger/{id}` dynamic amp limit (6–32 A) and `POST /v3/chargers/{id}/remote-action` (`action: 1` Resume / `2` Pause). | Verified closed-loop current modulation and automatic session pause/resume for Wallbox Pulsar Plus, Pulsar Max, Copper SB, and Commander 2. |
| **Webasto Next Charger**<br>Webasto | `webasto_charger` | MODBUS<br>Modbus TCP via LAN or Wi-Fi (Default Port: 502 | Active charging power (W, reg 5000), Delivered session energy (kWh, reg 5002). | Dynamic Charging Current Throttling (Amps, reg 5004, 6A to 32A). | Direct Modbus control: throttles Webasto wallbox within milliseconds to protect main grid fuses and prevent capacity tariff spikes while external backend handles corporate billing. |
| **Zaptec Pro/Go**<br>Zaptec | `zaptec_charger` | REST / HTTP<br>Cloud REST API (Zaptec OAuth2 authentication) | Active power (W), Cumulative energy (kWh), Session status. | Zaptec OAuth2 `/api/chargers/{id}/update` (`maxChargeCurrent` 0.1 A, `maxChargePhases` 1/3, Go 2 `threeToOnePhaseSwitchCurrent`), plus `/sendCommand/506` (Pause) & `507` (Resume). | Verified precision control for Zaptec Go, Go 2, and Pro installations with OCMF energy decoding, pause/resume, and 1P/3P solar switching. |
| **go-e Charger Gemini / HOMEfix / PRO**<br>go-e | `goe_charger` | REST / HTTP<br>Local HTTP REST API v2 (Default Port: 80 | Charging power (W), Session energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts), Authorization state. | Volatile RAM Charging Current Throttling (Amps, 6A to 32A via `/api/set?amx=X` + `frc` pause/resume — zero NVS flash wear), Phase Switching (1P / 3P via `/api/set?psm=X`). | High-speed local HTTP API: allows GEMS to switch dynamically between 1-phase (from 1.4 kW solar) and 3-phase (up to 22 kW) charging; throttles within milliseconds during domestic power spikes. |

| DEVICE & VENDOR | TEMPLATE ID | PROTOCOL & PORT | 🔍 READ OUT | 🎛️ CONTROLLED | ⚡ AUTOMATION ROLE |
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| **Generic Local REST EVSE Gateway**<br>Generic REST | `generic_rest_evse` | REST / HTTP<br>Local HTTP REST API (Default Port: 80 | Real-time Charging Power (W), Cumulative Session Energy (kWh), Charging Station State. | Dynamic Charging Current Throttling (Amps, 6A to 32A). | Distributed multi-charger orchestration and dynamic load management, balancing aggregated fleet charging against local grid connection limits. |
| **Kempower Fast Charging Infrastructure**<br>Kempower | `kempower_charger` | MODBUS<br>Modbus TCP via Kempower Central Controller (Default Port: 502 | Real-time Charging Power (W), Session Energy (kWh), 3-Phase Charging Currents (L1, L2, L3 in Amps). | Native local OCPP 1.6-J `SetChargingProfile` power/current throttling with automatic Modbus TCP register `200` (`0.1 A`) fallback. | Verified commercial fast-charging and fleet load management for Kempower Satellite and Station Chargers. |
| **MyEnergi Zappi EV Charger**<br>MyEnergi | `myenergi_zappi` | REST / HTTP<br>Local HTTP REST / CGI API (Default Port: 80 | Real-time Charging Power (W), Session Added Energy (kWh), Charging Status, Mode (Fast, Eco, Eco+). | Authenticated HTTP Digest CGI commands (`/cgi-zappi-mode-Z{sno}-{mode}-...`) with ASN redirect tracking, switching dynamically between Stop, Eco+, Eco, and Fast boost modes. | Verified local/hub control for myenergi Zappi v2/v2.1 wallboxes with full 3-phase telemetry, phase mode reporting, and active session throttling. |
| **SolaX Smart EV Charger**<br>SolaX | `solax_evcharger` | MODBUS<br>Modbus TCP via Wi-Fi/Ethernet (Default Port: 502 | Real-time Charging Power (W), Cumulative Session Energy (kWh), Phase Currents (L1, L2, L3 in Amps). | SolaX X1/X3-EVC Max Current (`0x0628` in `0.01 A` steps), Remote Mode (`0x060D=1`), Stop/Start (`0x0627`), and 1-Phase / 3-Phase Switching (`0xA105`). | Verified native Modbus TCP/RTU closed-loop control for SolaX Smart EV Chargers with 10 mA solar tracking and 1P/3P contactor switching. |
| **Teltonika TeltoCharge**<br>Teltonika Energy | `teltonika_teltocharge` | OCPP 1.6-J<br>Native OCPP 1.6-J WebSocket Server (Port: 8887 | Real-time Charging Power (W), Cumulative Session Energy (kWh), 3-Phase Currents (L1, L2, L3 in Amps), Voltages (V). | Dynamic Charging Current Throttling (Amps, 6A to 32A via SetChargingProfile). | Native OCPP integration for European residential installations; optimizes charging against Flemish capacity peaks and EPEX spot prices. |
| **Volt Time Source EV Charger**<br>Volt Time | `volttime_source` | OCPP 1.6-J<br>Modbus TCP / OCPP 1.6-J (Default Port: 502 | Real-time Charging Power (W), Cumulative Session Energy (kWh), Phase Currents (L1, L2, L3 in Amps). | Volt Time / Plugchoice API v3 dynamic amp limit & pause/resume with local Modbus TCP register `1020` fallback. | Verified active load balancing and solar surplus throttling for Volt Time Source and Plugchoice EV wallboxes. |
| **Wallbox Pulsar Plus / Commander (OCPP)**<br>Wallbox | `wallbox_ocpp` | OCPP 1.6-J<br>Native OCPP 1.6-J WebSocket Server (Port: 8887 | Real-time Charging Power (W), Cumulative Session Energy (kWh), 3-Phase Currents (L1, L2, L3 in Amps), Voltages (V). | Dynamic Charging Current Throttling (Amps, 6A to 32A via SetChargingProfile). | Direct zero-cloud local OCPP integration for Wallbox hardware; eliminates cloud dependency and latency for responsive home energy automation. |
| **cFos Power Brain Wallbox / Solar**<br>cFos | `cfos_powerbrain` | MODBUS<br>Modbus TCP (Default Port: 4701 | Real-Time Charging Power (W), Cumulative Imported Energy (kWh), 3-Phase Charging Currents (0.1 A), Active Phase Relay State. | 0.1 A Charging Current Limit (`8093`), Charge Enable (`8094`), and 1-Phase / 3-Phase Contactor Switching (`8087`). | High-precision 0.1 A solar surplus tracking, main breaker protection, and automated 1P/3P solar contactor switching. |

| DEVICE & VENDOR | TEMPLATE ID | PROTOCOL & PORT | 🔍 READ OUT | 🎛️ CONTROLLED | ⚡ AUTOMATION ROLE |
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| **FoxESS EV Charger A7300 / A011K / A022K**<br>FoxESS | `foxess_evcharger` | **MODBUS**<br>Modbus TCP / RTU (Default Port: 502 | Charging Power (W), Cumulative Energy (kWh), 3-Phase Currents (A), 3-Phase Voltages (V), Charger Status. | 0.1 kW Stepped Power Limit (`0x3002`: `1.4–4.1 kW` 1-Phase band, `4.2–22.0 kW` 3-Phase band, `0` Pause) & Auto Phase Switch (`0x300A`). | Seamless ecosystem companion for FoxESS hybrid inverters with fine-grained solar surplus modulation and 1P/3P switching. |
| **Heidelberg Energy Control / Amperfied connect.solar**<br>Amperfied (Heidelberg) | `heidelberg_ampere` | **MODBUS**<br>Modbus TCP (Port: 502 | Active Charging Power (W), Total Energy (kWh), 3-Phase Currents (0.1 A), 3-Phase Voltages (V), IEC Vehicle State, Active Phase Count. | 0.1 A Charging Current Limit (Holding `261`), Standby/Lock Management (`258`–`259`), and 1-Phase / 3-Phase Switching (`501` on `connect.solar`). | Sub-second 0.1 A current modulation for Flanders capacity peak shaving and seamless 1P/3P solar charging. |
| **openWB Pro / openWB series2 (HTTP & Modbus)**<br>openWB | `openwb_pro` | **REST / HTTP**<br>Local HTTP JSON/Form API (`/connect.php`) and openWB 2.0 Modbus TCP (Port: 1502 | Real-Time Charging Power (W), Imported Energy (kWh), 3-Phase Currents (A), 3-Phase Voltages (V), Plug & Charge State, Active Phases. | 0.1 A / 0.01 A Dynamic Current Limit (`ampere` / Holding `10171`) and Automated 1-Phase / 3-Phase Contactor Switching (`phasetarget` / `10180`). | Ultra-fine 10–100 mA solar surplus tracking and hardware 1P/3P phase switching with zero cloud dependency. |
| **Sungrow AC011E-01 / AC22E-01 EV Charger**<br>Sungrow | `sungrow_evcharger` | **MODBUS**<br>Modbus TCP / RTU (Default Port: 502 | Active Charging Power (W), Cumulative Energy (kWh), 3-Phase Currents (0.1 A), 3-Phase Voltages (0.1 V), Phase Switch Status. | 0.1 A Output Current Limit (`21202`), Start/Stop Control (`21211`), and 1-Phase / 3-Phase Switching (`21203`). | Direct local Modbus companion for Sungrow SH/SG installations with 0.1 A surplus regulation and 1P/3P switching. |
| **Tinkerforge WARP Charger 2 / 3 (Smart & Pro)**<br>Tinkerforge | `warp_charger` | **REST / HTTP**<br>Local HTTP REST API (`/evse/state` | Active Charging Power (W), Absolute Energy (kWh), 3-Phase Currents (A), 3-Phase Voltages (V), IEC 61851 State, 1P/3P Contactor State. | 1 mA Resolution External Current Limit (`6000–32000 mA`, `0` Pause) and WARP3 1-Phase / 3-Phase Contactor Switching (`phases_wanted`). | Industrial open-interface wallbox integration with 1 mA precision solar tracking and automated 1P/3P contactor control. |
| **Vestel EVC04 Home Smart / Mobility**<br>Vestel | `vestel_evc04` | **MODBUS**<br>Modbus TCP (Default Port: 502 | Total Active Power (W), Cumulative Energy (kWh), 3-Phase Currents (mA), 3-Phase Voltages (V), Chargepoint & Cable State. | Dynamic Charging Current Limit (`5004`, `0` or `6–32 A`), Fail-Safe Watchdog (`6000`), and 1-Phase / 3-Phase Switching (`405`). | Reliable local Modbus TCP load balancing, main fuse protection, and 1P/3P solar switching for Vestel EVC04 wallboxes. |
| **Hardy Barth cPµ1 / cPµ2 (Salia / eCB1) Wallbox**<br>Hardy Barth | `hardy_barth_charger` | **REST / HTTP**<br>Local HTTP REST API (Default Port: 80 | Real-Time Charging Power (W), Cumulative Session Energy (kWh), 3-Phase Charging Currents (`L1, L2, L3` in Amps) & Voltages (`L1, L2, L3` in Volts). | Dynamic Local REST Current Limit (`PUT /api/v1/chargecontrols/1/mode` with `manualmodeamp=6..32 A` or `mode=stop`; Salia `PUT /api/secc` `grid_current_limit` fallback). | 100% local zero-cloud current modulation for Hardy Barth Salia and eCB1 wallboxes: matches EV charging speed to rooftop solar surplus and throttles down during main breaker or capacity peaks. |

## 📊 4. Grid Smart Meters & Energy Analyzers (40 Devices)

Smart meter templates interface directly via P1 port serial (RJ12 cable at 115200 baud), network P1 bridges, or RS485/Modbus TCP meters. They supply net grid import/export power (W), lifetime energy totals (kWh), per-phase currents (L1, L2, L3), per-phase voltages, and the official 15-minute rolling average peak demand register required for Flemish capacity tariff calculations.

| DEVICE & VENDOR | TEMPLATE ID | PROTOCOL & PORT | 🔍 READ OUT | 🎛️ CONTROLLED | ⚡ AUTOMATION ROLE |
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| **ABB A43 / A44 / B23 / B24 Meter**<br>ABB | `abb_meter` | **MODBUS**<br>Modbus RTU via RS485-to-Ethernet Gateway or Modbus TCP (Default Port: 502 | Active grid import/export power (W), Total active import/export energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts), Frequency (Hz). | Read-only grid telemetry. | High-precision grid boundary metering; feeds continuous 3-phase currents into GEMS phase-load balancing engine to prevent tripping main service fuses; drives Flanders 15-minute capacity peak calculations. |
| **Acrel ADW300 Meter**<br>Acrel | `acrel_adw300` | **MODBUS**<br>Modbus TCP (Default Port: 502 | Total active power (W), Import/export energy (kWh), Per-phase active power (W), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts). | Read-only grid telemetry. | Monitors main grid connection or sub-distribution panels; informs GEMS about exact phase loads to modulate EV charger speeds and safeguard service fuses. |
| **B+GE-TECH WS100 / DS100 Meter**<br>B+GE-TECH | `bgetech_ws100` | **MODBUS**<br>Modbus TCP / RTU Gateway (Default Port: 502 | Real-time import/export power (W), Total active energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts). | Read-only grid telemetry. | Direct grid boundary metering; provides real-time wattage for closed-loop EV charging solar surplus tracking and Flanders peak shaving. |
| **Carlo Gavazzi EM112 / EM340 / EM540**<br>Carlo Gavazzi | `carlogavazzi_em340` | **MODBUS**<br>Modbus RTU via RS485 USB/Ethernet Gateway (Default Port: 502 | Net active power (W), Total import/export kWh, 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts). | Read-only grid telemetry. | Fast sub-second boundary meter (especially EM540); crucial for instant reactive throttling of EV chargers during household spikes to protect main fuses. |
| **Carlo Gavazzi EM24 (RS485/Ethernet)**<br>Carlo Gavazzi | `carlogavazzi_em24` | **MODBUS**<br>Modbus TCP / Modbus RTU Gateway (Default Port: 502 | Total active power (W, signed: + import, - export), Cumulative energy registers (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts). | Read-only grid telemetry. | Industry-standard utility meter used in premium solar and EVSE installations; provides deterministic 1-second telemetry for GEMS peak shaving and dynamic load balancing. |
| **ESPHome DSMR / DLMS Smart Meter**<br>ESPHome | `esphome_meter` | **REST / HTTP**<br>Local HTTP REST JSON API (Default Port: 80 | Real-time grid power (W), Total import/export energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts), Gas consumption (m³). | Read-only REST telemetry. | Integrates DIY open-hardware P1 readers into GEMS; delivers full digital telegram data over Wi-Fi with zero cloud dependencies. |
| **Eastron SDM120 / SDM220 / SDM230 (1-Phase Meter)**<br>Eastron | `eastron_sdm120_230` | **MODBUS**<br>Modbus RTU via RS485-to-Ethernet Bridge or USB dongle (Default Port: 502 | Active power (W), Total import/export energy (kWh), Phase current (Amps), Line voltage (Volts), Power factor. | Read-only telemetry poller. | Monitors single-phase grid connections, dedicated heat pump sub-circuits, or single-phase PV inverters; integrates into GEMS load analysis. |
| **Eastron SDM630 3-Phase Meter**<br>Eastron | `eastron_sdm630` | **MODBUS**<br>Modbus RTU via RS485 Gateway or Modbus TCP (Default Port: 502 | Net total active power (W, reg 0x0034), Total import/export energy (kWh), Individual phase currents (L1, L2, L3 in Amps, reg 0x0006), Individual phase voltages (L1, L2, L3 in Volts, reg 0x0000). | Read-only grid telemetry. | Widely deployed 3-phase meter; provides foundational telemetry for GEMS Flanders capacity tariff management, phase imbalance tracking, and solar surplus routing. |
| **Eastron SDM72D / SMART X96 3-Phase Meter**<br>Eastron | `eastron_sdm72_x96` | **MODBUS**<br>Modbus RTU via RS485-to-Ethernet Bridge (Default Port: 502 | Total active power (W), Active import/export energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts). | Read-only grid telemetry. | High-density DIN-rail energy monitoring; feeds live grid metrics into GEMS strategy loop to throttle chargers and log energy reports. |

| DEVICE & VENDOR | TEMPLATE ID | PROTOCOL & PORT | 🔍 READ OUT | 🎛️ CONTROLLED | ⚡ AUTOMATION ROLE |
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| **Eltako DSZ15DZMOD / DSZ16 Meter**<br>Eltako | `eltako_meter` | **MODBUS**<br>Modbus RTU via RS485 Gateway (Default Port: 502 | Active power (W), Import/export energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts). | Read-only grid telemetry. | German precision submetering; provides real-time phase currents and grid import/export for GEMS household overload protection. |
| **Finder 7M.24 / 7M.38 Meter**<br>Finder | `finder_7m` | **MODBUS**<br>Modbus RTU via RS485 Bridge (Default Port: 502 | Active power (W), Import/export active energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts). | Read-only grid telemetry. | Reliable MID-certified grid metering; feeds live electrical data into GEMS for closed-loop dynamic current regulation. |
| **HomeWizard Meter**<br>HomeWizard | `homewizard_meter` | **REST / HTTP**<br>Local HTTP REST API v1 (Default Port: 80 | Active grid power (W), Total import/export energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts), Coupled gas meter consumption (m³). | Read-only local REST API. | Plug-and-play Wi-Fi P1 reader: enables GEMS to poll digital meter telegrams every 1-2 seconds with zero cabling; supports full Flemish & Dutch capacity tariff tracking. |
| **Homey Energy Dongle (P1)**<br>Homey | `homey_p1` | **REST / HTTP**<br>Local Network UDP Broadcast / HTTP (Default Port: 80) | Active grid power (W), Total import/export energy (kWh), 3-Phase currents and voltages. | Read-only telemetry poller. | Captures local network telemetry broadcasts from Homey P1 dongles; streams live household grid flow into GEMS dashboard. |
| **Inepro PRO380-MOD / PRO1-MOD Meter**<br>Inepro | `inepro_pro380` | **MODBUS**<br>Modbus RTU via RS485 Gateway (Default Port: 502 | Total active power (W), Import/export kWh registers, 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts). | Read-only grid telemetry. | High-accuracy Dutch utility metering; supplies 3-phase currents for GEMS dynamic load balancing and fuse protection. |
| **Lovato DMG610 3-Phase Meter**<br>Lovato | `lovato_dmg610` | **MODBUS**<br>Modbus TCP / RTU (Default Port: 502 | Total active power (W), Import/export energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts). | Read-only grid telemetry. | Industrial-grade grid analyzer; provides rapid electrical metrics to protect main supply lines from transformer or fuse overload. |
| **Loxone Miniserver**<br>Loxone | `loxone_miniserver` | **REST / HTTP**<br>Local HTTP JSON API (Default Port: 80 / 443 | Aggregated household power (W), Submeter power values (W), Total energy counters (kWh). | Read-only REST telemetry poller. | Bridges smart home installations running Loxone Miniserver into GEMS; allows GEMS to utilize Loxone smart meter data as the primary grid reference. |
| **Niko Home Control (NHC II)**<br>Niko | `niko_nhc` | **REST / HTTP**<br>Local Hobby API / REST (Default Port: 8884 or 80 | Real-time grid import/export power (W), Sub-circuit power consumption (W), Cumulative energy (kWh). | Read-only local API. | Seamless integration with popular Belgian Niko Home Control installations; reads electricity clamp measurements to drive GEMS peak shaving. |
| **P1 Smart Meter (Network/TCP)**<br>Generic | `p1_network` | **P1 / SERIAL**<br>Raw TCP Socket stream (Default Port: 23 | Real-time active power (+ import / - export in Watts, OBIS 1.7.0 & 2.7.0), Cumulative tariff 1 & 2 energy (kWh, OBIS 1.8.1/2 & 2.8.1/2), 3-Phase currents (L1, L2, L3 in Amps, OBIS 31.7.0, 51.7.0, 71.7.0), 3-Phase voltages (L1, L2, L3 in Volts, OBIS 32.7.0, 52.7.0, 72.7.0), Flemish 15-minute peak demand register (W, OBIS 1.6.0), Gas consumption (m³, OBIS 0-1:24.2.1). | Read-only digital telegram stream. | Primary grid interface for remote P1 meters over LAN: decodes official Fluvius (Belgium) and DSMR 4.x/5.x (Netherlands) telegrams; powers GEMS predictive Flanders capacity tariff engine and adaptive peak ceiling. |

| DEVICE & VENDOR | TEMPLATE ID | PROTOCOL & PORT | 🔍 READ OUT | 🎛️ CONTROLLED | ⚡ AUTOMATION ROLE |
|---|---|---|---|---|---|
| **P1 Smart Meter (USB/Serial)**<br>Generic | `p1_serial` | **P1 / SERIAL**<br>Direct Serial TTY (/dev/ttyUSB0 or /dev/ttyAMA0 | Real-time active power (W, OBIS 1.7.0/2.7.0), Tariff 1/2 import/export energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts), Belgian quarterly maximum peak demand (W, OBIS 1.6.0), Gas meter volume (m³). | Read-only digital telegram stream. | Direct zero-latency hardware connection to Belgian/Dutch smart meters; provides 1-second CRC-validated telegrams; core foundation for sub-second EV load balancing and Flanders capacity peak avoidance. |
| **SMA Energy Meter 2.0 / Data Manager M (Modbus TCP)**<br>SMA | `sma_energymeter` | **MODBUS**<br>Modbus TCP (Default Port: 502 | Active grid import/export power (W), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts). | Active Power Curtailment Dispatch (commands SMA inverters via Data Manager). | Measures grid boundary power and acts as a gateway; informs GEMS about net exchange; executes active solar feed-in curtailment via SMA communications. |
| **Schneider Electric Acti9 iEM3000 / PM3200 Series**<br>Schneider Electric | `schneider_iem3000` | **MODBUS**<br>Modbus RTU via RS485 Gateway or Modbus TCP (Default Port: 502 | Total active power (W), Active energy import/export (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts). | Read-only grid telemetry. | Commercial and residential grade submetering; provides deterministic phase current telemetry to safeguard distribution boards from overloading. |
| **Shelly 3EM (Gen 1 3-Phase Meter)**<br>Shelly | `shelly_3em` | **REST / HTTP**<br>Local HTTP REST API (Default Port: 80 | Total active power (W), Per-phase power (W), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts), Cumulative returned/consumed energy (kWh). | Built-in 10A Contactor Output (Turn ON / Turn OFF via /relay/0?turn=on). | Reads high-speed 3-phase clamp measurements over Wi-Fi; built-in contactor can be automated to trigger external thermal loads or disconnect non-essential appliances during capacity peaks. |
| **Shelly Pro 3EM / Gen3 3EM-63T (3-Phase Meter)**<br>Shelly | `shelly_pro_3em` | **REST / HTTP**<br>Local Gen 2/3 HTTP RPC (Default Port: 80 | Net active power across phases (W), Per-phase active power (W), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts), 4-Quadrant energy counters (kWh). | Built-in Auxiliary Relay Output (Turn ON / Turn OFF via /rpc/Switch.Set). | High-reliability Ethernet-connected DIN meter: feeds instant phase telemetry into GEMS for sub-second EV current throttling; auxiliary relay output automates heat pump boost or domestic hot water. |
| **Siemens SENTRON (PAC2200 / 7KT1665 Meter)**<br>Siemens | `siemens_pac2200` | **MODBUS**<br>Modbus TCP (Default Port: 502 | Total active power (W), Cumulative import/export energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts). | Read-only grid telemetry. | Premium industrial grade metering; monitors connection capacity and streams telemetry to GEMS for active load shedding and ROI calculations. |
| **Socomec COUNTIS E24/E34/E44 / DIRIS A-40**<br>Socomec | `socomec_meter` | **MODBUS**<br>Modbus RTU via RS485 Gateway (Default Port: 502 | Total active power (W), Active energy import/export (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts). | Read-only grid telemetry. | Accurate 3-phase commercial and domestic metering; informs GEMS about phase load distribution to balance EV charging and avoid tripping supply fuses. |
| **WAGO 879 Series (879-3000 / 879-3020 / 879-3040)**<br>WAGO | `wago_879` | **MODBUS**<br>Modbus TCP (Default Port: 502 | Active power (W), Total active energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts). | Read-only grid telemetry. | Precision DIN-rail meter; streams 3-phase currents and voltages into GEMS for real-time fuse protection, solar surplus calculations, and energy flow visualization. |
| **Accuenergy AcuRev / Acuvim Series Meter**<br>Accuenergy | `accuenergy_meter` | **MODBUS**<br>Modbus TCP / RTU via RS485 Gateway (Default Port: 502 | Real-Time Net Grid Active Power (W), Total Energy (kWh), Phase Currents (L1, L2, L3 in Amps), Voltages (V). | Autonomous / Read-Only grid telemetry. | High-precision 3-phase grid boundary monitoring and phase load balancing for commercial and residential installations. |

| DEVICE & VENDOR | TEMPLATE ID | PROTOCOL & PORT | 🔍 READ OUT | 🎛️ CONTROLLED | ⚡ AUTOMATION ROLE |
|---|---|---|---|---|---|
| **Algodue UPM209 / UPM309 Series Meter**<br>Algodue | `algodue_meter` | **MODBUS**<br>Modbus RTU / TCP via RS485 Gateway (Default Port: 502 | Real-Time Net Grid Active Power (W), Total Energy (kWh), Phase Currents (L1, L2, L3 in Amps), Voltages (V). | Autonomous / Read-Only grid telemetry. | Continuous grid exchange measurement, power factor monitoring, and breaker overload protection. |
| **Chint DTSU666 / DTSU666-H 3-Phase Meter**<br>Chint | `chint_dtsu666` | **MODBUS**<br>Modbus RTU via RS485 / Modbus TCP Gateway (Default Port: 502 | Real-Time Net Grid Active Power (W), Total Import & Export Energy (kWh), Phase Currents (L1, L2, L3 in Amps), Voltages (V). | Autonomous / Read-Only grid telemetry. | Standard 3-phase bidirectional grid meter integration; drives solar surplus calculations and EV charger current modulation. |
| **Hager ECM / ECR / ECP Energy Meter**<br>Hager | `hager_meter` | **MODBUS**<br>Modbus RTU / TCP via Agardio / IP Gateway (Default Port: 502 | Real-Time Net Grid Power (W), Total Import Energy (kWh), Phase Currents (L1, L2, L3 in Amps), Voltages (V). | Autonomous / Read-Only grid telemetry. | DIN-rail residential distribution board monitoring, feeding sub-second phase telemetry into GEMS overload protection. |
| **Janitza UMG Series Power Analyzer**<br>Janitza | `janitza_umg` | **MODBUS**<br>Modbus TCP (Default Port: 502 | Real-Time Net Grid Power (W), Total Import Energy (kWh), Phase Currents (L1, L2, L3 in Amps), Line Voltages (V), Frequency (Hz). | Autonomous / Read-Only grid telemetry. | High-precision industrial power analyzer integration, capturing high-frequency grid flow and phase imbalance metrics. |
| **Kamstrup Energy Meter (Modbus Gateway)**<br>Kamstrup | `kamstrup_meter` | **MODBUS**<br>Modbus RTU / TCP via M-Bus / Modbus Gateway (Default Port: 502 | Real-Time Net Grid Power (W), Total Import Energy (kWh), Phase Currents (L1, L2, L3 in Amps), Voltages (V). | Autonomous / Read-Only grid telemetry. | Integrates utility electricity and district heating/cooling submeters into the GEMS overall energy overview. |
| **Landis+Gyr Industrial Utility Meter**<br>Landis+Gyr | `landisgyr_meter` | **MODBUS**<br>Modbus RTU / TCP via Communication Unit (Default Port: 502 | Real-Time Net Grid Power (W), Total Import & Export Energy (kWh), Phase Currents (L1, L2, L3 in Amps), Voltages (V). | Autonomous / Read-Only grid telemetry. | Monitors large commercial and industrial connection points to ensure dynamic peak shaving compliance. |
| **Phoenix Contact EMpro Energy Meter**<br>Phoenix Contact | `phoenix_empro` | **MODBUS**<br>Modbus TCP (Default Port: 502 | Real-Time Net Grid Power (W), Total Import Energy (kWh), Phase Currents (L1, L2, L3 in Amps), Voltages (V). | Autonomous / Read-Only grid telemetry. | Streams real-time 3-phase grid boundary measurements into GEMS for instantaneous solar tracking. |
| **Weidmüller Energy Meter (500 / 600 / 700)**<br>Weidmüller | `weidmuller_meter` | **MODBUS**<br>Modbus TCP (Default Port: 502 | Real-Time Net Grid Power (W), Total Import Energy (kWh), Phase Currents (L1, L2, L3 in Amps), Voltages (V). | Autonomous / Read-Only grid telemetry. | Industrial grade power monitoring for multi-tenant and facility sub-distribution boards. |
| **YouLess LS120 Pulse & P1 Meter**<br>YouLess | `youless_ls120` | **REST / HTTP**<br>Local HTTP REST API (Default Port: 80 | Real-Time Net Grid Power (W), Total Import Energy (kWh). | Autonomous / Read-Only grid telemetry. | Connects to digital smart meter P1 ports or legacy optical rotating disc meters over local LAN without cloud subscriptions. |
| **Fronius Smart Meter IP / Solar API v1**<br>Fronius | `fronius_smartmeter_ip` | **REST / HTTP**<br>Local HTTP Fronius Solar API v1 (`GET /solar_api/v1/GetMeterRealtimeData.cgi?Scope=Device&DeviceId=0`) | Signed Net Grid Power (W), Total Imported & Exported Energy (kWh), Signed Per-Phase Currents (L1, L2, L3 in Amps), Phase Voltages (V). | Autonomous / Read-Only | Zero-wiring local HTTP grid boundary metering for Fronius ecosystems, feeding signed per-phase currents into main fuse protection and peak shaving. |
| **Huawei EMMA-A02 / DTSU666-H Smart Power Sensor**<br>Huawei | `huawei_emma` | **MODBUS**<br>Modbus TCP / RTU (Port: 502 | Signed Grid Supply-Point Power (W), Total Import & Export Energy (kWh), Signed 3-Phase Currents (A), 3-Phase Voltages (V). | Autonomous / Read-Only | High-speed grid boundary telemetry for Huawei SUN2000 & LUNA2000 sites with EMMA-A02 or standalone RS485 power sensors. |
| **KOSTAL Smart Energy Meter (KSEM G1 / G2)**<br>KOSTAL | `kostal_ksem` | **MODBUS**<br>Modbus TCP (Default Port: 502 | Active Import & Export Power (0.1 W), Total Import & Export Energy (0.1 Wh → kWh), Directional Per-Phase Currents (A) & Voltages (V). | Autonomous / Read-Only | Sub-second Ethernet DIN-rail grid boundary metering with per-phase directional current signing for DLB and capacity peak shaving. |

| DEVICE & VENDOR | TEMPLATE ID | PROTOCOL & PORT | 🔍 READ OUT | 🎛️ CONTROLLED | ⚡ AUTOMATION ROLE |
|---|---|---|---|---|---|
| **Tasmota SML / OBIS Optical IR Reader & EcoTracker**<br>Tasmota | `tasmota_sml` | **REST / HTTP**<br>Local HTTP JSON API (`/cm?cmnd=Status%208` StatusSNS & EcoTracker `/v1/json`) | Real-Time Net Grid Power (W), Total Import & Export Energy Counters (kWh), Phase Currents (A) & Voltages (V). | Autonomous / Read-Only | Cloud-free optical IR smart meter reading for modern European digital utility meters lacking RJ12 P1 ports. |

## 🔌 5. Smart Relays & Load Actuators (12 Devices)

Actuator templates integrate potential-free contactors, smart DIN plugs, and SG-Ready heat pump inputs over local LAN REST/RPC or Modbus coils. GEMS automatically triggers domestic hot water (DHW) immersion heaters, heat pump boost states, or pool pumps when solar surplus or negative pricing thresholds are reached, with built-in minimum run-time hysteresis.

| DEVICE & VENDOR | TEMPLATE ID | PROTOCOL & PORT | 🔍 READ OUT | 🎛️ CONTROLLED | ⚡ AUTOMATION ROLE |
|---|---|---|---|---|---|
| **Generic HTTP Relay**<br>Generic | `generic_relay` | **REST / HTTP**<br>Local HTTP REST (Default Port: 80 / 8080 | Active power consumption of connected appliance (W, optional if meter reported). | Relay Contactor Output State (Turn ON / Turn OFF). | Automated contactor control: engages domestic hot water immersion heaters, pool circulation pumps, or resistive heating when solar export exceeds configurable thresholds (e.g., > 1500W for 2 minutes); enforces minimum run-time and off-time hysteresis. |
| **Shelly Plus 1PM / Pro 1PM / Plug S (Relay & Power)**<br>Shelly | `shelly_plus_1pm` | **REST / HTTP**<br>Local Gen 2/3 HTTP RPC (Default Port: 80 | Connected appliance active power (W), Cumulative energy consumption (kWh), Internal device temperature (°C), Supply voltage (V). | Relay Contactor Output State (Turn ON / Turn OFF via /rpc/Switch.Set). | SG-Ready Heat Pump Boost & Appliance Automation: drives Heat Pump SG-Ready Operating State 3 (thermal boost mode) or hot water boilers during solar surplus or negative spot price hours; tracks real-time load wattage in GEMS logger. |
| **Modbus Digital I/O Relay Module**<br>Modbus Relais / I/O | `modbus_relay` | **MODBUS**<br>Modbus TCP / RTU via RS485 (Default Port: 502 | Digital Input / Output Coil States (Open / Closed). | Relay Coil Actuation (Coil 1–8 Turn ON / OFF). | Industrial-grade contactor switching for high-current loads, three-phase thermal heaters, and heavy machinery peak shedding. |
| **SG-Ready Heat Pump (Daikin/Nibe/Bosch/Vaillant/Panasonic)**<br>SG-Ready (Daikin, Nibe, Bosch/Nefit, Vaillant, Panasonic, Mitsubishi, Stiebel Eltron, Wolf) | `sgready_heatpump` | **REST / HTTP**<br>Local HTTP REST / Relay Controller (Default Port: 80 | Operating State (Operating State 1: Lock / State 2: Normal / State 3: Boost / State 4: Fast Heat), Power Consumption (W). | SG-Ready Operating Mode (State 2 Normal vs. State 3 Thermal Boost). | Automated heat pump thermal storage: commands State 3 (increased thermal setpoint) during solar surplus or negative dynamic price hours to preheat domestic water and buffers; returns to State 2 during grid peaks. |
| **Shelly Plus 1 / Pro 1 (Dry Contact Relay)**<br>Shelly | `shelly_plus_1` | **REST / HTTP**<br>Local HTTP RPC (Default Port: 80 | Switch State (On / Off), Device Temperature (°C), Input State. | Dry Contact Relay Output State (Turn ON / Turn OFF). | Potential-free contact closure to trigger SG-Ready heat pump inputs, EV charger external inhibit terminals, or domestic circulation pumps. |
| **Shelly Plus 2PM / Pro 2PM (Dual Relay & Meter)**<br>Shelly | `shelly_plus_2pm` | **REST / HTTP**<br>Local HTTP RPC (Default Port: 80 | Channel 1 & Channel 2 Active Power (W), Cumulative Energy per channel (kWh), Supply Voltage (V). | Dual Relay Output States (Channel 0 & Channel 1 Independent On / Off). | Independent dual-circuit automation: controls and measures multiple resistive heating stages or controls dual-coil SG-Ready heat pump configurations (State 1 through State 4). |

| DEVICE & VENDOR | TEMPLATE ID | PROTOCOL & PORT | 🔍 READ OUT | 🎛️ CONTROLLED | ⚡ AUTOMATION ROLE |
|---|---|---|---|---|---|
| **Daikin Altherma 3 & HomeHub (Smart Grid Modbus)**<br>Daikin | `daikin_altherma` | **MODBUS**<br>Modbus TCP (Default Port: 502 | Instantaneous Heat Pump Electrical Power (W), DHW Tank Temperature (°C), Leaving Water Temperature (°C), Smart Grid Status. | Smart Grid Operating Mode (`53`: Free Running Normal vs Recommended On PV Surplus Boost) & DHW Temperature Setpoint (`2`). | Native Modbus TCP thermal storage automation for Daikin Altherma 3 heat pumps: converts excess PV and negative-price electricity into domestic hot water and space heating. |
| **my-PV AC•THOR / AC•THOR 9s / AC ELWA 2 Solar Diverter**<br>my-PV | `mypv_acthor` | **MODBUS**<br>Modbus TCP (Default Port: 502 | Actual Immersion Heating Power (W), Domestic Hot Water Tank Temperature (`0.1 °C`), Operating Status. | Stepless Linear Heating Power Setpoint (`1000`: `0–9000 W`), Relay Boost State, and Max Tank Temperature Setpoint (`1002`). | Proportional zero-export thermal storage: diverts exact watt-for-watt PV surplus into hot water buffers before any grid export or curtailment occurs. |
| **NIBE S-Series Heat Pump (S1155 / S1255 / S2125 / VVM)**<br>NIBE | `nibe_heatpump` | **MODBUS**<br>Modbus TCP (Default Port: 502 | Instantaneous Compressor Electrical Power (`2166` W), Cumulative Consumed Electrical Energy (`2285` kWh), DHW Top Temperature (`8` °C). | SG-Ready / Smart Price Luxury DHW Boost (`697` / `47`) and Comfort Temperature Setpoint (`707` `0.1 °C`). | Direct Wi-Fi/Ethernet Modbus TCP integration for NIBE S-Series heat pumps: boosts thermal storage during solar surplus and cheap EPEX hours. |
| **Stiebel Eltron / Tecalor ISG Web & Plus (WPM)**<br>Stiebel Eltron | `stiebel_eltron_isg` | **MODBUS**<br>Modbus TCP (Default Port: 502 | DHW Cylinder Temperature (`521` °C), Compressor & DHW Pump Status (`2500`), Cumulative Electrical Energy (kWh). | SG-Ready Energy Management Enable (`4000=1`), SG-Ready State 2 / State 3 PV Boost (`4001`–`4002`), and Comfort DHW Setpoint (`1509`). | Standardized Modbus TCP SG-Ready boost and comfort temperature control for Stiebel Eltron and Tecalor heat pumps. |
| **Tasmota Smart Plug / DIN Power Relay**<br>Tasmota | `tasmota_relay` | **REST / HTTP**<br>Local HTTP Command API (`/cm?cmnd=Power%20On\|Off` & `/cm?cmnd=Status%208`) | Active Load Power (W), Cumulative Energy (kWh), Supply Voltage (V), Load Current (A). | Relay Channel Output State (`Power{1..8}` `On` / `Off` with hysteresis protection). | 100% local cloud-free switching and power metering for resistive water heaters, pool pumps, e-bike chargers, and auxiliary relays. |
| **Askoma ASKOHEAT+ Modbus TCP PV Immersion Heater**<br>Askoma | `askoheat_diverter` | **MODBUS**<br>Modbus TCP (Default Port: 502 | Actual Immersion Heating Power (`300` W), Boiler Water Temperature (`302` `0.1 °C`), Active Heater Stage Bitmask (`301`), Cumulative Thermal Energy (`304` `0.1 kWh`). | Stepless Feed-In / Load Power Setpoint (`200`–`201` W), 7-Step Heater Stage Control (`100` Stages 0–7), Legionella Boost, and Maximum Water Temperature Setpoint (`202` `0.1 °C`). | Proportional 7-step and stepless solar thermal diversion for Askoma ASKOHEAT+ immersion heaters: converts excess solar generation and negative-price electricity into domestic hot water buffer storage. |

# 4. Detailed Hardware Specification by Category (All 168 Devices)

This section provides the complete technical breakdown for every single native hardware template supported by GEMS, detailing hardware models, connection interfaces, readable telemetry, controllable setpoints, and practical automation roles.

## ☀️ Category 1: Solar & Hybrid Inverters (37 Native Templates)

Inverter templates interface via Modbus TCP (SunSpec or vendor-specific registers) or local HTTP REST APIs. They continuously report photovoltaic (PV) generation, battery charge/discharge wattage, state of charge (SOC %), and grid flow. Advanced units support real-time active power curtailment for negative spot price protection and battery force-charge/discharge commands for day-ahead spot arbitrage.

### 1.1 Afore Hybrid Inverter `afore_inverter` MODBUS

**Manufacturer / Vendor:** Afore

**Communication Interface:** Modbus TCP via Ethernet / Wi-Fi dongle (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Afore AF-T (3-Phase) & AF-S (1-Phase) Series Hybrid Inverters (1kW – 30kW)

🔍 **TELEMETRY READ OUT**
Real-time solar PV generation (W), Battery charge/discharge power (W), Grid import/export power (W), Cumulative solar production (kWh), Battery State of Charge (SOC %), 3-Phase inverter output currents (L1, L2, L3 in Amps).

🎛️ **CONTROLLED SETPOINTS**
EMS Energy Mode & Forced Charge/Discharge Power Setpoints (Holding `3000`/`3002`), plus Active Power Derating Limit (`0–100%`).

⚡ **HANDLED & AUTOMATION ROLE**
Closed-loop hybrid management for Afore AF-S and AF-T inverters: dynamically charges/discharges storage for Flanders capacity peak shaving and EPEX spot arbitrage while curtailing PV export during negative price windows.

### 1.2 Alpha ESS Storion SMILE `alpha_ess_inverter` NETWORK TCP

**Manufacturer / Vendor:** Alpha ESS

**Communication Interface:** Modbus TCP via local network dongle (Default Port: 502 or 8502, Slave ID: 85)

**Supported Hardware Models:** Alpha ESS Storion SMILE5, SMILE-T10, SMILE-G3 Hybrid Inverter & Storage Systems

🔍 **TELEMETRY READ OUT**
Solar PV active power (W), Battery charging/discharging power (W), Net grid import/export (W), Cumulative energy (kWh), Battery SOC (%), 3-Phase currents (L1, L2, L3 in Amps).

🎛️ **CONTROLLED SETPOINTS**
Dispatch Mode Control (`0x084F`), Active Power Offset Setpoint (`0x0850`–`0x0851` with 32000 W bias), Dispatch SOC Target, and Grid Feed-In Power Limit.

⚡ **HANDLED & AUTOMATION ROLE**
Full closed-loop Alpha ESS Storion SMILE control: dispatches exact wattage charge/discharge setpoints for EPEX Day-Ahead arbitrage and Flemish capacity ceiling protection, plus active solar curtailment.

### 1.3 Anker SOLIX X1 Hybrid Inverter `anker_inverter` MODBUS

**Manufacturer / Vendor:** Anker

**Communication Interface:** Modbus TCP (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Anker SOLIX X1 Hybrid Energy Storage System (Single-Phase & Three-Phase)

🔍 **TELEMETRY READ OUT**
Solar PV power (W), Battery flow (W), Grid exchange power (W), Total lifetime energy (kWh), Battery SOC (%), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts).

🎛️ **CONTROLLED SETPOINTS**
Anker SOLIX X1 EMS Operating Mode and Forced Charge/Discharge Power Setpoints (W), plus Active Power Export Curtailment.

⚡ **HANDLED & AUTOMATION ROLE**
Closed-loop residential 3-phase hybrid control for Anker SOLIX X1: executes automated battery peak shaving, off-peak spot charging, and negative-price solar curtailment.

### 1.4 Deye / Sunsynk Hybrid Inverter `deye_inverter` MODBUS

**Manufacturer / Vendor:** Deye

**Communication Interface:** Modbus TCP via Ethernet LAN or Solarman/Deye Wi-Fi logger (Default Port: 502 or 8899, Slave ID: 1)

**Supported Hardware Models:** Deye SUN-(3-12)K-SG04LP3 / SUN-(5-16)K-SG01LP1, Sunsynk Hybrid Inverters

🔍 **TELEMETRY READ OUT**
Solar PV production (W), Battery charge/discharge power (W), Grid meter power (W), Lifetime energy (kWh), Battery SOC (%), 3-Phase inverter currents (L1, L2, L3 in Amps).

🎛️ **CONTROLLED SETPOINTS**
Time-of-Use Grid Charge & Discharge Power Setpoints, Minimum/Target SOC Windows, and Dynamic Max Solar Sell Power Limit (Holding `143` / `340`, LV & HV auto-scaled).

⚡ **HANDLED & AUTOMATION ROLE**
Full closed-loop control for Deye and Sunsynk LV/HV hybrid inverters: executes sub-second Flemish capacity peak shaving, Day-Ahead EPEX battery arbitrage, and zero-export / negative-price curtailment.

### 1.5 Enerlution Hybrid Inverter `enerlution_inverter` MODBUS

**Manufacturer / Vendor:** Enerlution

**Communication Interface:** Modbus TCP (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Enerlution Hybrid Energy Storage Inverters (L3-L12 Series)

🔍 **TELEMETRY READ OUT**
Solar PV power (W), Battery power (W), Grid power (W), Cumulative energy (kWh), Battery SOC (%), 3-Phase currents (L1, L2, L3 in Amps).

🎛️ **CONTROLLED SETPOINTS**
Active Power Curtailment Limit (W or % via Modbus registers 40400/40441).

⚡ **HANDLED & AUTOMATION ROLE**
Full active curtailment integration: automatically reduces inverter power output during negative EPEX Day-Ahead spot price hours (Netherlands Mode / Zero-Export) while respecting Flemish GSC subsidy offsets; manages solar surplus routing into battery and EV.

### 1.6 Enphase IQ Gateway / Envoy `enphase_envoy`

REST / HTTP

**Manufacturer / Vendor:** Enphase

**Communication Interface:** Local HTTP REST API via token/digest authentication (Default Port: 80 / 443, Endpoints: /api/v1/production, /ivp/livedata/status)

**Supported Hardware Models:** Enphase IQ Gateway, Envoy-S Standard / Metered, IQ System Controller (IQ Battery 3T/10T)

**🔍 TELEMETRY READ OUT**
Microinverter aggregated solar PV generation (W), IQ Battery charge/discharge wattage (W), Net grid import/export (W), Total lifetime generation (kWh), Battery SOC (%), Per-phase currents and voltages.

**🎛️ CONTROLLED SETPOINTS**
Local Enphase IQ Gateway HTTPS Bearer JWT dynamic power-export limiting and IQ Battery storage mode dispatch.

**⚡ HANDLED & AUTOMATION ROLE**
Local-first integration with Enphase IQ Gateway Metered and IQ Batteries: streams production/consumption/storage metrics and coordinates negative-price export limiting.

### 1.7 FoxESS H1 / H3 / H3-Pro Hybrid Inverter `foxess_inverter`

MODBUS

**Manufacturer / Vendor:** FoxESS

**Communication Interface:** Modbus TCP via LAN or RS485 Ethernet converter (Default Port: 502, Slave ID: 1 or 247)

**Supported Hardware Models:** FoxESS H1, AC1, H3, H3-Pro, AIO Series Hybrid Storage Inverters

**🔍 TELEMETRY READ OUT**
Solar PV generation (W), Battery power (W), Grid active power (W), Total energy (kWh), Battery SOC (%), 3-Phase currents (L1, L2, L3 in Amps).

**🎛️ CONTROLLED SETPOINTS**
FoxESS Work Mode (`41000` / `44000`), Remote EMS Power Setpoints (W), Min SOC Floor, and Active Export Power Limit.

**⚡ HANDLED & AUTOMATION ROLE**
Full closed-loop control across FoxESS H1, H3, and H3-Pro hybrid series: dispatches forced battery charging/discharging for spot arbitrage and capacity peak shaving alongside active solar curtailment.

### 1.8 Fronius Gen24 Inverter `fronius_inverter`

MODBUS

**Manufacturer / Vendor:** Fronius

**Communication Interface:** Modbus TCP via SunSpec protocol (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Fronius Primo Gen24, Symo Gen24 Plus, Symo Hybrid, Tauro (SunSpec Modbus)

**🔍 TELEMETRY READ OUT**
Solar PV generation (W), Battery charge/discharge wattage (W), Grid power flow (W), Cumulative energy (kWh), BYD / Fronius Battery SOC (%).

**🎛️ CONTROLLED SETPOINTS**
SunSpec Model 123 Active Power Limit (`WMaxLimPct` single-register FC06) and SunSpec Model 124 Storage Charge/Discharge Control (`StorCtl_Mod`, `OutWRte`, `InWRte`).

**⚡ HANDLED & AUTOMATION ROLE**
Full closed-loop Fronius Symo / Primo / Gen24 Plus control: regulates BYD Battery-Box HVS/HVM charge/discharge rates for spot arbitrage and peak shaving, and curtails PV output during negative market prices.

### 1.9 GoodWe Inverter `goodwe_inverter`

MODBUS

**Manufacturer / Vendor:** GoodWe

**Communication Interface:** Modbus TCP / UDP via LAN or Wi-Fi+LAN kit (Default Port: 502 or 8899, Slave ID: 247)

**Supported Hardware Models:** GoodWe ET, EH, ES, EM, Lynx Home F Series Hybrid Inverters (ARM/DSP)

**🔍 TELEMETRY READ OUT**
Solar PV power (W), Battery charge/discharge power (W), Net grid flow (W), Lifetime energy (kWh), Battery SOC (%).

**🎛️ CONTROLLED SETPOINTS**
GoodWe ET/EH/BT/BH EMS Power Mode (`47511` modes `1`/`2`/`8`/`9`), EMS Power Setpoint (`47512` W), and Grid Export Power Limit Enable & Value (`47509`–`47510`).

**⚡ HANDLED & AUTOMATION ROLE**
Verified closed-loop hybrid control for GoodWe ET/EH/BT/BH inverters: executes automated battery peak shaving, Day-Ahead spot arbitrage, and dynamic PV export curtailment.

### 1.10 Growatt Inverter `growatt_inverter`

MODBUS

**Manufacturer / Vendor:** Growatt

**Communication Interface:** Modbus TCP via ShineLAN-X Ethernet dongle (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Growatt SPH, MIN, MOD, MID Series Hybrid and Grid-Tied Inverters (ShineLAN-X)

**🔍 TELEMETRY READ OUT**
Solar PV active power (W), Battery power (W), Grid feed-in/consumption (W), Cumulative kWh, Battery SOC (%).

**🎛️ CONTROLLED SETPOINTS**
Growatt SPH / MOD TL3-XH Priority Mode (`1044` / `3049`), Charge/Discharge Rate & Stop SOC (`1070`–`1092`), and Active Power Rate / Export Limit (`3`, `122`–`123`).

**⚡ HANDLED & AUTOMATION ROLE**
Verified closed-loop control for Growatt SPH and MOD hybrid/string inverters: manages forced battery charging, peak discharge, and negative spot price PV curtailment.

### 1.11 Huawei Hybrid Inverter `huawei_inverter`

MODBUS

**Manufacturer / Vendor:** Huawei

**Communication Interface:** Modbus TCP via SDongleA-05 (Ethernet/WLAN) or SmartLogger3000 (Default Port: 502, Slave ID: 1 or 2)

**Supported Hardware Models:** Huawei SUN2000 Series (2KTL-6KTL-L1, 3KTL-10KTL-M1, 12KTL-25KTL-M2/M5, 30KTL-100KTL-M3, MAP0, MB0) with LUNA2000 Battery (5kWh-30kWh)

**🔍 TELEMETRY READ OUT**
Active solar PV generation (W, reg 32080), LUNA2000 Battery charge/discharge power (W, reg 37001), Grid import/export power via DTSU666-H (W, reg 37113), Cumulative generation (kWh, reg 32106), Battery SOC (%, reg 37004).

**🎛 CONTROLLED SETPOINTS**
Forced Battery Charge / Discharge Power Setpoint (W, reg 40118/40120), Active Power Curtailment Limit (W or % of rated power, reg 40118/40125).

**⚡ HANDLED & AUTOMATION ROLE**
Full bidirectional battery and curtailment control: commands LUNA2000 to discharge during household peak spikes (Flanders capacity peak shaving); commands grid force-charging during negative/cheap EPEX spot hours; curtails active PV output during negative export prices or zero-export mandates.

### 1.12 LG ESS Home (8 / 10 / 15) `lg_ess`

REST / HTTP

**Manufacturer / Vendor:** LG

**Communication Interface:** Local HTTP REST API (Default Port: 80 / 443, JSON endpoints: /v1/user/setting/network)

**Supported Hardware Models:** LG ESS Home 8, Home 10, Home 15 Hybrid Energy Storage Systems

**🔍 TELEMETRY READ OUT**
Solar PV production (W), Battery power flow (W), Grid import/export (W), Cumulative energy (kWh), Battery SOC (%).

**🎛 CONTROLLED SETPOINTS**
Local LG ESS Home `/v1/user/setting/batt` forced-charge / self-consumption control and PCS power curtailment.

**⚡ HANDLED & AUTOMATION ROLE**
Local zero-cloud HTTPS integration for LG ESS Home 8 / 10 / 15 systems with automated off-peak battery charging and grid flow balancing.

### 1.13 Marstek Venus / Jupiter Storage `marstek_storage`

MODBUS

**Manufacturer / Vendor:** Marstek

**Communication Interface:** Modbus TCP (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Marstek Venus Series, Jupiter Hybrid All-In-One Battery Energy Storage

**🔍 TELEMETRY READ OUT**
Solar PV power (W), Battery charge/discharge power (W), Grid exchange power (W), Total energy (kWh), Battery SOC (%).

**🎛 CONTROLLED SETPOINTS**
Marstek Venus C / E / Jupiter Passive Mode Enable (`42000 = 0x55AA`), Direction Control (`42010`), and Exact Charge/Discharge Watt Setpoints (`42020`–`42021`).

**⚡ HANDLED & AUTOMATION ROLE**
Full closed-loop AC-coupled storage control for Marstek Venus C/E and Jupiter batteries: absorbs excess rooftop solar, shaves domestic capacity peaks, and executes Day-Ahead spot arbitrage.

### 1.14 SAJ eSolar (H1 / H2 / R5 / R6 Hybrid Inverter) `saj_hybrid`

MODBUS

**Manufacturer / Vendor:** SAJ

**Communication Interface:** Modbus TCP (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** SAJ eSolar H1, H2, R5, R6 Series Hybrid & Storage Inverters (B1/B2 Batteries)

**🔍 TELEMETRY READ OUT**
Solar PV generation (W), Battery power (W), Grid active power (W), Total energy (kWh), Battery SOC (%), 3-Phase currents (L1, L2, L3 in Amps).

**🎛 CONTROLLED SETPOINTS**
SAJ H1 / H2 / R5 / R6 AppMode / Passive Charge & Discharge Wattage Setpoints and Active Power Derating Limit (`0.1%`).

**⚡ HANDLED & AUTOMATION ROLE**
Closed-loop hybrid management for SAJ H1/H2 systems: coordinates 3-phase PV curtailment, forced off-peak battery charging, and peak demand shaving.

### 1.15 SMA Sunny Boy / Tripower / Storage Inverter `sma_inverter`

MODBUS

**Manufacturer / Vendor:** SMA

**Communication Interface:** Modbus TCP via Speedwire / SunSpec (Default Port: 502, Slave ID: 126 or 3)

**Supported Hardware Models:** SMA Sunny Boy (1.5-6.0), Sunny Tripower (3.0-10.0, X, Smart Energy), Sunny Island, Sunny Boy Storage

**🔍 TELEMETRY READ OUT**
Solar PV generation (W, reg 30775), Battery charge/discharge power (W, reg 31393), Grid active power (W, reg 30865), Total energy (kWh, reg 30513), Battery SOC (%, reg 30845), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts).

**🎛 CONTROLLED SETPOINTS**
SMA Active Power Limit (`WModCfg` / `WMaxLimPct`) and Bidirectional Battery Control across both Sunny Island / Sunny Boy Storage (`40151=802`, `40149`) and Sunny Tripower Smart Energy (`40236`, `40793`–`40801`).

**⚡ HANDLED & AUTOMATION ROLE**
Complete closed-loop solar curtailment and battery arbitrage across SMA Sunny Boy, Sunny Tripower, Sunny Tripower Smart Energy (STP-SE), and Sunny Boy Storage.

### 1.16 Senergy (SE Series / Hybrid Inverter) `senergy_inverter`

MODBUS

**Manufacturer / Vendor:** Senergy

**Communication Interface:** Modbus TCP (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Senergy SE Series Hybrid & Storage Inverters (3kW – 15kW)

**🔍 TELEMETRY READ OUT**
Solar PV active power (W), Battery charge/discharge power (W), Grid exchange (W), Cumulative energy (kWh), Battery SOC (%).

**🎛 CONTROLLED SETPOINTS**
Senergy SE Hybrid Passive Mode Charge/Discharge Power Setpoints (W) and Active Power Output Limit (%).

**⚡ HANDLED & AUTOMATION ROLE**
Closed-loop control for Senergy hybrid inverters: automates battery peak shaving, off-peak charging, and negative-price PV curtailment.

### 1.17 Sigenergy SigenStor (Hybrid Inverter & Storage) `sigenergy_hybrid`

MODBUS

**Manufacturer / Vendor:** Sigenergy

**Communication Interface:** Modbus TCP via Sigen Gateway / Inverter (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Sigenergy SigenStor 5-in-1 Energy Storage System (Inverter, Battery, EVDC/EVAC)

**🔍 TELEMETRY READ OUT**
Solar PV power (W), Sigen Battery flow (W), Grid exchange power (W), Total energy (kWh), Sigen Battery SOC (%), 3-Phase currents (L1, L2, L3 in Amps).

**🎛️ CONTROLLED SETPOINTS**
SigenStor Remote EMS Enable (`40029`), Operating Mode (`40031`), Max Charge/Discharge Power Setpoints (`40032`–`40034` U32 W), and PV Curtailment Limit (`40038`).

**⚡ HANDLED & AUTOMATION ROLE**
High-precision closed-loop EMS control for Sigenergy SigenStor 5-in-1 energy systems: executes sub-second capacity peak shaving, EPEX spot arbitrage, and PV export curtailment.

### 1.18 SofarSolar (HYD / G3 / AMASS Hybrid Inverter) `sofarsolar_hybrid`

MODBUS

**Manufacturer / Vendor:** SofarSolar

**Communication Interface:** Modbus TCP via Ethernet/Wi-Fi stick (Default Port: 8899 or 502, Slave ID: 1)

**Supported Hardware Models:** SofarSolar HYD 3000-6000-ES, HYD 5K-20KTL-3PH, G3 Series, AMASS StoreEdge Batteries

**🔍 TELEMETRY READ OUT**
Solar PV power (W), Battery charge/discharge power (W), Grid power flow (W), Lifetime energy (kWh), Battery SOC (%), 3-Phase currents (L1, L2, L3 in Amps).

**🎛️ CONTROLLED SETPOINTS**
SofarSolar HYD 5K–20KTL-3PH Passive Storage Setpoints (`0x1187` mode, `0x1189`–`0x118D` signed S32 W) and Active Power / Anti-Reflux Export Limit.

**⚡ HANDLED & AUTOMATION ROLE**
Verified closed-loop control for SofarSolar HYD 3PH G3 and ME3000SP hybrid systems: regulates battery charge/discharge wattage and solar curtailment.

### 1.19 SolaX Inverter `solax_inverter`

MODBUS

**Manufacturer / Vendor:** SolaX

**Communication Interface:** Modbus TCP via Pocket LAN or Pocket Wi-Fi (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** SolaX Power X1-Hybrid, X3-Hybrid Gen 2/3/4, X1/X3-Fit, Triple Power Batteries

**🔍 TELEMETRY READ OUT**
Solar PV active power (W), Battery power (W), Grid active power (W), Total energy (kWh), Battery SOC (%).

**🎛️ CONTROLLED SETPOINTS**
SolaX X1/X3-Hybrid G4 Remote Control Power Mode (`0x007C`–`0x0080`, exact active power target W + watchdog) with `0x001F` Manual Mode fallback, plus Export Control User Limit (`0x0042`).

**⚡ HANDLED & AUTOMATION ROLE**
Verified sub-second closed-loop control for SolaX X1/X3-Hybrid and Fit systems: executes dynamic solar self-consumption, spot-price arbitrage, and PV curtailment.

### 1.20 SolarEdge 3-Phase Inverter `solaredge_3phase`

MODBUS

**Manufacturer / Vendor:** SolarEdge

**Communication Interface:** Modbus TCP via SunSpec protocol (Default Port: 1502 or 502, Slave ID: 1)

**Supported Hardware Models:** SolarEdge 3-Phase Inverters (SE3K – SE33.3K, StorEdge 3-Phase with SetApp)

**🔍 TELEMETRY READ OUT**
Solar PV generation (W), Battery charge/discharge power (W), Grid meter power (W), Total energy (kWh), Connected Energy Bank SOC (%).

**🎛️ CONTROLLED SETPOINTS**
SolarEdge StorEdge Control Mode `4` (Remote), Charge/Discharge Limit (`0xE00E`/`0xE010` float32 W), Mode Command (`0xE00D`), and Advanced Power Control (`0xF001` / SunSpec M123).

**⚡ HANDLED & AUTOMATION ROLE**
Full closed-loop 3-phase SolarEdge Home Hub & StorEdge control: modulates home battery charge/discharge wattage and curtails PV generation during negative spot prices.

### 1.21 SolarEdge Inverter `solaredge_inverter`

MODBUS

**Manufacturer / Vendor:** SolarEdge

**Communication Interface:** Modbus TCP via SunSpec protocol (Default Port: 1502 or 502, Slave ID: 1)

**Supported Hardware Models:** SolarEdge 1-Phase HD-Wave Inverters (SE2200H – SE6000H, StorEdge 1-Phase)

**🔍 TELEMETRY READ OUT**
Solar PV generation (W), Battery charge/discharge power (W), Grid active power (W), Total energy (kWh), Battery SOC (%).

**🎛️ CONTROLLED SETPOINTS**
SolarEdge StorEdge Control Mode `4` (Remote), Charge/Discharge Limit (`0xE00E`/`0xE010` float32 W), Mode Command (`0xE00D`), and Advanced Power Control (`0xF001` / SunSpec M123).

**⚡ HANDLED & AUTOMATION ROLE**
Full closed-loop 1-phase SolarEdge HD-Wave & Home Hub control: regulates StorEdge / Home Battery charge and discharge setpoints and curtails solar production when required.

### 1.22 Solis Hybrid & Retrofit Inverters (Primary Reference) `solis_hybrid`

⭐ Reference Flagship | MODBUS

**Manufacturer / Vendor:** Solis

**Communication Interface:** Modbus TCP via direct internal Ethernet / Ginlong Wi-Fi/LAN stick (Default Port: 502 or 8899, Slave ID: 1)

**Supported Hardware Models:** Solis S5-EH1P, S6-EH1P Series (Single-Phase Hybrid), S6-EH3P Series (Three-Phase Hybrid 5kW-20kW), Solis RAI-3K-48V-5G, S6-EA1P3.6K-L, S6-EA1P5K-L Series (AC-Coupled Storage Retrofit)

**🔍 TELEMETRY READ OUT**

Solar PV Active Generation Power (W, reg 33079), Battery Charge/Discharge Power (W, reg 33149), Battery SOC (%, reg 33139), Grid Meter Active Power (W, reg 33130), Total Cumulative Energy Yield (kWh, reg 33029), 3-Phase AC Currents (0.1A, regs 33073-33075), 3-Phase Voltages (0.1V, regs 33076-33078).

**🎛 CONTROLLED SETPOINTS**

Active Power Curtailment Mode and Limit (W / % via holding registers 3051/3052), Forced Battery Charging & Discharging Setpoint (W via registers 43110/43135/43136).

**⚡ HANDLED & AUTOMATION ROLE**

Designated primary reference platform for hybrid solar and battery energy storage. Executes sub-second closed-loop active power curtailment during negative spot price hours (Netherlands Mode / Zero-Export), manages automatic solar surplus sinking into battery storage, dynamically throttles charging during Flanders capacity tariff peak shaving, and orchestrates spot market arbitrage (EPEX Day-Ahead price force charging and peak discharging). Solis Retrofit models operate seamlessly as AC-coupled storage units with dedicated battery control and zero solar PV string dependency.

### 1.23 Sungrow Inverter `sungrow_inverter`

MODBUS

**Manufacturer / Vendor:** Sungrow

**Communication Interface:** Modbus TCP via direct internal LAN RJ45 port (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Sungrow SH5.0RT, SH6.0RT, SH8.0RT, SH10RT, SH15T, SH20T Hybrid Inverters (SBR096-SBR256)

**🔍 TELEMETRY READ OUT**

Solar PV generation (W, reg 5016), Battery charge/discharge power (W, reg 13021), Grid exchange power (W, reg 13009), Cumulative generation (kWh, reg 5003), SBR Battery SOC (%, reg 13022).

**🎛 CONTROLLED SETPOINTS**

Sungrow SH-RS / SH-RT / SH-T EMS Mode Selection (`13049`), Charge/Discharge Command (`13050` `0xAA`/`0xBB`/`0xCC`), Power Setpoint (`13051` W), and Active Power Limitation (`5006`–`5007`).

**⚡ HANDLED & AUTOMATION ROLE**

Verified high-speed closed-loop control for Sungrow SH hybrid and SG string inverters: executes sub-second peak shaving, EPEX battery arbitrage, and PV curtailment.

### 1.24 Victron Inverter/GX `victron_inverter`

MODBUS

**Manufacturer / Vendor:** Victron

**Communication Interface:** Modbus TCP via Venus OS (Default Port: 502, Unit ID: 100 for System Calc / 228 for Inverter)

**Supported Hardware Models:** Victron Energy MultiPlus-II, Quattro, Cerbo GX, Ekrano GX, Color Control GX (Venus OS)

**🔍 TELEMETRY READ OUT**

Solar PV DC/AC power (W, reg 811), Battery charge/discharge power (W, reg 842), Net AC grid import/export (W, reg 820), Total energy (kWh), Battery SOC (%, reg 843).

**🎛 CONTROLLED SETPOINTS**

Victron Venus OS / Cerbo GX ESS Grid Power Setpoint (`2700` S16 / `2703` S32 W), ESS Discharge Disable, and Signed Max PV Feed-In Power Limit (`2706` in 100 W steps).

**⚡ HANDLED & AUTOMATION ROLE**

Deep closed-loop integration with Victron MultiPlus-II / Quattro GX ESS installations: steers AC grid setpoint for battery arbitrage/peak shaving and curtails excess feed-in.

### 1.25 ABB / Fimer Solar & Hybrid Inverter `abb_fimer_inverter`

MODBUS

**Manufacturer / Vendor:** ABB / Fimer

**Communication Interface:** Modbus SunSpec over TCP/RTU (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** ABB UNO-DM-PLUS, REACT 2, TRIO, PVI, and Fimer string/hybrid inverters

**🔍 TELEMETRY READ OUT**

Solar PV Active Power (W, SunSpec standard register 40083), Battery State of Charge (SOC %), Battery Charge / Discharge Power (W), Grid Feed-in Power (W), Cumulative Generation Energy (kWh).

**🎛 CONTROLLED SETPOINTS**

SunSpec Model 123 Active Power Curtailment (`WMaxLimPct`, `WMaxLim_Ena`) and SunSpec Model 124 Storage Charge/Discharge Rate Control (`StorCtl_Mod`, `OutWRte`, `InWRte`).

**⚡ HANDLED & AUTOMATION ROLE**

Full closed-loop SunSpec control for ABB / Fimer PV and REACT 2 hybrid installations: executes negative spot price solar curtailment, zero-export limiting, and automated battery peak shaving and Day-Ahead spot arbitrage.

### 1.26 Autarco Solar Inverter `autarco_inverter`

NETWORK TCP

**Manufacturer / Vendor:** Autarco

**Communication Interface:** Modbus TCP via local network dongle/gateway (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Autarco SX, MX, and LX series single-phase and three-phase solar inverters

**🔍 TELEMETRY READ OUT**

Solar PV Active Power (W), Total Daily & Lifetime Generation (kWh), Grid AC Voltage (V), AC Frequency (Hz).

**🎛 CONTROLLED SETPOINTS**

Remote Control Battery Charge/Discharge Power Setpoints (`43135`–`43136`) and Active Power Curtailment Limit (`43052` / `0–100%`).

**⚡ HANDLED & AUTOMATION ROLE**

Full closed-loop hybrid inverter and PV curtailment control for Autarco S2.x / MH / LX systems: coordinates solar self-consumption, spot-price battery arbitrage, and Flemish capacity peak shaving.

### 1.27 Chisage ESS Hybrid Inverter `chisage_ess`

MODBUS

**Manufacturer / Vendor:** Chisage ESS

**Communication Interface:** Modbus TCP / RTU via RS485 Ethernet Gateway (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Chisage ESS Commercial, Industrial & Residential Hybrid Inverters (3kW – 50kW)

**🔍 TELEMETRY READ OUT**
Solar PV Active Power (W), Battery Charge / Discharge Power (W), Battery SOC (%), Grid Import / Export Power (W), Cumulative Inverter Energy (kWh).

**🎛️ CONTROLLED SETPOINTS**
Monitor-Only (Read-Only Telemetry — closed register control map).

**⚡ HANDLED & AUTOMATION ROLE**
Integrates commercial, industrial, and residential Chisage ESS hybrid solar and battery telemetry into GEMS site load balancing.

### 1.28 Delta Electronics Solar Inverter `delta_inverter`

MODBUS

**Manufacturer / Vendor:** Delta Electronics

**Communication Interface:** Modbus SunSpec over TCP (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Delta M6A, M8A, M10A, M15A, M50A, and RPI commercial/residential series

**🔍 TELEMETRY READ OUT**
Solar PV Generation Power (W), DC String Voltages and Currents, Cumulative Generation (kWh), Grid AC Power (W).

**🎛️ CONTROLLED SETPOINTS**
SunSpec Model 123 Active Power Curtailment (`WMaxLimPct`, `WMaxLim_Ena`) and SunSpec Model 124 Basic Storage Control.

**⚡ HANDLED & AUTOMATION ROLE**
Standardized SunSpec closed-loop telemetry and active power curtailment for Delta residential and commercial solar/hybrid inverters.

### 1.29 HYXiPOWER Hybrid Inverter `hyxipower_inverter`

MODBUS

**Manufacturer / Vendor:** HYXiPOWER

**Communication Interface:** Modbus TCP via Wi-Fi/Ethernet dongle (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** HYXiPOWER 1-Phase & 3-Phase Residential & Commercial Hybrid Inverters

**🔍 TELEMETRY READ OUT**
Solar PV Generation (W), Battery Charge / Discharge Power (W), Battery SOC (%), Grid Exchange (W), Cumulative Energy (kWh).

**🎛️ CONTROLLED SETPOINTS**
Autonomous / Read-Only telemetry poller.

**⚡ HANDLED & AUTOMATION ROLE**
Streams PV production and battery storage metrics to the GEMS flow chart and telemetry pipeline.

### 1.30 KSTAR BluE Hybrid Inverter `kstar_inverter`

MODBUS

**Manufacturer / Vendor:** KSTAR

**Communication Interface:** Modbus RTU / TCP via RS485 Ethernet converter (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** KSTAR BluE-S, BluE-H3, and BluE-G residential and C&I hybrid systems

**🔍 TELEMETRY READ OUT**
Solar PV Active Power (W), Battery Charge / Discharge Power (W), Battery SOC (%), Grid Power Flow (W), Phase Currents (L1, L2, L3 in Amps).

**🎛️ CONTROLLED SETPOINTS**
Autonomous / Read-Only telemetry poller.

**⚡ HANDLED & AUTOMATION ROLE**
Feeds hybrid solar and battery metrics into the GEMS load balancer and multi-tariff cost optimization engine.

### 1.31 Solinteg Integ M Hybrid Inverter `solinteg_inverter`

MODBUS

**Manufacturer / Vendor:** Solinteg

**Communication Interface:** Modbus RTU / TCP via LAN/Wi-Fi stick (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Solinteg Integ M OHT/OMT Series (3kW – 50kW) 3-Phase Hybrid Inverters

**🔍 TELEMETRY READ OUT**
Solar PV Active Power (W), Battery Charge / Discharge Power (W), Battery SOC (%), Grid Net Exchange Power (W), Cumulative Generation (kWh).

**🎛️ CONTROLLED SETPOINTS**
Solinteg Integ M series Remote EMS Mode, Charge/Discharge Power Setpoints, and Active Power Limit.

**⚡ HANDLED & AUTOMATION ROLE**
Verified closed-loop 3-phase high-voltage hybrid control for Solinteg Integ M (4–50 kW) commercial and residential inverters.

### 1.32 Solplanet / AISWEI Inverter `solplanet_inverter`

MODBUS

**Manufacturer / Vendor:** Solplanet (AISWEI)

**Communication Interface:** Modbus RTU / TCP via Ethernet/Wi-Fi Ai-Dongle (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Solplanet ASW H-S2, ASW H-T2, ASW-LT-G2/G3 Single and 3-Phase Inverters

**🔍 TELEMETRY READ OUT**
Solar PV Active Power (W), Total Yield Energy (kWh), AC Voltage and Current per phase, Grid Import/Export (W).

**🎛️ CONTROLLED SETPOINTS**
AISWEI / Solplanet Battery Working Mode (`1103` wire / `41104` doc), Signed Charge/Discharge Power Setpoint (`1152` wire S16 W), and Active Power Control (`4000` / `5402` `0.01%`).

**⚡ HANDLED & AUTOMATION ROLE**
Verified closed-loop control for Solplanet ASW H-T2/T3 and KACO blueplanet hybrid NH3 inverters: dispatches exact Watt storage setpoints and dynamic PV curtailment.

### 1.33 APsystems EZ1-M / EZ1-D Local Microinverter `apsystems_ez1`

REST / HTTP

**Manufacturer / Vendor:** APsystems

**Communication Interface:** Local HTTP JSON API (Default Port: 8050, Endpoints: `/getOutputData`, `/setMaxPower`)

**Supported Hardware Models:** APsystems EZ1-M, EZ1-D Wi-Fi Balcony & Rooftop Microinverters (Local Mode)

**🔍 TELEMETRY READ OUT**
Channel 1 & Channel 2 PV Active Power (W), Lifetime Cumulative Generation Energy (kWh).

**🎛️ CONTROLLED SETPOINTS**
Direct Local Max AC Output Power Setpoint (`/setMaxPower?p=30..800` W).

**⚡ HANDLED & AUTOMATION ROLE**
100% cloud-free local microinverter telemetry and stepless active power curtailment during negative EPEX spot prices or zero-export operation.

### 1.34 FENECON Home / OpenEMS Edge (FEMS) `fenecon_fems`

REST / HTTP

**Manufacturer / Vendor:** FENECON

**Communication Interface:** Local HTTP REST API (Default Port: 8084, `/rest/channel/_sum/*` & `/rest/channel/ess0/*`)

**Supported Hardware Models:** FENECON Home 6, Home 10, Home 15, Home 20, Home 30 & OpenEMS Edge Controllers

**🔍 TELEMETRY READ OUT**
Solar PV Production Power (W), Cumulative PV Energy (kWh), Battery Charge/Discharge Power (W), Battery SOC (%), Grid Active Power (W).

**🎛️ CONTROLLED SETPOINTS**
Exact ESS Charge / Discharge Active Power Setpoint (`ess0/SetActivePowerEquals` W) and PV Active Power Curtailment (`pvInverter0/SetActivePowerLimit` W).

**⚡ HANDLED & AUTOMATION ROLE**
Full closed-loop local integration for FENECON Home & OpenEMS systems: coordinates EPEX Day-Ahead battery arbitrage, Flemish capacity ceiling peak shaving, and negative-price curtailment.

### 1.35 Hoymiles DTU-Pro / OpenDTU Microinverter `hoymiles_dtu`

REST / HTTP

**Manufacturer / Vendor:** Hoymiles

**Communication Interface:** Modbus TCP (Default Port: 502, Unit ID: 1) or Local OpenDTU HTTP REST (`/api/livedata/status`, `/api/limit/config`)

**Supported Hardware Models:** Hoymiles HM / HMS / HMT Series Microinverters via DTU-Pro, DTU-Pro-S, OpenDTU, or AhoyDTU

**🔍 TELEMETRY READ OUT**
Aggregated & Per-Port Solar PV Active Power (W), Cumulative Solar Yield (kWh).

**🎛️ CONTROLLED SETPOINTS**
Active Power Output Limit (`0xC006` % on DTU-Pro Modbus; volatile RAM `limit_type=1` % on OpenDTU without EEPROM wear).

**⚡ HANDLED & AUTOMATION ROLE**
Zero-cloud microinverter monitoring and dynamic PV curtailment during negative spot market hours.

### 1.36 KOSTAL Plenticore Plus / G2 / G3 Hybrid `kostal_plenticore`

MODBUS

**Manufacturer / Vendor:** KOSTAL

**Communication Interface:** Modbus TCP (Default Port: 1502, Unit ID: 71, Auto-detected CDAB/ABCD IEEE-754 Float32)

**Supported Hardware Models:** KOSTAL Plenticore Plus, Plenticore BI, Plenticore G2, and Plenticore G3 Hybrid Inverters

**🔍 TELEMETRY READ OUT**
Total DC PV Power (W), Cumulative Yield (kWh), Battery Charge/Discharge Power (W), Battery SOC (%), Grid Sensor Power (W), 3-Phase Voltages & Currents.

**🎛️ CONTROLLED SETPOINTS**
External Battery Management Signed Power Setpoint (Holding `1034` Float32 W) and SunSpec Model 123 Active Power Curtailment (`WMaxLimPct`).

**⚡ HANDLED & AUTOMATION ROLE**
Full closed-loop control for KOSTAL Plenticore hybrid installations: executes battery peak shaving, Day-Ahead spot arbitrage, and negative-price solar curtailment.

### 1.37 KACO blueplanet hybrid NH3 / gridsave TL3 `kaco_inverter`

MODBUS

**Manufacturer / Vendor:** KACO

**Communication Interface:** Modbus TCP / RTU (Default Port: 502, Unit ID: 1, SunSpec Models 103/123/124 & NH3 Storage Block `40200`–`40203`)

**Supported Hardware Models:** KACO blueplanet hybrid 6.0 / 8.0 / 10.0 / 12.0 NH3, blueplanet TL3, and gridsave 50.0 / 92.0 TL3-S Commercial Storage Inverters

**🔍 TELEMETRY READ OUT**
Solar PV Active Power (W), Cumulative Energy (kWh), Battery Charge/Discharge Power (W), Battery SOC (%), Live BMS Charge/Discharge Power Limits (`40202`–`40203` W), 3-Phase Currents & Voltages.

**🎛️ CONTROLLED SETPOINTS**
SunSpec Model 123 Active Power Curtailment (`WMaxLimPct`, `WMaxLim_Ena`) and Model 124 Storage Control (`WChaMax`, `WChaGra`, `WDisChaGra`, `StorCtl_Mod`, `WChaMaxRvrtTms=300s` Watchdog) plus NH3 EMS fallback (`40210`–`40212`).

**⚡ HANDLED & AUTOMATION ROLE**
Full closed-loop control for KACO blueplanet hybrid NH3 and commercial gridsave TL3 installations: executes battery peak shaving, EPEX Day-Ahead spot arbitrage, and negative-price solar curtailment.

## 🔋 Category 2: Dedicated Battery Energy Storage Systems (BESS) (18 Native Templates)

Dedicated BESS integrations interface directly with stationary battery controllers and BMS gateways over Modbus TCP or local LAN APIs. They provide deep pack telemetry (SOC %, voltage, cell temperature, power flow) and support dynamic power limits, scheduled force-charging during cheap/negative hours, and autonomous peak shaving for commercial and residential tariffs.

### 2.1 Siemens Junelight Smart Battery `siemens_junelight`

MODBUS

**Manufacturer / Vendor:** Siemens

**Communication Interface:** Modbus TCP (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Siemens Junelight Smart Battery Energy Storage System (3.3kWh – 19.8kWh)

🔍 **TELEMETRY READ OUT**
Battery charge/discharge power (W), Battery State of Charge (SOC %), Net grid power flow (W), Internal battery status.

🎛️ **CONTROLLED SETPOINTS**
Siemens Junelight Smart Battery EMS Operating Mode and Forced Charge/Discharge Power Setpoints (W).

⚡ **HANDLED & AUTOMATION ROLE**
Verified integration with Siemens Junelight residential storage for solar self-consumption, peak shaving, and off-peak charging.

### 2.2 Solarwatt (EnergyManager / MyReserve Matrix) `solarwatt_storage`

REST / HTTP

**Manufacturer / Vendor:** Solarwatt

**Communication Interface:** Local HTTP REST API (Default Port: 80, JSON endpoint: /rest/kiwigrid/private/items)

**Supported Hardware Models:** Solarwatt EnergyManager Pro, MyReserve Matrix Battery Storage System

🔍 **TELEMETRY READ OUT**
Battery power flow (W), Battery SOC (%), Solar generation (W), Grid exchange (W), Cumulative energy (kWh).

🎛️ **CONTROLLED SETPOINTS**
Solarwatt EnergyManager / Reserve local `/rest/items` battery charge/discharge control and telemetry ingestion.

⚡ **HANDLED & AUTOMATION ROLE**
Verified zero-cloud local LAN integration for Solarwatt EnergyManager Pro and MyReserve / Battery flex storage.

### 2.3 sonnenBatterie (eco 6 / 8 / 10 / hybrid) `sonnen_batterie`

REST / HTTP

**Manufacturer / Vendor:** Sonnen

**Communication Interface:** Local HTTP REST API v1/v2 (Default Port: 80, JSON endpoint: /api/v2/status)

**Supported Hardware Models:** sonnenBatterie eco 6, eco 8, eco 8.2, eco 10, sonnenBatterie hybrid 9.53

🔍 **TELEMETRY READ OUT**
Battery charge/discharge wattage (W), Battery SOC (%), Solar PV production (W), Grid feed-in / consumption (W).

🎛️ **CONTROLLED SETPOINTS**
sonnenBatterie Local API v2 `PUT /api/v2/configurations` (`EM_OperatingMode=1` Manual / `2` Self-Consumption) and `POST /api/v2/setpoint/charge|discharge/{W}`.

⚡ **HANDLED & AUTOMATION ROLE**
Complete closed-loop control for sonnenBatterie eco 8 / 10 / performance: switches cleanly into Manual API mode during forced arbitrage or peak shaving and restores autonomous self-consumption afterward.

### 2.4 Duracell Energy Residential Storage `duracell_energy`

MODBUS

**Manufacturer / Vendor:** Duracell Energy

**Communication Interface:** Modbus RTU / TCP via Gateway (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Duracell Energy Dura5, DuraCharge, and Durai storage systems

🔍 **TELEMETRY READ OUT**
Battery Charge / Discharge Power (W), Battery State of Charge (SOC %), Grid Power Flow (W), Inverter AC Power (W).

🎛️ **CONTROLLED SETPOINTS**
Monitor-Only (Read-Only Telemetry).

⚡ **HANDLED & AUTOMATION ROLE**
Monitors Duracell residential battery SOC and power exchange for site load balancing and EV solar surplus routing.

### 2.5 Dyness Battery Storage System `dyness_battery`

MODBUS

**Manufacturer / Vendor:** Dyness

**Communication Interface:** Modbus RTU / TCP via Gateway / Inverter BMS Port (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Dyness Tower, Powerbox F, DL5.0C, and A48100 residential lithium batteries

🔍 **TELEMETRY READ OUT**
Battery Charge / Discharge Power (W), Battery State of Charge (SOC %), Total Pack Voltage (V), Current (A), Cell Temperatures (°C).

🎛️ **CONTROLLED SETPOINTS**
Dyness Tower / Powerbox / Junior Box Modbus EMS Operating Mode and Charge/Discharge Power Limits (W).

⚡ **HANDLED & AUTOMATION ROLE**
Verified BMS & PCS storage integration for Dyness HV/LV battery stacks with closed-loop charge/discharge dispatch.

### 2.6 Elecnova Commercial & Industrial BESS `elecnova_bess` MODBUS

**Manufacturer / Vendor:** Elecnova

**Communication Interface:** Modbus TCP via Industrial Gateway (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Elecnova ECO Series Commercial & Industrial Outdoor Cabinet BESS (50kW – 500kW)

**🔍 TELEMETRY READ OUT**
Battery Active Power (W), Battery State of Charge (SOC %), Grid Exchange Power (W), Cumulative In / Out Energy (kWh).

**🎛️ CONTROLLED SETPOINTS**
Monitor-Only (Read-Only Telemetry).

**⚡ HANDLED & AUTOMATION ROLE**
Monitors commercial BESS power flow, state of charge, and cumulative energy on the live dashboard.

### 2.7 Goneo Residential Storage System `goneo_storage` MODBUS

**Manufacturer / Vendor:** Goneo

**Communication Interface:** Modbus TCP / RTU via Wi-Fi/Ethernet Module (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Goneo Residential All-in-One Energy Storage Systems

**🔍 TELEMETRY READ OUT**
Battery Charge / Discharge Power (W), Battery SOC (%), Grid Power Flow (W), Total Energy (kWh).

**🎛️ CONTROLLED SETPOINTS**
Monitor-Only (Read-Only Telemetry).

**⚡ HANDLED & AUTOMATION ROLE**
Monitors residential battery storage SOC and power exchange for coordinated EV and relay load balancing.

### 2.8 Mastervolt MLI Ultra / Combi CZone `mastervolt_storage` MODBUS

**Manufacturer / Vendor:** Mastervolt

**Communication Interface:** Modbus RTU / TCP via CZone Bridge / MasterBus gateway (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Mastervolt MLI Ultra Lithium-ion series, Mass Combi Ultra, CZone Modbus Bridge

**🔍 TELEMETRY READ OUT**
Battery Power (W), Battery SOC (%), Inverter AC Output Power (W), DC Bus Voltage (V), Battery Current (A).

**🎛️ CONTROLLED SETPOINTS**
Monitor-Only (Read-Only Telemetry).

**⚡ HANDLED & AUTOMATION ROLE**
Provides off-grid, marine, and residential Mastervolt / CZone battery system telemetry monitoring.

### 2.9 Power Sonic C&I Energy Storage `powersonic_bess` MODBUS

**Manufacturer / Vendor:** Power Sonic

**Communication Interface:** Modbus TCP via BMS/PCS Controller (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Power Sonic PS-C&I and PowerBlock Commercial & Industrial BESS

**🔍 TELEMETRY READ OUT**
Battery Active Power (W), Battery State of Charge (SOC %), Grid Power Flow (W), Fault Status Registers.

**🎛️ CONTROLLED SETPOINTS**
Monitor-Only (Read-Only Telemetry).

**⚡ HANDLED & AUTOMATION ROLE**
Monitors commercial BESS power flow and state of charge on the live GEMS dashboard.

### 2.10 Poweroad Residential & C&I Storage `poweroad_storage` MODBUS

**Manufacturer / Vendor:** Poweroad

**Communication Interface:** Modbus RTU / TCP via Gateway (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Poweroad ERA, Centauri, and H-ES residential & commercial energy storage systems

**🔍 TELEMETRY READ OUT**
Battery Power (W), Battery State of Charge (SOC %), Grid Flow (W), Cumulative In / Out Energy (kWh).

**🎛️ CONTROLLED SETPOINTS**
Monitor-Only (Read-Only Telemetry).

**⚡ HANDLED & AUTOMATION ROLE**
Monitors residential and light-commercial Poweroad battery storage telemetry.

### 2.11 Pylontech Battery Storage (US / Force) `pylontech_storage` MODBUS

**Manufacturer / Vendor:** Pylontech

**Communication Interface:** Modbus RTU / TCP via Inverter Gateway or SC0500/SC1000 controller (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Pylontech US2000, US3000, US5000, Force-H1, Force-H2, and Pelio series

**🔍 TELEMETRY READ OUT**
Battery Charge / Discharge Power (W), State of Charge (SOC %), Pack Voltage (V), Pack Current (A), SOH (%).

**🎛️ CONTROLLED SETPOINTS**
Pylontech US2000/US3000/US5000 & Force-H1/H2 BMS/EMS Charge and Discharge Power Setpoints via Modbus TCP/RTU.

**⚡ HANDLED & AUTOMATION ROLE**
Verified closed-loop telemetry and dispatch for Pylontech low-voltage and high-voltage stationary storage racks.

### 2.12 Socomec SUNSYS C-System / HES BESS `socomec_bess` MODBUS

**Manufacturer / Vendor:** Socomec

**Communication Interface:** Modbus TCP via Diris Digiware / BESS Controller (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Socomec SUNSYS C-System and SUNSYS HES L / XL Industrial Energy Storage Systems

**🔍 TELEMETRY READ OUT**
Battery Active Power (W), Battery SOC (%), Live BMS Max Charge/Discharge Limits (`0xC008`–`0xC009` W), Grid Active Power Flow (W), Cumulative Energy (kWh).

**🎛️ CONTROLLED SETPOINTS**
Socomec SUNSYS PCS Remote Active Power Control (`0xC100` Mode, `0xC101`–`0xC102` S32 0.1 kW Setpoint, `0xC103=300s` Hardware Watchdog) with SunSpec Model 124 Storage Control fallback.

**⚡ HANDLED & AUTOMATION ROLE**
Verified closed-loop C&I battery control for Socomec SUNSYS HES L/XL and PCS+: executes industrial peak shaving, capacity ceiling protection, and EPEX spot market arbitrage.

### 2.13 Sunwoda AtSmart Energy Storage `sunwoda_energy` MODBUS

**Manufacturer / Vendor:** Sunwoda Energy

**Communication Interface:** Modbus TCP via BMS Communication Hub (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Sunwoda AtSmart Residential Battery System and Noah C&I BESS

**🔍 TELEMETRY READ OUT**
Battery Charge / Discharge Power (W), Battery State of Charge (SOC %), Grid Power Flow (W), Temperature (°C).

**🎛️ CONTROLLED SETPOINTS**
Monitor-Only (Read-Only Telemetry).

**⚡ HANDLED & AUTOMATION ROLE**
Monitors Sunwoda battery state of charge and power exchange on the live GEMS dashboard.

### 2.14 Sessy Smart Battery (AC Home Storage) `sessy_battery` REST / HTTP

**Manufacturer / Vendor:** Sessy

**Communication Interface:** Local HTTP REST API with Basic Auth (`/api/v1/power/status`, `/api/v1/power/active_strategy`, `/api/v1/power/setpoint`)

**Supported Hardware Models:** Sessy 5 kWh Modular AC-Coupled Retrofit Home Battery

**🔍 TELEMETRY READ OUT**
Battery Charge/Discharge Power (W), State of Charge (SOC %), Exported Energy (kWh), Per-Phase RMS Voltages (V) & Currents (A).

**🎛️ CONTROLLED SETPOINTS**
Dynamic Strategy Selection (`POWER_STRATEGY_API` vs `POWER_STRATEGY_NOM`) and Signed Watt Setpoint (`-W` Charge / `+W` Discharge).

**⚡ HANDLED & AUTOMATION ROLE**
Zero-cloud local control of Dutch-engineered Sessy retrofit batteries for solar self-consumption, Flanders capacity peak shaving, and EPEX spot arbitrage.

### 2.15 Tesla Powerwall 2 / 3 & Backup Gateway (Local HTTPS) `tesla_powerwall` REST / HTTP

**Manufacturer / Vendor:** Tesla Energy

**Communication Interface:** Local HTTPS Gateway API (`POST /api/login/Basic`, `GET /api/meters/aggregates`, `GET /api/system_status/soe`)

**Supported Hardware Models:** Tesla Powerwall 2, Powerwall+, Powerwall 3 & Backup Gateway 2 (Stationary Home Storage)

**🔍 TELEMETRY READ OUT**
Battery Power (W), Usable State of Energy (SOC %), Site Grid Power (W), Solar Power (W), Cumulative Energy (kWh), 3-Phase Site Currents & Voltage.

**🎛️ CONTROLLED SETPOINTS**
Monitor-Only (Local Gateway v23+ firmware enforces read-only local telemetry).

**⚡ HANDLED & AUTOMATION ROLE**
High-speed local HTTPS ingestion of stationary Powerwall site, solar, and battery flows to drive EV charging surplus regulation and capacity peak protection.

### 2.16 VARTA pulse / neo / element Home Storage `varta_storage` MODBUS

**Manufacturer / Vendor:** VARTA

**Communication Interface:** Modbus TCP (Default Port: 502, Unit ID: 255, Holding Registers `1066`–`1102` & `2065`–`2066`)

**Supported Hardware Models:** VARTA pulse, VARTA pulse neo, VARTA element, VARTA element backup, VARTA link

**🔍 TELEMETRY READ OUT**
Battery Charge/Discharge Power (W), Battery SOC (%), Cumulative Charged Energy (kWh), Grid Sensor Power (W), Attached PV Sensor Power (W).

**🎛️ CONTROLLED SETPOINTS**
VARTA EMS Active Power Control (`2065` Command: Charge/Discharge/Auto, `2066` Power W; fallback `1074`).

**⚡ HANDLED & AUTOMATION ROLE**
Closed-loop local Modbus TCP control for VARTA AC-coupled storage systems: automates off-peak spot charging and household peak shaving.

### 2.17 Zendure SolarFlow AC / Hyper 2000 `zendure_solarflow` REST / HTTP

**Manufacturer / Vendor:** Zendure

**Communication Interface:** Local HTTP JSON API (`GET /properties/report`, `POST /properties/write`)

**Supported Hardware Models:** Zendure Hyper 2000, SolarFlow 800, SolarFlow AB1000/AB2000, SolarFlow 2400 AC

**🔍 TELEMETRY READ OUT**
Solar Input Power (W), Battery Pack Charge & Discharge Power (W), State of Charge (`electricLevel` %).

**🎛️ CONTROLLED SETPOINTS**
Smart AC Mode (`acMode` Charge/Discharge) and Stepless `inputLimit` / `outputLimit` Watt Setpoints (`0–2400 W`).

**⚡ HANDLED & AUTOMATION ROLE**
100% local zero-cloud control for plug-in and AC-coupled Zendure storage: absorbs surplus PV and discharges to cover household evening peaks.

## 2.18 Sinexcel PWS Bi-Directional PCS / Commercial BESS `sinexcel_bess`

MODBUS

**Manufacturer / Vendor:** Sinexcel

**Communication Interface:** Modbus TCP / RTU (Default Port: 502, Unit ID: 1, Holding `100`–`113` & `600`–`610`)

**Supported Hardware Models:** Sinexcel PWS1-30K / 50K / 100K / 250K / 500K / 630K Bi-Directional PCS & Commercial BESS Racks

### 🔍 TELEMETRY READ OUT

Battery Active Power (`106` 0.1 kW), Battery SOC (`109` 0.1%), Cumulative Discharged Energy (`110`–`111` 0.1 kWh), Live BMS Max Charge/Discharge Power Limits (`112`–`113` 0.1 kW), 3-Phase Voltages & Currents.

### 🎛️ CONTROLLED SETPOINTS

Remote PCS Start/Standby (`600`), Constant-Power PQ Mode (`601`), Signed Active Power Setpoint (`602` S16 0.1 kW), and 60-Second Fail-Safe Watchdog (`610`).

### ⚡ HANDLED & AUTOMATION ROLE

Closed-loop industrial and C&I battery control for Sinexcel PWS storage converters: automates high-power peak shaving, grid transformer capacity protection, and Day-Ahead spot arbitrage.

## ⚡ Category 3: EV Charging Stations & Wallboxes (61 Native Templates)

EVSE integrations communicate via native local OCPP 1.6-J over WebSockets, high-speed Modbus TCP/RTU, or local REST APIs. They receive dynamic charge current setpoints (throttling 6A to 32A) calculated every second by the GEMS control loop. Supported multi-contactor units also feature opt-in 1-phase to 3-phase switching with a mandatory 20-second anti-welding cooldown.

### 3.1 ABB Charger `abb_charger`

MODBUS

**Manufacturer / Vendor:** ABB

**Communication Interface:** Modbus TCP via Ethernet / Wi-Fi (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** ABB Terra AC Wallbox (TAC-W7, TAC-W11, TAC-W22 Series)

🔍 **TELEMETRY READ OUT**
Active charging power (W, reg 0x401C), Cumulative session energy (kWh, reg 0x401E).

🎛️ **CONTROLLED SETPOINTS**
Dynamic Charging Current Throttling (Amps, reg 0x4100, 6A to 32A in 1mA steps).

⚡ **HANDLED & AUTOMATION ROLE**
Sub-second current throttling over Modbus TCP: dynamically modulates charging current based on solar surplus (Eco mode) and throttles down immediately during capacity peaks (Flanders mode); coexists seamlessly with third-party billing/reimbursement backends.

### 3.2 ABL eM4 Single/Twin (SBCx) `abl_em4_charger`

MODBUS

**Manufacturer / Vendor:** ABL

**Communication Interface:** Modbus TCP (Default Port: 502, Unit ID: 255 or 1)

**Supported Hardware Models:** ABL Wallbox eM4 Single & eM4 Twin (with internal SBCx Controller)

🔍 **TELEMETRY READ OUT**
Active charging power (W, reg 0x300D), Total energy (kWh, reg 0x300F).

🎛️ **CONTROLLED SETPOINTS**
Dynamic Charging Current Throttling (Amps, reg 0x3001, 6A to 32A).

⚡ **HANDLED & AUTOMATION ROLE**
Direct Modbus control: throttles single or dual socket charging currents to match rooftop solar surplus and Flanders capacity limits without disconnecting corporate billing.

### 3.3 ABL eMH1 / eMH2 / eMH3 `abl_emh_charger`

MODBUS

**Manufacturer / Vendor:** ABL

**Communication Interface:** Modbus RTU over TCP Bridge (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** ABL Wallbox eMH1, eMH2, eMH3 Series (with RS485-to-Ethernet Bridge)

🔍 **TELEMETRY READ OUT**
Real-time charging power (W), Session energy (kWh).

🎛️ **CONTROLLED SETPOINTS**
Charging Current Throttling (Amps, 6A to 32A).

⚡ **HANDLED & AUTOMATION ROLE**
Integrates legacy and modern ABL eMH chargers; provides automated solar-following charging and peak shaving.

### 3.4 AUTEL AC MaxiCharger `autel_charger`

OCPP 1.6-J

**Manufacturer / Vendor:** AUTEL

**Communication Interface:** Native OCPP 1.6-J WebSocket Server (Port: 8887, ws://<EMS-IP>:8887/<ChargePointID>)

**Supported Hardware Models:** Autel MaxiCharger AC Compact / MaxiCharger AC Wallbox (7.4kW, 11kW, 22kW)

🔍 **TELEMETRY READ OUT**
Real-time charging power (W), Session energy (kWh), Vehicle connection state, 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts).

🎛️ **CONTROLLED SETPOINTS**
Dynamic Current Throttling via SetChargingProfile (6A to 32A), Phase Switching (1P / 3P).

⚡ **HANDLED & AUTOMATION ROLE**
Autonomous residential charging over native OCPP: adjusts charging speed to solar excess and halts charging during peak tariff periods.

### 3.5 Alfen Charger `alfen_charger`

MODBUS

**Manufacturer / Vendor:** Alfen

**Communication Interface:** Modbus TCP / Active Load Balancing license (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Alfen Eve Single Pro-line, Eve Single S-line, Eve Double Pro-line (Direct Modbus TCP)

🔍 **TELEMETRY READ OUT**
Active charging power (W), Total energy (kWh), Per-phase charging currents (Amps).

🎛️ **CONTROLLED SETPOINTS**
Modbus Slave Max Current (`1210` IEEE-754 Float32 Amps with station watchdog) and Contactor Phase Switching (`1215`: `1` = 1-Phase, `3` = 3-Phase).

⚡ **HANDLED & AUTOMATION ROLE**
Preferred Benelux Modbus TCP integration for Alfen Eve Single & Double Pro-line: sub-second solar surplus tracking, main fuse protection, and automated 1P/3P switching.

### 3.6 Alpitronic Hypercharger `alpitronic_charger`

OCPP 1.6-J

**Manufacturer / Vendor:** Alpitronic

**Communication Interface:** Modbus TCP / OCPP 1.6-J (Default Port: 502 or 8887, Unit ID: 1)

**Supported Hardware Models:** Alpitronic Hypercharger HYC50, HYC150, HYC300, HYC400 DC Fast Chargers

🔍 **TELEMETRY READ OUT**
Active DC charging power (kW/W), Cumulative delivered energy (kWh), Connector status.

🎛️ **CONTROLLED SETPOINTS**
Dynamic Charging Power Limit (kW / Amps setpoints).

⚡ **HANDLED & AUTOMATION ROLE**
Dynamically limits DC fast charging output to stay strictly below site grid connection capacity and Flemish peak limits during high-demand periods.

### 3.7 BMW i Wallbox (Connect/Pro) `bmw_charger`

MODBUS

**Manufacturer / Vendor:** BMW

**Communication Interface:** Modbus TCP via LAN (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** BMW i Wallbox Connect, BMW i Wallbox Pro (Modbus TCP / UDP)

🔍 **TELEMETRY READ OUT**
Charging power (W), Session energy (kWh).

🎛️ **CONTROLLED SETPOINTS**
Dynamic Current Setpoint (Amps, 6A to 32A).

⚡ **HANDLED & AUTOMATION ROLE**
Local LAN current regulation; coordinates charging with home solar availability and off-peak EPEX price hours.

### 3.8 Bender Charger `bender_charger`

MODBUS

**Manufacturer / Vendor:** Bender

**Communication Interface:** Modbus TCP (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Bender CC612, CC613 Charge Controllers (embedded in various OEM wallboxes)

🔍 **TELEMETRY READ OUT**
Active charging power (W), Session energy (kWh), Pilot state.

🎛️ **CONTROLLED SETPOINTS**
Bender CC612 / CC613 HEMSw Limit (`1001` 0.1 A on FW >= 4.70, fallback `1000` 1 A) and Dynamic 1-Phase / 3-Phase Switching (`1002`).

⚡ **HANDLED & AUTOMATION ROLE**
Direct local Modbus TCP control for Bender CC612/CC613 controllers (EBEE, Compleo, Mennekes OEMs) with sub-amp throttling and 1P/3P solar switching.

### 3.9 Blitzpower / Phoenix CharX `blitzpower_charger`

MODBUS

**Manufacturer / Vendor:** Blitzpower

**Communication Interface:** Modbus TCP (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Blitzpower AC Wallbox Series (Phoenix Contact CHARX architecture)

🔍 **TELEMETRY READ OUT**
Charging power (W), Cumulative energy (kWh).

🎛️ **CONTROLLED SETPOINTS**
Charging Current Throttling (Amps, 6A to 32A).

⚡ **HANDLED & AUTOMATION ROLE**
Fast Modbus register regulation: modulates current setpoints for solar surplus charging and peak shaving.

### 3.10 Delta AC MAX / SLIM / Ultra Fast Charger `delta_charger`

MODBUS

**Manufacturer / Vendor:** Delta

**Communication Interface:** Modbus TCP (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Delta AC MAX Basic/Smart (7.4kW/11kW/22kW), Delta SLIM, Ultra Fast EV Chargers

🔍 **TELEMETRY READ OUT**
Active charging power (W), Total energy (kWh).

🎛️ **CONTROLLED SETPOINTS**
Charging Current Setpoint (Amps, 6A to 32A).

⚡ **HANDLED & AUTOMATION ROLE**
Modulates Delta wallbox current setpoint via Modbus TCP; optimizes charging speed to absorb solar production without importing expensive grid power.

### 3.11 EN+ AC EV Charger `enplus_charger`

OCPP 1.6-J

**Manufacturer / Vendor:** EN+

**Communication Interface:** Native OCPP 1.6-J WebSocket Server (Port: 8887)

**Supported Hardware Models:** EN+ AC EV Charger Series (Home / Commercial Series)

🔍 **TELEMETRY READ OUT**
Charging power (W), Energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages.

🎛️ **CONTROLLED SETPOINTS**
Charging Current Throttling (Amps, 6A to 32A), Phase Switching (1P / 3P).

⚡ **HANDLED & AUTOMATION ROLE**
Autonomous solar tracking and peak shaving via OCPP SetChargingProfile commands.

### 3.12 ETEK EKEPC2-C/S Charge Controller `etek_charger`

MODBUS

**Manufacturer / Vendor:** ETEK

**Communication Interface:** Modbus TCP via Gateway (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** ETEK EKEPC2-C / EKEPC2-S Intelligent Charge Controller (Modbus RTU/TCP)

🔍 **TELEMETRY READ OUT**
Charging power (W), Session energy (kWh).

🎛️ **CONTROLLED SETPOINTS**
Dynamic Current Throttling (Amps, 6A to 32A).

⚡ **HANDLED & AUTOMATION ROLE**
Direct register writes to adjust charge rate based on real-time household power balance.

### 3.13 EVBox Elvi / Livo / BusinessLine `evbox_charger`

OCPP 1.6-J

**Manufacturer / Vendor:** EVBox

**Communication Interface:** Native OCPP 1.6-J WebSocket Server (Port: 8887, ws://<EMS-IP>:8887/<ChargePointID>)

**Supported Hardware Models:** EVBox Elvi, EVBox Livo, EVBox BusinessLine (OCPP 1.6-J)

**🔍 TELEMETRY READ OUT**
Real-time charging power (W), Total delivered energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts), Connector status.

**🎛️ CONTROLLED SETPOINTS**
Dynamic Current Throttling via SetChargingProfile (6A to 32A), Phase Switching (1P / 3P).

**⚡ HANDLED & AUTOMATION ROLE**
Full native OCPP support for EVBox: dynamically throttles charging current in 1A increments to absorb rooftop solar and prevent capacity tariff penalties.

### 3.14 Easee Charger `easee_charger`

REST / HTTP

**Manufacturer / Vendor:** Easee

**Communication Interface:** Easee Cloud REST API (Cloud credentials authentication)

**Supported Hardware Models:** Easee Home, Easee Charge (Cloud REST API)

**🔍 TELEMETRY READ OUT**
Active charging power (W), Total session energy (kWh), Charger status.

**🎛️ CONTROLLED SETPOINTS**
Easee API dynamic circuit/charger current (6–32 A), session `pause_charging` / `resume_charging`, and automatic `phaseMode` 1-Phase / 3-Phase switching.

**⚡ HANDLED & AUTOMATION ROLE**
Verified smart charging control for Easee Home, Charge, and Charge Lite/Up: modulates charging amps, pauses/resumes cleanly, and switches between 1-phase and 3-phase solar modes.

### 3.15 Enovates SingleWallBox `enovates_charger`

MODBUS

**Manufacturer / Vendor:** Enovates

**Communication Interface:** Modbus TCP via LAN (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Enovates SingleWallBox (Modbus TCP)

**🔍 TELEMETRY READ OUT**
Charging power (W), Total energy (kWh).

**🎛️ CONTROLLED SETPOINTS**
Dynamic Current Setpoint (Amps, 6A to 32A).

**⚡ HANDLED & AUTOMATION ROLE**
Direct local Modbus control; operates in harmony with external CPO billing platforms while GEMS manages local load balancing and peak shaving.

### 3.16 Ensto Chago Wallbox / Pro `ensto_charger`

MODBUS

**Manufacturer / Vendor:** Ensto

**Communication Interface:** Modbus TCP (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Ensto Chago Wallbox, Ensto One, Ensto Pro Series (Modbus TCP)

**🔍 TELEMETRY READ OUT**
Active charging power (W), Total energy (kWh).

**🎛️ CONTROLLED SETPOINTS**
Charging Current Throttling (Amps, 6A to 32A).

**⚡ HANDLED & AUTOMATION ROLE**
High-reliability Scandinavian commercial/home charger; throttles charge current to respect domestic supply limits and solar surplus.

### 3.17 Etrel INCH / sonnenCharger `etrel_charger`

MODBUS

**Manufacturer / Vendor:** Etrel

**Communication Interface:** Modbus TCP via Ethernet (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Etrel INCH Pro, Etrel INCH Lite, sonnenCharger (Modbus TCP)

**🔍 TELEMETRY READ OUT**
Active power (W), Session energy (kWh).

**🎛️ CONTROLLED SETPOINTS**
Dynamic Current Setpoint (Amps, 6A to 32A).

**⚡ HANDLED & AUTOMATION ROLE**
Local Modbus TCP regulation; optimizes charging to match sonnen or third-party battery states and rooftop solar.

### 3.18 Fronius Wattpilot (Home / Go) `fronius_wattpilot`

OCPP 1.6-J

**Manufacturer / Vendor:** Fronius

**Communication Interface:** Native OCPP 1.6-J WebSocket Server (Port: 8887)

**Supported Hardware Models:** Fronius Wattpilot Home (11J/22J), Fronius Wattpilot Go (11J/22J)

**🔍 TELEMETRY READ OUT**
Active power (W), Energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts).

**🎛️ CONTROLLED SETPOINTS**
Dynamic Current Throttling (Amps, 6A to 32A), Phase Switching (1P / 3P).

**⚡ HANDLED & AUTOMATION ROLE**
Operates Wattpilot in zero-cloud mode directly connected to GEMS; supports dynamic 1-phase / 3-phase switching and solar surplus tracking.

### 3.19 GoodWe HCA Series EV Wallbox (Gen 2) `goodwe_charger` MODBUS

**Manufacturer / Vendor:** GoodWe
**Communication Interface:** Modbus TCP (Default Port: 502, Slave ID: 1)
**Supported Hardware Models:** GoodWe HCA Series EV Wallbox (Gen 2 Modbus TCP)

🔍 **TELEMETRY READ OUT**
Charging power (W), Session energy (kWh).

🎛️ **CONTROLLED SETPOINTS**
Charging Current Setpoint (Amps, 6A to 32A).

⚡ **HANDLED & AUTOMATION ROLE**
Coordinates GoodWe wallboxes with GoodWe hybrid inverters; prioritizes battery charging before routing surplus into the EV.

### 3.20 Hager witty (one / plus / pro / park 2) `hager_charger` OCPP 1.6-J

**Manufacturer / Vendor:** Hager
**Communication Interface:** Native OCPP 1.6-J WebSocket Server (Port: 8887)
**Supported Hardware Models:** Hager witty, witty one, witty plus, witty pro, witty park 2 (OCPP 1.6-J)

🔍 **TELEMETRY READ OUT**
Active power (W), Session energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages.

🎛️ **CONTROLLED SETPOINTS**
Dynamic Current Throttling (Amps, 6A to 32A), Phase Switching (1P / 3P).

⚡ **HANDLED & AUTOMATION ROLE**
Zero-cloud residential management of Hager witty wallboxes; solar excess following and capacity tariff peak shaving.

### 3.21 Huawei FusionCharge (SCharger-7KS / 22KT) `huawei_scharger` OCPP 1.6-J

**Manufacturer / Vendor:** Huawei
**Communication Interface:** Native OCPP 1.6-J WebSocket Server (Port: 8887, ws://<EMS-IP>:8887/<ChargePointID>)
**Supported Hardware Models:** Huawei FusionCharge AC Wallbox (SCharger-7KS Single-Phase, SCharger-22KT Three-Phase)

🔍 **TELEMETRY READ OUT**
Active charging power (W), Session energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts), Plug state.

🎛️ **CONTROLLED SETPOINTS**
Dynamic Current Throttling via SetChargingProfile (6A to 32A), Phase Switching (1P / 3P).

⚡ **HANDLED & AUTOMATION ROLE**
Direct local control of Huawei FusionCharge without requiring FusionSolar cloud: coordinates perfectly with Huawei SUN2000 inverters and LUNA2000 batteries to ensure the car only charges from 100% pure solar.

### 3.22 KEBA KeContact P30/P40 `keba_charger` MODBUS

**Manufacturer / Vendor:** KEBA
**Communication Interface:** Modbus TCP via LAN (Default Port: 502, Unit ID: 255)
**Supported Hardware Models:** KEBA KeContact P30 (c-series, x-series), KeContact P40 Series (Modbus TCP / UDP)

🔍 **TELEMETRY READ OUT**
Active charging power (W, reg 1020), Total energy (kWh, reg 1036).

🎛️ **CONTROLLED SETPOINTS**
KEBA KeContact P30 x/c-series & P40 Charging Current in milliamps (`5004`), Enable (`5014`), and KeContact S10 Phase Switching (`5050=3`, `5052=0/1`).

⚡ **HANDLED & AUTOMATION ROLE**
High-precision 1 mA Modbus TCP control and 1P/3P phase switching for KEBA KeContact P30 and P40 installations.

### 3.23 Mennekes AMTRON (4You / 4Business / ACU / AMEDIO) `mennekes_ocpp` OCPP 1.6-J

**Manufacturer / Vendor:** Mennekes
**Communication Interface:** Native OCPP 1.6-J WebSocket Server (Port: 8887)
**Supported Hardware Models:** Mennekes AMTRON 4You, AMTRON 4Business, AMTRON Professional, AMEDIO (OCPP 1.6-J)

🔍 **TELEMETRY READ OUT**
Active power (W), Total energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages.

🎛️ **CONTROLLED SETPOINTS**
Dynamic Current Throttling via SetChargingProfile (6A to 32A), Phase Switching (1P / 3P).

⚡ **HANDLED & AUTOMATION ROLE**
Universal OCPP control for modern Mennekes product lines; enables solar-only charging and Flemish capacity peak prevention.

### 3.24 Mennekes AMTRON (Xtra / Premium / HCC3 Modbus) `mennekes_modbus` MODBUS

**Manufacturer / Vendor:** Mennekes
**Communication Interface:** Modbus TCP via LAN (Default Port: 502, Slave ID: 1)
**Supported Hardware Models:** Mennekes AMTRON Xtra, AMTRON Premium, HCC3 Controller Series (Modbus TCP)

🔍 **TELEMETRY READ OUT**
Active charging power (W), Cumulative energy (kWh).

🎛️ **CONTROLLED SETPOINTS**
Dynamic Charging Current Throttling (Amps, 6A to 32A).

⚡ **HANDLED & AUTOMATION ROLE**
Direct local Modbus control for Mennekes premium hardware; eliminates cloud lag for instant peak shaving.

### 3.25 Mercedes-Benz Wallbox `mercedes_wallbox` OCPP 1.6-J

**Manufacturer / Vendor:** Mercedes-Benz
**Communication Interface:** Native OCPP 1.6-J WebSocket Server (Port: 8887)
**Supported Hardware Models:** Mercedes-Benz Wallbox (Home AC Charger, up to 22kW)

🔍 **TELEMETRY READ OUT**
Active power (W), Energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages.

🎛️ **CONTROLLED SETPOINTS**
Charging Current Throttling (Amps, 6A to 32A), Phase Switching (1P / 3P).

⚡ **HANDLED & AUTOMATION ROLE**
Autonomous home charging for Mercedes-Benz Wallbox; integrates into GEMS dynamic load balancing.

### 3.26 NRGkick (Gen 2) `nrgkick_gen2` MODBUS

**Manufacturer / Vendor:** NRGkick

**Communication Interface:** Modbus TCP via local Wi-Fi (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** NRGkick Gen 2 Mobile & Wall-Mounted Smart EV Charger (Local Modbus TCP / Wi-Fi)

**🔍 TELEMETRY READ OUT**
Real-time charging power (W), Energy delivered (kWh).

**🎛️ CONTROLLED SETPOINTS**
Charging Current Throttling (Amps, 6A to 32A), Phase Switching (1P / 3P).

**⚡ HANDLED & AUTOMATION ROLE**
Local Modbus TCP regulation for mobile/wallbox NRGkick; automatically modulates current setpoints based on solar excess.

### 3.27 OCPP Charger `ocpp_charger` OCPP 1.6-J

**Manufacturer / Vendor:** Generic

**Communication Interface:** Native OCPP 1.6-J WebSocket Server (`ws://<EMS-IP>:8887/<ChargePointID>` or `ws://<EMS-IP>:8080/api/ocpp/<ChargePointID>`)

**Supported Hardware Models:** Universal Generic OCPP 1.6-J Smart EV Charger (Any OCPP 1.6-J compliant hardware)

**🔍 TELEMETRY READ OUT**
Active power (W), Delivered energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts), Plug and transaction status.

**🎛️ CONTROLLED SETPOINTS**
Standardized OCPP 1.6-J SetChargingProfile (Current Setpoints: 0A pause and 6A to 32A).

**⚡ HANDLED & AUTOMATION ROLE**
Vendor-agnostic central system: supports hundreds of compliant commercial and residential wallboxes worldwide; coordinates dynamic charging profiles according to GEMS site strategy.

### 3.28 Peblar EV Charger (Home / Business / Plus) `peblar_charger` MODBUS

**Manufacturer / Vendor:** Peblar

**Communication Interface:** Modbus TCP (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Peblar Home, Peblar Business, Peblar Plus (Modbus TCP Direct)

**🔍 TELEMETRY READ OUT**
Real-time charging power (W), Cumulative session energy (kWh).

**🎛️ CONTROLLED SETPOINTS**
Dynamic Current Setpoint (Amps, 6A to 32A in 0.1A steps), Phase Switching (1P / 3P).

**⚡ HANDLED & AUTOMATION ROLE**
Dutch engineering wallbox: allows GEMS to command sub-second current adjustments over Modbus TCP while Pebble/Peblar cloud handles business split-billing.

### 3.29 Phoenix Contact Charx Charger `phoenix_charger` MODBUS

**Manufacturer / Vendor:** Phoenix Contact

**Communication Interface:** Modbus TCP (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Phoenix Contact CHARX SEC / EV-CC-AC1-M3 Series (Modbus TCP)

**🔍 TELEMETRY READ OUT**
Charging power (W), Total session energy (kWh).

**🎛️ CONTROLLED SETPOINTS**
Phoenix Contact CHARX SEC-3000 / EV-CC Modbus Charging Current Limit and Charge Enable Coil/Register.

**⚡ HANDLED & AUTOMATION ROLE**
Industrial-grade local Modbus TCP load balancing and solar surplus charging for Phoenix Contact CHARX controllers.

### 3.30 Phoenix Contact EV-CC / CHARX (Modbus TCP) `phoenix_contact_evcc` MODBUS

**Manufacturer / Vendor:** Phoenix Contact

**Communication Interface:** Modbus TCP (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Phoenix Contact EV-CC-AC1-M3 / CHARX Control Modular (Modbus TCP)

**🔍 TELEMETRY READ OUT**
Active charging power (W), Total energy (kWh).

**🎛️ CONTROLLED SETPOINTS**
Dynamic Current Throttling (Amps, 6A to 32A), Phase Switching (1P / 3P).

**⚡ HANDLED & AUTOMATION ROLE**
Direct register writes to adjust charge rates for peak shaving and solar surplus.

### 3.31 External CPO Wallbox (OCPP 1.6-J) `generic_cpo_ocpp` OCPP 1.6-J

**Manufacturer / Vendor:** Generic OCPP

**Communication Interface:** Native OCPP 1.6-J WebSocket Server (`ws://<EMS-IP>:8080/api/ocpp/<ChargePointID>`)

**Supported Hardware Models:** AC Wallbox Series compatible with OCPP 1.6-J and External CPO Reimbursement

**🔍 TELEMETRY READ OUT**
Active power (W), Delivered energy (kWh), 3-Phase currents and voltages.

**🎛️ CONTROLLED SETPOINTS**
Dynamic Current Throttling (Amps, 6A to 32A).

**⚡ HANDLED & AUTOMATION ROLE**
Local smart charging and straight-through OCPP proxy load arbitration with external CPO billing platforms based on solar excess and fuse limits.

### 3.32 Porsche Wallbox / Mobile Charger Connect `porsche_wallbox`

OCPP 1.6-J

**Manufacturer / Vendor:** Porsche

**Communication Interface:** Native OCPP 1.6-J WebSocket Server (Port: 8887)

**Supported Hardware Models:** Porsche Mobile Charger Connect, Porsche Wallbox (OCPP 1.6-J)

| 🔍 TELEMETRY READ OUT | 🎛️ CONTROLLED SETPOINTS | ⚡ HANDLED & AUTOMATION ROLE |
|---|---|---|
| Active charging power (W), Delivered energy (kWh), 3-Phase currents and voltages. | Dynamic Current Throttling (Amps, 6A to 32A), Phase Switching (1P / 3P). | Connects Porsche chargers directly to GEMS without subscription; enforces Flanders peak limits. |

### 3.33 RAEDIAN Gemini AC Wallbox `raedian_gemini`

⭐ Reference Flagship · MODBUS

**Manufacturer / Vendor:** RAEDIAN

**Communication Interface:** Modbus TCP via LAN (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** RAEDIAN Gemini Dual-Socket AC Commercial & Fleet Wallbox (Up to 2× 22kW, 44kW total)

| 🔍 TELEMETRY READ OUT | 🎛️ CONTROLLED SETPOINTS | ⚡ HANDLED & AUTOMATION ROLE |
|---|---|---|
| Combined active power (W), Individual Socket 1 & Socket 2 charging power (W), Per-socket energy (kWh), Individual 3-phase currents for Socket 1 (L1, L2, L3) and Socket 2 (L1, L2, L3 in Amps), Per-socket plug status. | Continuous Dynamic Current Modulation (0A to 64A total range, dynamically split across Socket 1 & Socket 2), Individual Socket Mode (Eco / Fast / Off). | Advanced dual-socket load management: balances charging speeds between two vehicles simultaneously; dynamically adjusts combined current from 0A to 64A; protects building electrical service fuses while maximizing solar self-consumption. |

### 3.34 RAEDIAN NEO / NEX AC Wallbox `raedian_charger`

⭐ Reference Flagship · MODBUS

**Manufacturer / Vendor:** RAEDIAN

**Communication Interface:** Modbus TCP via LAN or Wi-Fi (Default Port: 502, Slave ID: 1 or 2)

**Supported Hardware Models:** RAEDIAN NEO Series, RAEDIAN NEX Series AC Smart Wallbox (7.4kW, 11kW, 22kW)

| 🔍 TELEMETRY READ OUT | 🎛️ CONTROLLED SETPOINTS | ⚡ HANDLED & AUTOMATION ROLE |
|---|---|---|
| Real-time active charging power (W), Session energy (kWh), 3-Phase charging currents (L1, L2, L3 in Amps), 3-Phase charging voltages (L1, L2, L3 in Volts), Socket status & error codes. | Continuous Dynamic Current Throttling (Amps, 6A to 32A in 0.1A increments), Automated Session Pause & Resume. | Preferred Modbus TCP flagship integration: single-read batching polls all 3-phase currents and voltages in a single request; sub-second current throttling responds instantly to solar surplus and Flanders capacity tariff peaks. |

### 3.35 SMA EV Charger (7.4 / 22 / eCharger) `sma_evcharger`

REST / HTTP

**Manufacturer / Vendor:** SMA

**Communication Interface:** Local HTTP REST API (Default Port: 80 / 443)

**Supported Hardware Models:** SMA EV Charger 7.4, SMA EV Charger 22

| 🔍 TELEMETRY READ OUT | 🎛️ CONTROLLED SETPOINTS | ⚡ HANDLED & AUTOMATION ROLE |
|---|---|---|
| Active power (W), Cumulative energy (kWh), 3-Phase currents (L1, L2, L3 in Amps). | Monitor-Only (Read-Only Telemetry — SMA EV Charger requires SEMP/SHM2 for active current control). | Monitors SMA EV Charger real-time power draw and session energy for household load accounting. |

### 3.36 Schneider Electric EVlink (Wallbox / Pro AC) `schneider_evlink`

OCPP 1.6-J

**Manufacturer / Vendor:** Schneider Electric

**Communication Interface:** Native OCPP 1.6-J WebSocket Server (Port: 8887)

**Supported Hardware Models:** Schneider Electric EVlink Wallbox, EVlink Pro AC, EVlink Smart Wallbox (OCPP 1.6-J)

| 🔍 TELEMETRY READ OUT | 🎛️ CONTROLLED SETPOINTS | ⚡ HANDLED & AUTOMATION ROLE |
|---|---|---|
| Active charging power (W), Energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages. | Dynamic Current Throttling via SetChargingProfile (6A to 32A), Phase Switching (1P / 3P). | Native OCPP setpoint regulation; provides smooth solar surplus charging and capacity peak avoidance. |

### 3.37 Siemens VersiCharge GEN3 Wallbox `siemens_versicharge`

OCPP 1.6-J

**Manufacturer / Vendor:** Siemens

**Communication Interface:** Native OCPP 1.6-J WebSocket Server (Port: 8887, ws://<EMS-IP>:8887/<ChargePointID>)

**Supported Hardware Models:** Siemens VersiCharge GEN3 AC Smart Wallbox (Commercial & Residential 7.4kW - 22kW)

| 🔍 TELEMETRY READ OUT | 🎛️ CONTROLLED SETPOINTS | ⚡ HANDLED & AUTOMATION ROLE |
|---|---|---|
| Real-time power (W), Cumulative energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages. | Dynamic Current Throttling (Amps, 6A to 32A), Phase Switching (1P / 3P). | Connects Siemens VersiCharge GEN3 to GEMS over OCPP; executes Flanders peak shaving and solar surplus matching. |

### 3.38 Sigenergy SigenStor EVAC / EVDC Charger `sigenergy_evac`

MODBUS

**Manufacturer / Vendor:** Sigenergy

**Communication Interface:** Modbus TCP via Sigen Gateway (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Sigenergy Sigen EVAC AC Wallbox, Sigen EVDC Bidirectional Fast Charger

🔍 **TELEMETRY READ OUT**
Charging power (W), Session energy (kWh), 3-Phase currents and voltages.

🎛️ **CONTROLLED SETPOINTS**
Dynamic Current Throttling (Amps, 6A to 32A), Phase Switching (1P / 3P).

⚡ **HANDLED & AUTOMATION ROLE**
Direct Modbus control over Sigen EV charging; coordinates with SigenStor battery to absorb excess solar.

### 3.39 SmartEVSE v2 / v3 (HTTP / Modbus) `smartevse_charger`

REST / HTTP

**Manufacturer / Vendor:** SmartEVSE

**Communication Interface:** Local HTTP REST API / Modbus TCP (Default Port: 80, REST endpoint: /add)

**Supported Hardware Models:** SmartEVSE v2, SmartEVSE v3 Open-Source EV Charging Controller

🔍 **TELEMETRY READ OUT**
Active power (W), Total energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts), Pilot status.

🎛️ **CONTROLLED SETPOINTS**
Dynamic Current Throttling (Amps, 6A to 32A via /add?current=X), Phase Switching (1P / 3P).

⚡ **HANDLED & AUTOMATION ROLE**
Local REST setpoint modulation; allows open-source charging hardware to run 100% solar surplus charging without cloud accounts.

### 3.40 SolarEdge EV Charger `solaredge_evse`

MODBUS

**Manufacturer / Vendor:** SolarEdge

**Communication Interface:** Modbus TCP via SolarEdge inverter or standalone module (Default Port: 502 / 1502, Slave ID: 1)

**Supported Hardware Models:** SolarEdge Home EV Charger (Integrated Inverter-Charger & Standalone AC Wallbox)

🔍 **TELEMETRY READ OUT**
Active charging power (W), Cumulative energy (kWh).

🎛️ **CONTROLLED SETPOINTS**
Dynamic Current Setpoint (Amps, 6A to 32A).

⚡ **HANDLED & AUTOMATION ROLE**
Synchronizes charging rate with SolarEdge inverter production; protects household main fuses.

### 3.41 Veton / Phoenix CharX `veton_charger`

MODBUS

**Manufacturer / Vendor:** Veton

**Communication Interface:** Modbus TCP via internal Phoenix Contact CHARX controller (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Veton One, Veton Two, Veton Wallbox Series (Design Luxury Chargers)

🔍 **TELEMETRY READ OUT**
Active charging power (W), Session energy (kWh).

🎛️ **CONTROLLED SETPOINTS**
Dynamic Charging Current Throttling (Amps, 6A to 32A).

⚡ **HANDLED & AUTOMATION ROLE**
Direct Modbus TCP register writes; provides seamless solar surplus charging and peak shaving for premium architectural Veton chargers.

### 3.42 Volkswagen / Elli / Skoda / SEAT (ID. Charger Connect / Pro) `elli_charger`

OCPP 1.6-J

**Manufacturer / Vendor:** Volkswagen / Elli

**Communication Interface:** Native OCPP 1.6-J WebSocket Server (Port: 8887, ws://<EMS-IP>:8887/<ChargePointID>)

**Supported Hardware Models:** Volkswagen ID. Charger Connect / Pro, Elli Charger, Skoda iV Charger, SEAT Charger

🔍 **TELEMETRY READ OUT**
Active charging power (W), Total session energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts), Plug status.

🎛️ **CONTROLLED SETPOINTS**
Dynamic Current Throttling via SetChargingProfile (6A to 32A), Phase Switching (1P / 3P).

⚡ **HANDLED & AUTOMATION ROLE**
Direct native OCPP server integration: allows VW Group ID. Chargers to charge purely on excess solar power and avoid Flanders capacity tariff peaks without requiring expensive manufacturer cloud subscriptions.

### 3.43 Wallbox Pulsar/Commander `wallbox_charger`

REST / HTTP

**Manufacturer / Vendor:** Wallbox

**Communication Interface:** Cloud REST API (Wallbox Cloud Credentials Authentication)

**Supported Hardware Models:** Wallbox Pulsar Plus, Pulsar Max, Commander 2, Copper SB (Wallbox Cloud REST API)

🔍 **TELEMETRY READ OUT**
Active charging power (W), Total energy (kWh), Charger status.

🎛️ **CONTROLLED SETPOINTS**
Wallbox API `PUT /v2/charger/{id}` dynamic amp limit (6–32 A) and `POST /v3/chargers/{id}/remote-action` (`action: 1` Resume / `2` Pause).

⚡ **HANDLED & AUTOMATION ROLE**
Verified closed-loop current modulation and automatic session pause/resume for Wallbox Pulsar Plus, Pulsar Max, Copper SB, and Commander 2.

### 3.44 Webasto Next Charger `webasto_charger` MODBUS

**Manufacturer / Vendor:** Webasto

**Communication Interface:** Modbus TCP via LAN or Wi-Fi (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Webasto Next, Webasto Live, Webasto Unite (Modbus TCP Direct)

**🔍 TELEMETRY READ OUT**
Active charging power (W, reg 5000), Delivered session energy (kWh, reg 5002).

**🎛️ CONTROLLED SETPOINTS**
Dynamic Charging Current Throttling (Amps, reg 5004, 6A to 32A).

**⚡ HANDLED & AUTOMATION ROLE**
Direct Modbus control: throttles Webasto wallbox within milliseconds to protect main grid fuses and prevent capacity tariff spikes while external backend handles corporate billing.

### 3.45 Zaptec Pro/Go `zaptec_charger` REST / HTTP

**Manufacturer / Vendor:** Zaptec

**Communication Interface:** Cloud REST API (Zaptec OAuth2 authentication)

**Supported Hardware Models:** Zaptec Go, Zaptec Pro (Zaptec Cloud REST API)

**🔍 TELEMETRY READ OUT**
Active power (W), Cumulative energy (kWh), Session status.

**🎛️ CONTROLLED SETPOINTS**
Zaptec OAuth2 `/api/chargers/{id}/update` (`maxChargeCurrent` 0.1 A, `maxChargePhases` 1/3, Go 2 `threeToOnePhaseSwitchCurrent`), plus `/sendCommand/506` (Pause) & `507` (Resume).

**⚡ HANDLED & AUTOMATION ROLE**
Verified precision control for Zaptec Go, Go 2, and Pro installations with OCMF energy decoding, pause/resume, and 1P/3P solar switching.

### 3.46 go-e Charger Gemini / HOMEfix / PRO `goe_charger` REST / HTTP

**Manufacturer / Vendor:** go-e

**Communication Interface:** Local HTTP REST API v2 (Default Port: 80, JSON endpoints: /api/status, /api/set)

**Supported Hardware Models:** go-e Charger Gemini, Gemini flex, HOMEfix, HOME+ (Local HTTP API v2)

**🔍 TELEMETRY READ OUT**
Charging power (W), Session energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts), Authorization state.

**🎛️ CONTROLLED SETPOINTS**
Volatile RAM Charging Current Throttling (Amps, 6A to 32A via `/api/set?amx=X` + `frc` pause/resume — zero NVS flash wear), Phase Switching (1P / 3P via `/api/set?psm=X`).

**⚡ HANDLED & AUTOMATION ROLE**
High-speed local HTTP API: allows GEMS to switch dynamically between 1-phase (from 1.4 kW solar) and 3-phase (up to 22 kW) charging; throttles within milliseconds during domestic power spikes.

### 3.47 Generic Local REST EVSE Gateway `generic_rest_evse` REST / HTTP

**Manufacturer / Vendor:** Generic REST

**Communication Interface:** Local HTTP REST API (Default Port: 80, REST endpoint: /api/v1)

**Supported Hardware Models:** Open Local REST EVSE Controllers and Load Gateways

**🔍 TELEMETRY READ OUT**
Real-time Charging Power (W), Cumulative Session Energy (kWh), Charging Station State.

**🎛️ CONTROLLED SETPOINTS**
Dynamic Charging Current Throttling (Amps, 6A to 32A).

**⚡ HANDLED & AUTOMATION ROLE**
Distributed multi-charger orchestration and dynamic load management, balancing aggregated fleet charging against local grid connection limits.

### 3.48 Kempower Fast Charging Infrastructure `kempower_charger` MODBUS

**Manufacturer / Vendor:** Kempower

**Communication Interface:** Modbus TCP via Kempower Central Controller (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Kempower C-Series and S-Series Fast Charging Satellites & Power Units

**🔍 TELEMETRY READ OUT**
Real-time Charging Power (W), Session Energy (kWh), 3-Phase Charging Currents (L1, L2, L3 in Amps).

**🎛️ CONTROLLED SETPOINTS**
Native local OCPP 1.6-J `SetChargingProfile` power/current throttling with automatic Modbus TCP register `200` (`0.1 A`) fallback.

**⚡ HANDLED & AUTOMATION ROLE**
Verified commercial fast-charging and fleet load management for Kempower Satellite and Station Chargers.

### 3.49 MyEnergi Zappi EV Charger `myenergi_zappi` REST / HTTP

**Manufacturer / Vendor:** MyEnergi

**Communication Interface:** Local HTTP REST / CGI API (Default Port: 80, Endpoint: /cgi-jstatus)

**Supported Hardware Models:** MyEnergi Zappi v2 and v2.1 with Hub / built-in Wi-Fi

**🔍 TELEMETRY READ OUT**
Real-time Charging Power (W), Session Added Energy (kWh), Charging Status, Mode (Fast, Eco, Eco+).

**🎛️ CONTROLLED SETPOINTS**
Authenticated HTTP Digest CGI commands (`/cgi-zappi-mode-Z{sno}-{mode}-...`) with ASN redirect tracking, switching dynamically between Stop, Eco+, Eco, and Fast boost modes.

**⚡ HANDLED & AUTOMATION ROLE**
Verified local/hub control for myenergi Zappi v2/v2.1 wallboxes with full 3-phase telemetry, phase mode reporting, and active session throttling.

### 3.50 SolaX Smart EV Charger `solax_evcharger`

MODBUS

**Manufacturer / Vendor:** SolaX

**Communication Interface:** Modbus TCP via Wi-Fi/Ethernet (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** SolaX Smart EV Charger X1-EVC / X3-EVC (7.2kW – 22kW)

**🔍 TELEMETRY READ OUT**
Real-time Charging Power (W), Cumulative Session Energy (kWh), Phase Currents (L1, L2, L3 in Amps).

**🎛️ CONTROLLED SETPOINTS**
SolaX X1/X3-EVC Max Current (`0x0628` in `0.01 A` steps), Remote Mode (`0x060D=1`), Stop/Start (`0x0627`), and 1-Phase / 3-Phase Switching (`0xA105`).

**⚡ HANDLED & AUTOMATION ROLE**
Verified native Modbus TCP/RTU closed-loop control for SolaX Smart EV Chargers with 10 mA solar tracking and 1P/3P contactor switching.

### 3.51 Teltonika TeltoCharge `teltonika_teltocharge`

OCPP 1.6-J

**Manufacturer / Vendor:** Teltonika Energy

**Communication Interface:** Native OCPP 1.6-J WebSocket Server (Port: 8887, ws://<EMS-IP>:8887/<ChargePointID>)

**Supported Hardware Models:** Teltonika TeltoCharge EVC1 Series (7.4kW – 22kW)

**🔍 TELEMETRY READ OUT**
Real-time Charging Power (W), Cumulative Session Energy (kWh), 3-Phase Currents (L1, L2, L3 in Amps), Voltages (V).

**🎛️ CONTROLLED SETPOINTS**
Dynamic Charging Current Throttling (Amps, 6A to 32A via SetChargingProfile).

**⚡ HANDLED & AUTOMATION ROLE**
Native OCPP integration for European residential installations; optimizes charging against Flemish capacity peaks and EPEX spot prices.

### 3.52 Volt Time Source EV Charger `volttime_source`

OCPP 1.6-J

**Manufacturer / Vendor:** Volt Time

**Communication Interface:** Modbus TCP / OCPP 1.6-J (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Volt Time Source and Source Pro Smart Charging Stations

**🔍 TELEMETRY READ OUT**
Real-time Charging Power (W), Cumulative Session Energy (kWh), Phase Currents (L1, L2, L3 in Amps).

**🎛️ CONTROLLED SETPOINTS**
Volt Time / Plugchoice API v3 dynamic amp limit & pause/resume with local Modbus TCP register `1020` fallback.

**⚡ HANDLED & AUTOMATION ROLE**
Verified active load balancing and solar surplus throttling for Volt Time Source and Plugchoice EV wallboxes.

### 3.53 Wallbox Pulsar Plus / Commander (OCPP) `wallbox_ocpp`

OCPP 1.6-J

**Manufacturer / Vendor:** Wallbox

**Communication Interface:** Native OCPP 1.6-J WebSocket Server (Port: 8887, ws://<EMS-IP>:8887/<ChargePointID>)

**Supported Hardware Models:** Wallbox Pulsar Plus, Pulsar Max, Commander 2 / 2s

**🔍 TELEMETRY READ OUT**
Real-time Charging Power (W), Cumulative Session Energy (kWh), 3-Phase Currents (L1, L2, L3 in Amps), Voltages (V).

**🎛️ CONTROLLED SETPOINTS**
Dynamic Charging Current Throttling (Amps, 6A to 32A via SetChargingProfile).

**⚡ HANDLED & AUTOMATION ROLE**
Direct zero-cloud local OCPP integration for Wallbox hardware; eliminates cloud dependency and latency for responsive home energy automation.

### 3.54 cFos Power Brain Wallbox / Solar `cfos_powerbrain`

MODBUS

**Manufacturer / Vendor:** cFos

**Communication Interface:** Modbus TCP (Default Port: 4701, EVSE Unit ID: 1, Holding Registers `8058`–`8094`)

**Supported Hardware Models:** cFos Power Brain Wallbox 11kW / 22kW, cFos Power Brain Solar (1P/3P Relay)

**🔍 TELEMETRY READ OUT**
Real-Time Charging Power (W), Cumulative Imported Energy (kWh), 3-Phase Charging Currents (0.1 A), Active Phase Relay State.

**🎛️ CONTROLLED SETPOINTS**
0.1 A Charging Current Limit (`8093`), Charge Enable (`8094`), and 1-Phase / 3-Phase Contactor Switching (`8087`).

**⚡ HANDLED & AUTOMATION ROLE**
High-precision 0.1 A solar surplus tracking, main breaker protection, and automated 1P/3P solar contactor switching.

### 3.55 FoxESS EV Charger A7300 / A011K / A022K `foxess_evcharger`

MODBUS

**Manufacturer / Vendor:** FoxESS

**Communication Interface:** Modbus TCP / RTU (Default Port: 502, Unit ID: 1, Holding `0x1003`–`0x1017` & `0x3002`)

**Supported Hardware Models:** FoxESS A7300 (7.3 kW), A011K (11 kW), and A022K (22 kW) Smart AC Wallboxes

**🔍 TELEMETRY READ OUT**
Charging Power (W), Cumulative Energy (kWh), 3-Phase Currents (A), 3-Phase Voltages (V), Charger Status.

**🎛️ CONTROLLED SETPOINTS**
0.1 kW Stepped Power Limit (`0x3002`: `1.4–4.1 kW` 1-Phase band, `4.2–22.0 kW` 3-Phase band, `0` Pause) & Auto Phase Switch (`0x300A`).

**⚡ HANDLED & AUTOMATION ROLE**
Seamless ecosystem companion for FoxESS hybrid inverters with fine-grained solar surplus modulation and 1P/3P switching.

### 3.56 Heidelberg Energy Control / Amperfied connect.solar `heidelberg_ampere` MODBUS

**Manufacturer / Vendor:** Amperfied (Heidelberg)

**Communication Interface:** Modbus TCP (Port: 502, Unit ID: 255) & Modbus RTU RS485 (19200 8E1, Unit ID: 1)

**Supported Hardware Models:** Heidelberg Energy Control, Amperfied Wallbox connect.home, connect.solar, connect.business

**🔍 TELEMETRY READ OUT**
Active Charging Power (W), Total Energy (kWh), 3-Phase Currents (0.1 A), 3-Phase Voltages (V), IEC Vehicle State, Active Phase Count.

**🎛️ CONTROLLED SETPOINTS**
0.1 A Charging Current Limit (Holding `261`), Standby/Lock Management (`258`–`259`), and 1-Phase / 3-Phase Switching (`501` on `connect.solar`).

**⚡ HANDLED & AUTOMATION ROLE**
Sub-second 0.1 A current modulation for Flanders capacity peak shaving and seamless 1P/3P solar charging.

### 3.57 openWB Pro / openWB series2 (HTTP & Modbus) `openwb_pro` REST / HTTP

**Manufacturer / Vendor:** openWB

**Communication Interface:** Local HTTP JSON/Form API (`/connect.php`) and openWB 2.0 Modbus TCP (Port: 1502, Unit ID: 1)

**Supported Hardware Models:** openWB Pro, openWB series2 standard+, custom, duo (2.0 Modbus TCP & Pro HTTP)

**🔍 TELEMETRY READ OUT**
Real-Time Charging Power (W), Imported Energy (kWh), 3-Phase Currents (A), 3-Phase Voltages (V), Plug & Charge State, Active Phases.

**🎛️ CONTROLLED SETPOINTS**
0.1 A / 0.01 A Dynamic Current Limit (`ampere` / Holding `10171`) and Automated 1-Phase / 3-Phase Contactor Switching (`phasetarget` / `10180`).

**⚡ HANDLED & AUTOMATION ROLE**
Ultra-fine 10–100 mA solar surplus tracking and hardware 1P/3P phase switching with zero cloud dependency.

### 3.58 Sungrow AC011E-01 / AC22E-01 EV Charger `sungrow_evcharger` MODBUS

**Manufacturer / Vendor:** Sungrow

**Communication Interface:** Modbus TCP / RTU (Default Port: 502, Default Unit ID: 248, Input `21269`–`21316` & Holding `21202`–`21211`)

**Supported Hardware Models:** Sungrow AC011E-01 (11 kW) and AC22E-01 (22 kW) 3-Phase Smart Wallboxes

**🔍 TELEMETRY READ OUT**
Active Charging Power (W), Cumulative Energy (kWh), 3-Phase Currents (0.1 A), 3-Phase Voltages (0.1 V), Phase Switch Status.

**🎛️ CONTROLLED SETPOINTS**
0.1 A Output Current Limit (`21202`), Start/Stop Control (`21211`), and 1-Phase / 3-Phase Switching (`21203`).

**⚡ HANDLED & AUTOMATION ROLE**
Direct local Modbus companion for Sungrow SH/SG installations with 0.1 A surplus regulation and 1P/3P switching.

### 3.59 Tinkerforge WARP Charger 2 / 3 (Smart & Pro) `warp_charger` REST / HTTP

**Manufacturer / Vendor:** Tinkerforge

**Communication Interface:** Local HTTP REST API (`/evse/state`, `/meter/values`, `/evse/external_current`, `/power_manager/external_control`)

**Supported Hardware Models:** Tinkerforge WARP Charger 2 Smart/Pro & WARP Charger 3 Smart/Pro (11 kW / 22 kW)

**🔍 TELEMETRY READ OUT**
Active Charging Power (W), Absolute Energy (kWh), 3-Phase Currents (A), 3-Phase Voltages (V), IEC 61851 State, 1P/3P Contactor State.

**🎛️ CONTROLLED SETPOINTS**
1 mA Resolution External Current Limit (`6000`–`32000 mA`, `0` Pause) and WARP3 1-Phase / 3-Phase Contactor Switching (`phases_wanted`).

**⚡ HANDLED & AUTOMATION ROLE**
Industrial open-interface wallbox integration with 1 mA precision solar tracking and automated 1P/3P contactor control.

### 3.60 Vestel EVC04 Home Smart / Mobility `vestel_evc04` MODBUS

**Manufacturer / Vendor:** Vestel

**Communication Interface:** Modbus TCP (Default Port: 502, Default Unit ID: 255, Lifebit Watchdog `6000=1`)

**Supported Hardware Models:** Vestel EVC04-AC11 / EVC04-AC22 Home Smart, Mobility, and Compatible OEM Wallboxes

**🔍 TELEMETRY READ OUT**
Total Active Power (W), Cumulative Energy (kWh), 3-Phase Currents (mA), 3-Phase Voltages (V), Chargepoint & Cable State.

**🎛️ CONTROLLED SETPOINTS**
Dynamic Charging Current Limit (`5004`, `0` or `6–32 A`), Fail-Safe Watchdog (`6000`), and 1-Phase / 3-Phase Switching (`405`).

**⚡ HANDLED & AUTOMATION ROLE**
Reliable local Modbus TCP load balancing, main fuse protection, and 1P/3P solar switching for Vestel EVC04 wallboxes.

### 3.61 Hardy Barth cPμ1 / cPμ2 (Salia / eCB1) Wallbox `hardy_barth_charger` REST / HTTP

**Manufacturer / Vendor:** Hardy Barth

**Communication Interface:** Local HTTP REST API (Default Port: 80, `/api/v1/chargecontrols/1` & `/api/secc`)

**Supported Hardware Models:** Hardy Barth cPμ1 (eCB1), cPμ2 (Salia PLCC), Wallbox cPH1 / cPH2 / cPP2 Series

**🔍 TELEMETRY READ OUT**
Real-Time Charging Power (W), Cumulative Session Energy (kWh), 3-Phase Charging Currents (`L1, L2, L3` in Amps) & Voltages (`L1, L2, L3` in Volts).

**🎛️ CONTROLLED SETPOINTS**
Dynamic Local REST Current Limit (`PUT /api/v1/chargecontrols/1/mode` with `manualmodeamp=6..32 A` or `mode=stop`; Salia `PUT /api/secc` `grid_current_limit` fallback).

**⚡ HANDLED & AUTOMATION ROLE**
100% local zero-cloud current modulation for Hardy Barth Salia and eCB1 wallboxes: matches EV charging speed to rooftop solar surplus and throttles down during main breaker or capacity peaks.

## 📊 Category 4: Grid Smart Meters & Energy Analyzers (40 Native Templates)

Smart meter templates interface directly via P1 port serial (RJ12 cable at 115200 baud), network P1 bridges, or RS485/Modbus TCP meters. They supply net grid import/export power (W), lifetime energy totals (kWh), per-phase currents (L1, L2, L3), per-phase voltages, and the official 15-minute rolling average peak demand register required for Flemish capacity tariff calculations.

### 4.1 ABB A43 / A44 / B23 / B24 Meter `abb_meter`

MODBUS

**Manufacturer / Vendor:** ABB

**Communication Interface:** Modbus RTU via RS485-to-Ethernet Gateway or Modbus TCP (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** ABB A43, A44 (3-Phase Direct/CT), B23, B24 (3-Phase Compact) Modbus Meters

🔍 **TELEMETRY READ OUT**
Active grid import/export power (W), Total active import/export energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts), Frequency (Hz).

🎛️ **CONTROLLED SETPOINTS**
Read-only grid telemetry.

⚡ **HANDLED & AUTOMATION ROLE**
High-precision grid boundary metering; feeds continuous 3-phase currents into GEMS phase-load balancing engine to prevent tripping main service fuses; drives Flanders 15-minute capacity peak calculations.

### 4.2 Acrel ADW300 Meter `acrel_adw300`

MODBUS

**Manufacturer / Vendor:** Acrel

**Communication Interface:** Modbus TCP (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Acrel ADW300 Series Wireless / Ethernet 3-Phase Multi-Function Energy Meter

🔍 **TELEMETRY READ OUT**
Total active power (W), Import/export energy (kWh), Per-phase active power (W), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts).

🎛️ **CONTROLLED SETPOINTS**
Read-only grid telemetry.

⚡ **HANDLED & AUTOMATION ROLE**
Monitors main grid connection or sub-distribution panels; informs GEMS about exact phase loads to modulate EV charger speeds and safeguard service fuses.

### 4.3 B+GE-TECH WS100 / DS100 Meter `bgetech_ws100`

MODBUS

**Manufacturer / Vendor:** B+GE-TECH

**Communication Interface:** Modbus TCP / RTU Gateway (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** B+GE-TECH WS100, DS100 Series 3-Phase DIN-Rail Modbus Smart Meters

🔍 **TELEMETRY READ OUT**
Real-time import/export power (W), Total active energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts).

🎛️ **CONTROLLED SETPOINTS**
Read-only grid telemetry.

⚡ **HANDLED & AUTOMATION ROLE**
Direct grid boundary metering; provides real-time wattage for closed-loop EV charging solar surplus tracking and Flanders peak shaving.

### 4.4 Carlo Gavazzi EM112 / EM340 / EM540 `carlogavazzi_em340`

MODBUS

**Manufacturer / Vendor:** Carlo Gavazzi

**Communication Interface:** Modbus RTU via RS485 USB/Ethernet Gateway (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Carlo Gavazzi EM112 (1-Phase), EM340 (3-Phase Direct), EM540 (Ultra-Fast 3-Phase) Meters

🔍 **TELEMETRY READ OUT**
Net active power (W), Total import/export kWh, 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts).

🎛️ **CONTROLLED SETPOINTS**
Read-only grid telemetry.

⚡ **HANDLED & AUTOMATION ROLE**
Fast sub-second boundary meter (especially EM540); crucial for instant reactive throttling of EV chargers during household spikes to protect main fuses.

### 4.5 Carlo Gavazzi EM24 (RS485/Ethernet) `carlogavazzi_em24`

MODBUS

**Manufacturer / Vendor:** Carlo Gavazzi

**Communication Interface:** Modbus TCP / Modbus RTU Gateway (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Carlo Gavazzi EM24 Series 3-Phase Energy Meter (RS485 & Ethernet LAN versions)

🔍 **TELEMETRY READ OUT**
Total active power (W, signed: + import, - export), Cumulative energy registers (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts).

🎛️ **CONTROLLED SETPOINTS**
Read-only grid telemetry.

⚡ **HANDLED & AUTOMATION ROLE**
Industry-standard utility meter used in premium solar and EVSE installations; provides deterministic 1-second telemetry for GEMS peak shaving and dynamic load balancing.

### 4.6 ESPHome DSMR / DLMS Smart Meter `esphome_meter`

REST / HTTP

**Manufacturer / Vendor:** ESPHome

**Communication Interface:** Local HTTP REST JSON API (Default Port: 80, Endpoint: /sensor/grid_power or /text_sensor/dsmr)

**Supported Hardware Models:** ESPHome Smart Meter Gateway (P1 DSMR Reader / DLMS / SML IR Optical Heads on ESP8266/ESP32)

**🔍 TELEMETRY READ OUT**
Real-time grid power (W), Total import/export energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts), Gas consumption (m³).

**🎛️ CONTROLLED SETPOINTS**
Read-only REST telemetry.

**⚡ HANDLED & AUTOMATION ROLE**
Integrates DIY open-hardware P1 readers into GEMS; delivers full digital telegram data over Wi-Fi with zero cloud dependencies.

### 4.7 Eastron SDM120 / SDM220 / SDM230 (1-Phase Meter) `eastron_sdm120_230`

MODBUS

**Manufacturer / Vendor:** Eastron

**Communication Interface:** Modbus RTU via RS485-to-Ethernet Bridge or USB dongle (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Eastron SDM120, SDM220, SDM230 Series 1-Phase DIN-Rail Modbus Smart Meters

**🔍 TELEMETRY READ OUT**
Active power (W), Total import/export energy (kWh), Phase current (Amps), Line voltage (Volts), Power factor.

**🎛️ CONTROLLED SETPOINTS**
Read-only telemetry poller.

**⚡ HANDLED & AUTOMATION ROLE**
Monitors single-phase grid connections, dedicated heat pump sub-circuits, or single-phase PV inverters; integrates into GEMS load analysis.

### 4.8 Eastron SDM630 3-Phase Meter `eastron_sdm630`

MODBUS

**Manufacturer / Vendor:** Eastron

**Communication Interface:** Modbus RTU via RS485 Gateway or Modbus TCP (Default Port: 502, Slave ID: 1, Baud: 9600)

**Supported Hardware Models:** Eastron SDM630 Modbus V2 / V3 / MCT 3-Phase Multifunction Energy Meter

**🔍 TELEMETRY READ OUT**
Net total active power (W, reg 0x0034), Total import/export energy (kWh), Individual phase currents (L1, L2, L3 in Amps, reg 0x0006), Individual phase voltages (L1, L2, L3 in Volts, reg 0x0000).

**🎛️ CONTROLLED SETPOINTS**
Read-only grid telemetry.

**⚡ HANDLED & AUTOMATION ROLE**
Widely deployed 3-phase meter; provides foundational telemetry for GEMS Flanders capacity tariff management, phase imbalance tracking, and solar surplus routing.

### 4.9 Eastron SDM72D / SMART X96 3-Phase Meter `eastron_sdm72_x96`

MODBUS

**Manufacturer / Vendor:** Eastron

**Communication Interface:** Modbus RTU via RS485-to-Ethernet Bridge (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Eastron SDM72D-Modbus, SDM72DR, SMART X96-1 / X96-5 Multifunction Meters

**🔍 TELEMETRY READ OUT**
Total active power (W), Active import/export energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts).

**🎛️ CONTROLLED SETPOINTS**
Read-only grid telemetry.

**⚡ HANDLED & AUTOMATION ROLE**
High-density DIN-rail energy monitoring; feeds live grid metrics into GEMS strategy loop to throttle chargers and log energy reports.

### 4.10 Eltako DSZ15DZMOD / DSZ16 Meter `eltako_meter`

MODBUS

**Manufacturer / Vendor:** Eltako

**Communication Interface:** Modbus RTU via RS485 Gateway (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Eltako DSZ15DZMOD-3×80A, DSZ16 Series 3-Phase Modbus Energy Meters

**🔍 TELEMETRY READ OUT**
Active power (W), Import/export energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts).

**🎛️ CONTROLLED SETPOINTS**
Read-only grid telemetry.

**⚡ HANDLED & AUTOMATION ROLE**
German precision submetering; provides real-time phase currents and grid import/export for GEMS household overload protection.

### 4.11 Finder 7M.24 / 7M.38 Meter `finder_7m`

MODBUS

**Manufacturer / Vendor:** Finder

**Communication Interface:** Modbus RTU via RS485 Bridge (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Finder 7M Series (7M.24 1-Phase, 7M.38 3-Phase Bi-directional Modbus Meters)

**🔍 TELEMETRY READ OUT**
Active power (W), Import/export active energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts).

**🎛️ CONTROLLED SETPOINTS**
Read-only grid telemetry.

**⚡ HANDLED & AUTOMATION ROLE**
Reliable MID-certified grid metering; feeds live electrical data into GEMS for closed-loop dynamic current regulation.

### 4.12 HomeWizard Meter `homewizard_meter`

REST / HTTP

**Manufacturer / Vendor:** HomeWizard

**Communication Interface:** Local HTTP REST API v1 (Default Port: 80, JSON Endpoint: /api/v1/data)

**Supported Hardware Models:** HomeWizard Wi-Fi P1 Meter (HWE-P1), HomeWizard Wi-Fi kWh Meter (1-Phase & 3-Phase)

**🔍 TELEMETRY READ OUT**
Active grid power (W), Total import/export energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts), Coupled gas meter consumption (m³).

**🎛️ CONTROLLED SETPOINTS**
Read-only local REST API.

**⚡ HANDLED & AUTOMATION ROLE**
Plug-and-play Wi-Fi P1 reader: enables GEMS to poll digital meter telegrams every 1-2 seconds with zero cabling; supports full Flemish & Dutch capacity tariff tracking.

### 4.13 Homey Energy Dongle (P1) `homey_p1`

REST / HTTP

**Manufacturer / Vendor:** Homey

**Communication Interface:** Local Network UDP Broadcast / HTTP (Default Port: 80)

**Supported Hardware Models:** Homey Energy Dongle (P1 Wi-Fi Gateway)

**🔍 TELEMETRY READ OUT**
Active grid power (W), Total import/export energy (kWh), 3-Phase currents and voltages.

**🎛️ CONTROLLED SETPOINTS**
Read-only telemetry poller.

**⚡ HANDLED & AUTOMATION ROLE**
Captures local network telemetry broadcasts from Homey P1 dongles; streams live household grid flow into GEMS dashboard.

### 4.14 Inepro PRO380-MOD / PRO1-MOD Meter `inepro_pro380`

MODBUS

**Manufacturer / Vendor:** Inepro

**Communication Interface:** Modbus RTU via RS485 Gateway (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Inepro PRO380-MOD, PRO1-MOD Series 3-Phase & 1-Phase DIN-Rail Modbus Meters

**🔍 TELEMETRY READ OUT**
Total active power (W), Import/export kWh registers, 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts).

**🎛️ CONTROLLED SETPOINTS**
Read-only grid telemetry.

**⚡ HANDLED & AUTOMATION ROLE**
High-accuracy Dutch utility metering; supplies 3-phase currents for GEMS dynamic load balancing and fuse protection.

### 4.15 Lovato DMG610 3-Phase Meter `lovato_dmg610`

MODBUS

**Manufacturer / Vendor:** Lovato

**Communication Interface:** Modbus TCP / RTU (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Lovato Electric DMG610, DMG800 Series Digital Multimeters

**🔍 TELEMETRY READ OUT**
Total active power (W), Import/export energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts).

**🎛️ CONTROLLED SETPOINTS**
Read-only grid telemetry.

**⚡ HANDLED & AUTOMATION ROLE**
Industrial-grade grid analyzer; provides rapid electrical metrics to protect main supply lines from transformer or fuse overload.

### 4.16 Loxone Miniserver `loxone_miniserver`

REST / HTTP

**Manufacturer / Vendor:** Loxone

**Communication Interface:** Local HTTP JSON API (Default Port: 80 / 443, Endpoints: /jdev/sfs/getstatus)

**Supported Hardware Models:** Loxone Miniserver Gen 1 / Gen 2, Miniserver Compact (HTTP JSON API)

**🔍 TELEMETRY READ OUT**
Aggregated household power (W), Submeter power values (W), Total energy counters (kWh).

**🎛️ CONTROLLED SETPOINTS**
Read-only REST telemetry poller.

**⚡ HANDLED & AUTOMATION ROLE**
Bridges smart home installations running Loxone Miniserver into GEMS; allows GEMS to utilize Loxone smart meter data as the primary grid reference.

### 4.17 Niko Home Control (NHC II) `niko_nhc`

REST / HTTP

**Manufacturer / Vendor:** Niko

**Communication Interface:** Local Hobby API / REST (Default Port: 8884 or 80, JSON Endpoint: /hobby/v1/devices)

**Supported Hardware Models:** Niko Home Control II (Connected Controller with Energy Module / Electricity Clamps)

**🔍 TELEMETRY READ OUT**
Real-time grid import/export power (W), Sub-circuit power consumption (W), Cumulative energy (kWh).

**🎛️ CONTROLLED SETPOINTS**
Read-only local API.

**⚡ HANDLED & AUTOMATION ROLE**
Seamless integration with popular Belgian Niko Home Control installations; reads electricity clamp measurements to drive GEMS peak shaving.

### 4.18 P1 Smart Meter (Network/TCP) `p1_network`

P1 / SERIAL

**Manufacturer / Vendor:** Generic

**Communication Interface:** Raw TCP Socket stream (Default Port: 23, 8088, or 8234, Raw DSMR telegrams)

**Supported Hardware Models:** P1 Smart Meter Network Bridge (TCP/Ethernet P1 Readers, Telnet Bridges, ESP-P1, Ser2Net)

**🔍 TELEMETRY READ OUT**
Real-time active power (+ import / - export in Watts, OBIS 1.7.0 & 2.7.0), Cumulative tariff 1 & 2 energy (kWh, OBIS 1.8.1/2 & 2.8.1/2), 3-Phase currents (L1, L2, L3 in Amps, OBIS 31.7.0, 51.7.0, 71.7.0), 3-Phase voltages (L1, L2, L3 in Volts, OBIS 32.7.0, 52.7.0, 72.7.0), Flemish 15-minute peak demand register (W, OBIS 1.6.0), Gas consumption (m³, OBIS 0-1:24.2.1).

**🎛️ CONTROLLED SETPOINTS**
Read-only digital telegram stream.

**⚡ HANDLED & AUTOMATION ROLE**
Primary grid interface for remote P1 meters over LAN: decodes official Fluvius (Belgium) and DSMR 4.x/5.x (Netherlands) telegrams; powers GEMS predictive Flanders capacity tariff engine and adaptive peak ceiling.

### 4.19 P1 Smart Meter (USB/Serial) `p1_serial`

P1 / SERIAL

**Manufacturer / Vendor:** Generic

**Communication Interface:** Direct Serial TTY (/dev/ttyUSB0 or /dev/ttyAMA0, Baud: 115200 for DSMR 4/5, 9600 for DSMR 2/3)

**Supported Hardware Models:** P1 Smart Meter Direct Serial Cable (USB-to-RJ12 FTDI / CP2102 Cable on Raspberry Pi USB)

**🔍 TELEMETRY READ OUT**
Real-time active power (W, OBIS 1.7.0/2.7.0), Tariff 1/2 import/export energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts), Belgian quarterly maximum peak demand (W, OBIS 1.6.0), Gas meter volume (m³).

**🎛️ CONTROLLED SETPOINTS**
Read-only digital telegram stream.

**⚡ HANDLED & AUTOMATION ROLE**
Direct zero-latency hardware connection to Belgian/Dutch smart meters; provides 1-second CRC-validated telegrams; core foundation for sub-second EV load balancing and Flanders capacity peak avoidance.

### 4.20 SMA Energy Meter 2.0 / Data Manager M (Modbus TCP) `sma_energymeter`

MODBUS

**Manufacturer / Vendor:** SMA

**Communication Interface:** Modbus TCP (Default Port: 502, Slave ID: 2 or 126)

**Supported Hardware Models:** SMA Energy Meter 2.0, SMA Data Manager M (Modbus TCP Speedwire)

**🔍 TELEMETRY READ OUT**
Active grid import/export power (W), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts).

**🎛️ CONTROLLED SETPOINTS**
Active Power Curtailment Dispatch (commands SMA inverters via Data Manager).

**⚡ HANDLED & AUTOMATION ROLE**
Measures grid boundary power and acts as a gateway; informs GEMS about net exchange; executes active solar feed-in curtailment via SMA communications.

### 4.21 Schneider Electric Acti9 iEM3000 / PM3200 Series `schneider_iem3000`

MODBUS

**Manufacturer / Vendor:** Schneider Electric

**Communication Interface:** Modbus RTU via RS485 Gateway or Modbus TCP (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Schneider Electric Acti9 iEM3150, iEM3155, iEM3255, PowerLogic PM3200 Series

**🔍 TELEMETRY READ OUT**
Total active power (W), Active energy import/export (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts).

**🎛️ CONTROLLED SETPOINTS**
Read-only grid telemetry.

**⚡ HANDLED & AUTOMATION ROLE**
Commercial and residential grade submetering; provides deterministic phase current telemetry to safeguard distribution boards from overloading.

### 4.22 Shelly 3EM (Gen 1 3-Phase Meter) `shelly_3em`

REST / HTTP

**Manufacturer / Vendor:** Shelly

**Communication Interface:** Local HTTP REST API (Default Port: 80, JSON Endpoint: /status)

**Supported Hardware Models:** Shelly 3EM (Gen 1 3-Phase Wi-Fi Energy Meter with 120A CT Clamps)

**🔍 TELEMETRY READ OUT**
Total active power (W), Per-phase power (W), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts), Cumulative returned/consumed energy (kWh).

**🎛️ CONTROLLED SETPOINTS**
Built-in 10A Contactor Output (Turn ON / Turn OFF via /relay/0?turn=on).

**⚡ HANDLED & AUTOMATION ROLE**
Reads high-speed 3-phase clamp measurements over Wi-Fi; built-in contactor can be automated to trigger external thermal loads or disconnect non-essential appliances during capacity peaks.

### 4.23 Shelly Pro 3EM / Gen3 3EM-63T (3-Phase Meter) `shelly_pro_3em`

REST / HTTP

**Manufacturer / Vendor:** Shelly

**Communication Interface:** Local Gen 2/3 HTTP RPC (Default Port: 80, JSON Endpoint: /rpc/EM.GetStatus)

**Supported Hardware Models:** Shelly Pro 3EM, Shelly Pro 3EM-400, Shelly Gen3 3EM-63T (DIN-Rail Ethernet/Wi-Fi)

**🔍 TELEMETRY READ OUT**
Net active power across phases (W), Per-phase active power (W), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts), 4-Quadrant energy counters (kWh).

**🎛️ CONTROLLED SETPOINTS**
Built-in Auxiliary Relay Output (Turn ON / Turn OFF via /rpc/Switch.Set).

**⚡ HANDLED & AUTOMATION ROLE**
High-reliability Ethernet-connected DIN meter: feeds instant phase telemetry into GEMS for sub-second EV current throttling; auxiliary relay output automates heat pump boost or domestic hot water.

### 4.24 Siemens SENTRON (PAC2200 / 7KT1665 Meter) `siemens_pac2200` MODBUS

**Manufacturer / Vendor:** Siemens

**Communication Interface:** Modbus TCP (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Siemens SENTRON PAC2200, 7KT1665 Series 3-Phase Modbus Smart Meters

**🔍 TELEMETRY READ OUT**
Total active power (W), Cumulative import/export energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts).

**🎛️ CONTROLLED SETPOINTS**
Read-only grid telemetry.

**⚡ HANDLED & AUTOMATION ROLE**
Premium industrial grade metering; monitors connection capacity and streams telemetry to GEMS for active load shedding and ROI calculations.

### 4.25 Socomec COUNTIS E24/E34/E44 / DIRIS A-40 `socomec_meter` MODBUS

**Manufacturer / Vendor:** Socomec

**Communication Interface:** Modbus RTU via RS485 Gateway (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Socomec COUNTIS E24, E34, E44 Series, DIRIS A-40 Multifunction Energy Meters

**🔍 TELEMETRY READ OUT**
Total active power (W), Active energy import/export (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts).

**🎛️ CONTROLLED SETPOINTS**
Read-only grid telemetry.

**⚡ HANDLED & AUTOMATION ROLE**
Accurate 3-phase commercial and domestic metering; informs GEMS about phase load distribution to balance EV charging and avoid tripping supply fuses.

### 4.26 WAGO 879 Series (879-3000 / 879-3020 / 879-3040) `wago_879` MODBUS

**Manufacturer / Vendor:** WAGO

**Communication Interface:** Modbus TCP (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** WAGO 879 Series (879-3000, 879-3020, 879-3040 3-Phase DIN-Rail Energy Meters)

**🔍 TELEMETRY READ OUT**
Active power (W), Total active energy (kWh), 3-Phase currents (L1, L2, L3 in Amps), 3-Phase voltages (L1, L2, L3 in Volts).

**🎛️ CONTROLLED SETPOINTS**
Read-only grid telemetry.

**⚡ HANDLED & AUTOMATION ROLE**
Precision DIN-rail meter; streams 3-phase currents and voltages into GEMS for real-time fuse protection, solar surplus calculations, and energy flow visualization.

### 4.27 Accuenergy AcuRev / Acuvim Series Meter `accuenergy_meter` MODBUS

**Manufacturer / Vendor:** Accuenergy

**Communication Interface:** Modbus TCP / RTU via RS485 Gateway (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Accuenergy Acuvim II, AcuRev 1310, AcuRev 2100 Multi-Circuit Submeter

**🔍 TELEMETRY READ OUT**
Real-Time Net Grid Active Power (W), Total Energy (kWh), Phase Currents (L1, L2, L3 in Amps), Voltages (V).

**🎛️ CONTROLLED SETPOINTS**
Autonomous / Read-Only grid telemetry.

**⚡ HANDLED & AUTOMATION ROLE**
High-precision 3-phase grid boundary monitoring and phase load balancing for commercial and residential installations.

### 4.28 Algodue UPM209 / UPM309 Series Meter `algodue_meter` MODBUS

**Manufacturer / Vendor:** Algodue

**Communication Interface:** Modbus RTU / TCP via RS485 Gateway (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Algodue UPM209, UPM309 3-Phase Network Analyzers

**🔍 TELEMETRY READ OUT**
Real-Time Net Grid Active Power (W), Total Energy (kWh), Phase Currents (L1, L2, L3 in Amps), Voltages (V).

**🎛️ CONTROLLED SETPOINTS**
Autonomous / Read-Only grid telemetry.

**⚡ HANDLED & AUTOMATION ROLE**
Continuous grid exchange measurement, power factor monitoring, and breaker overload protection.

### 4.29 Chint DTSU666 / DTSU666-H 3-Phase Meter `chint_dtsu666` MODBUS

**Manufacturer / Vendor:** Chint

**Communication Interface:** Modbus RTU via RS485 / Modbus TCP Gateway (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Chint DTSU666, DTSU666-H 3-Phase Direct & CT Smart Meters

**🔍 TELEMETRY READ OUT**
Real-Time Net Grid Active Power (W), Total Import & Export Energy (kWh), Phase Currents (L1, L2, L3 in Amps), Voltages (V).

**🎛️ CONTROLLED SETPOINTS**
Autonomous / Read-Only grid telemetry.

**⚡ HANDLED & AUTOMATION ROLE**
Standard 3-phase bidirectional grid meter integration; drives solar surplus calculations and EV charger current modulation.

### 4.30 Hager ECM / ECR / ECP Energy Meter `hager_meter`

MODBUS

**Manufacturer / Vendor:** Hager

**Communication Interface:** Modbus RTU / TCP via Agardio / IP Gateway (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Hager ECM, ECR, ECP, and SM103E 1-Phase and 3-Phase Energy Meters

🔍 **TELEMETRY READ OUT**
Real-Time Net Grid Power (W), Total Import Energy (kWh), Phase Currents (L1, L2, L3 in Amps), Voltages (V).

🎛️ **CONTROLLED SETPOINTS**
Autonomous / Read-Only grid telemetry.

⚡ **HANDLED & AUTOMATION ROLE**
DIN-rail residential distribution board monitoring, feeding sub-second phase telemetry into GEMS overload protection.

### 4.31 Janitza UMG Series Power Analyzer `janitza_umg`

MODBUS

**Manufacturer / Vendor:** Janitza

**Communication Interface:** Modbus TCP (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Janitza UMG 96RM, UMG 604, UMG 509 Power Quality Analyzers

🔍 **TELEMETRY READ OUT**
Real-Time Net Grid Power (W), Total Import Energy (kWh), Phase Currents (L1, L2, L3 in Amps), Line Voltages (V), Frequency (Hz).

🎛️ **CONTROLLED SETPOINTS**
Autonomous / Read-Only grid telemetry.

⚡ **HANDLED & AUTOMATION ROLE**
High-precision industrial power analyzer integration, capturing high-frequency grid flow and phase imbalance metrics.

### 4.32 Kamstrup Energy Meter (Modbus Gateway) `kamstrup_meter`

MODBUS

**Manufacturer / Vendor:** Kamstrup

**Communication Interface:** Modbus RTU / TCP via M-Bus / Modbus Gateway (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Kamstrup OMNIPOWER and MULTICAL Electricity & Thermal Meters

🔍 **TELEMETRY READ OUT**
Real-Time Net Grid Power (W), Total Import Energy (kWh), Phase Currents (L1, L2, L3 in Amps), Voltages (V).

🎛️ **CONTROLLED SETPOINTS**
Autonomous / Read-Only grid telemetry.

⚡ **HANDLED & AUTOMATION ROLE**
Integrates utility electricity and district heating/cooling submeters into the GEMS overall energy overview.

### 4.33 Landis+Gyr Industrial Utility Meter `landisgyr_meter`

MODBUS

**Manufacturer / Vendor:** Landis+Gyr

**Communication Interface:** Modbus RTU / TCP via Communication Unit (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Landis+Gyr E550, E650, E850 Industrial & Commercial Meters

🔍 **TELEMETRY READ OUT**
Real-Time Net Grid Power (W), Total Import & Export Energy (kWh), Phase Currents (L1, L2, L3 in Amps), Voltages (V).

🎛️ **CONTROLLED SETPOINTS**
Autonomous / Read-Only grid telemetry.

⚡ **HANDLED & AUTOMATION ROLE**
Monitors large commercial and industrial connection points to ensure dynamic peak shaving compliance.

### 4.34 Phoenix Contact EMpro Energy Meter `phoenix_empro`

MODBUS

**Manufacturer / Vendor:** Phoenix Contact

**Communication Interface:** Modbus TCP (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Phoenix Contact EMpro EEM-MA370, EEM-MB370 Series

🔍 **TELEMETRY READ OUT**
Real-Time Net Grid Power (W), Total Import Energy (kWh), Phase Currents (L1, L2, L3 in Amps), Voltages (V).

🎛️ **CONTROLLED SETPOINTS**
Autonomous / Read-Only grid telemetry.

⚡ **HANDLED & AUTOMATION ROLE**
Streams real-time 3-phase grid boundary measurements into GEMS for instantaneous solar tracking.

### 4.35 Weidmüller Energy Meter (500 / 600 / 700) `weidmuller_meter`

MODBUS

**Manufacturer / Vendor:** Weidmüller

**Communication Interface:** Modbus TCP (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Weidmüller Energy Meter 525, 610, 700 Series

🔍 **TELEMETRY READ OUT**
Real-Time Net Grid Power (W), Total Import Energy (kWh), Phase Currents (L1, L2, L3 in Amps), Voltages (V).

🎛️ **CONTROLLED SETPOINTS**
Autonomous / Read-Only grid telemetry.

⚡ **HANDLED & AUTOMATION ROLE**
Industrial grade power monitoring for multi-tenant and facility sub-distribution boards.

### 4.36 YouLess LS120 Pulse & P1 Meter `youless_ls120`

REST / HTTP

**Manufacturer / Vendor:** YouLess

**Communication Interface:** Local HTTP REST API (Default Port: 80, Endpoint: /a?f=j)

**Supported Hardware Models:** YouLess LS120 Optical Pulse & P1 Dongle

🔍 **TELEMETRY READ OUT**
Real-Time Net Grid Power (W), Total Import Energy (kWh).

🎛️ **CONTROLLED SETPOINTS**
Autonomous / Read-Only grid telemetry.

⚡ **HANDLED & AUTOMATION ROLE**
Connects to digital smart meter P1 ports or legacy optical rotating disc meters over local LAN without cloud subscriptions.

### 4.37 Fronius Smart Meter IP / Solar API v1 `fronius_smartmeter_ip`

REST / HTTP

**Manufacturer / Vendor:** Fronius

**Communication Interface:** Local HTTP Fronius Solar API v1 (`GET /solar_api/v1/GetMeterRealtimeData.cgi?Scope=Device&DeviceId=0`)

**Supported Hardware Models:** Fronius Smart Meter IP, TS 65A-3, TS 5kA-3, TS 100A-1 (Direct IP or via Fronius Inverter)

🔍 **TELEMETRY READ OUT**
Signed Net Grid Power (W), Total Imported & Exported Energy (kWh), Signed Per-Phase Currents (L1, L2, L3 in Amps), Phase Voltages (V).

🎛️ **CONTROLLED SETPOINTS**
Autonomous / Read-Only

⚡ **HANDLED & AUTOMATION ROLE**
Zero-wiring local HTTP grid boundary metering for Fronius ecosystems, feeding signed per-phase currents into main fuse protection and peak shaving.

### 4.38 Huawei EMMA-A02 / DTSU666-H Smart Power Sensor `huawei_emma`

MODBUS

**Manufacturer / Vendor:** Huawei

**Communication Interface:** Modbus TCP / RTU (Port: 502, Holding `31633`–`31666` on EMMA-A02 & `2102`–`2175` Float32 on DTSU666-H)

**Supported Hardware Models:** Huawei EMMA-A02 Energy Management Assistant, Huawei Smart Power Sensor DTSU666-H / DDSU666-H

🔍 **TELEMETRY READ OUT**
Signed Grid Supply-Point Power (W), Total Import & Export Energy (kWh), Signed 3-Phase Currents (A), 3-Phase Voltages (V).

🎛️ **CONTROLLED SETPOINTS**
Autonomous / Read-Only

⚡ **HANDLED & AUTOMATION ROLE**
High-speed grid boundary telemetry for Huawei SUN2000 & LUNA2000 sites with EMMA-A02 or standalone RS485 power sensors.

### 4.39 KOSTAL Smart Energy Meter (KSEM G1 / G2) `kostal_ksem`

MODBUS

**Manufacturer / Vendor:** KOSTAL

**Communication Interface:** Modbus TCP (Default Port: 502, Default Unit ID: 71, Holding Registers `0`–`143` & `512`–`519`)

**Supported Hardware Models:** KOSTAL Smart Energy Meter G1 & G2 (DIN-Rail Ethernet / RS485 Grid Meter)

🔍 **TELEMETRY READ OUT**
Active Import & Export Power (0.1 W), Total Import & Export Energy (0.1 Wh → kWh), Directional Per-Phase Currents (A) & Voltages (V).

🎛️ **CONTROLLED SETPOINTS**
Autonomous / Read-Only

⚡ **HANDLED & AUTOMATION ROLE**
Sub-second Ethernet DIN-rail grid boundary metering with per-phase directional current signing for DLB and capacity peak shaving.

### 4.40 Tasmota SML / OBIS Optical IR Reader & EcoTracker `tasmota_sml`

REST / HTTP

**Manufacturer / Vendor:** Tasmota

**Communication Interface:** Local HTTP JSON API (`/cm?cmnd=Status%208` StatusSNS & EcoTracker `/v1/json`)

**Supported Hardware Models:** Tasmota SML / OBIS / DLMS Optical IR Heads (Hichi, BitShake) & everHome EcoTracker

🔍 **TELEMETRY READ OUT**
Real-Time Net Grid Power (W), Total Import & Export Energy Counters (kWh), Phase Currents (A) & Voltages (V).

🎛️ **CONTROLLED SETPOINTS**
Autonomous / Read-Only

⚡ **HANDLED & AUTOMATION ROLE**
Cloud-free optical IR smart meter reading for modern European digital utility meters lacking RJ12 P1 ports.

## 🔌 Category 5: Smart Relays & Load Actuators (12 Native Templates)

Actuator templates integrate potential-free contactors, smart DIN plugs, and SG-Ready heat pump inputs over local LAN REST/RPC or Modbus coils. GEMS automatically triggers domestic hot water (DHW) immersion heaters, heat pump boost states, or pool pumps when solar surplus or negative pricing thresholds are reached, with built-in minimum run-time hysteresis.

### 5.1 Generic HTTP Relay `generic_relay`

REST / HTTP

**Manufacturer / Vendor:** Generic

**Communication Interface:** Local HTTP REST (Default Port: 80 / 8080, Configurable URL endpoints for State and Toggle)

**Supported Hardware Models:** Generic HTTP REST Relay (ESP8266, ESP32, Tasmota, Shelly Gen1, Custom Microcontrollers)

🔍 **TELEMETRY READ OUT**
Active power consumption of connected appliance (W, optional if meter reported).

🎛️ **CONTROLLED SETPOINTS**
Relay Contactor Output State (Turn ON / Turn OFF).

⚡ **HANDLED & AUTOMATION ROLE**
Automated contactor control: engages domestic hot water immersion heaters, pool circulation pumps, or resistive heating when solar export exceeds configurable thresholds (e.g., > 1500W for 2 minutes); enforces minimum run-time and off-time hysteresis.

### 5.2 Shelly Plus 1PM / Pro 1PM / Plug S (Relay & Power) `shelly_plus_1pm`

REST / HTTP

**Manufacturer / Vendor:** Shelly

**Communication Interface:** Local Gen 2/3 HTTP RPC (Default Port: 80, Endpoints: /rpc/Switch.Set, /rpc/Switch.GetStatus)

**Supported Hardware Models:** Shelly Plus 1PM, Shelly Pro 1PM, Shelly Plug S, Shelly Pro 2PM (Gen 2 / Gen 3 Mongoose OS)

🔍 **TELEMETRY READ OUT**
Connected appliance active power (W), Cumulative energy consumption (kWh), Internal device temperature (°C), Supply voltage (V).

🎛️ **CONTROLLED SETPOINTS**
Relay Contactor Output State (Turn ON / Turn OFF via /rpc/Switch.Set).

⚡ **HANDLED & AUTOMATION ROLE**
SG-Ready Heat Pump Boost & Appliance Automation: drives Heat Pump SG-Ready Operating State 3 (thermal boost mode) or hot water boilers during solar surplus or negative spot price hours; tracks real-time load wattage in GEMS logger.

### 5.3 Modbus Digital I/O Relay Module `modbus_relay`

MODBUS

**Manufacturer / Vendor:** Modbus Relais / I/O

**Communication Interface:** Modbus TCP / RTU via RS485 (Default Port: 502, Slave ID: 1)

**Supported Hardware Models:** Waveshare Modbus RTU/TCP Relay, ADAM-4000/6000, Moxa ioLogik, Kincony Modbus Relay

🔍 **TELEMETRY READ OUT**
Digital Input / Output Coil States (Open / Closed).

🎛️ **CONTROLLED SETPOINTS**
Relay Coil Actuation (Coil 1–8 Turn ON / OFF).

⚡ **HANDLED & AUTOMATION ROLE**
Industrial-grade contactor switching for high-current loads, three-phase thermal heaters, and heavy machinery peak shedding.

### 5.4 SG-Ready Heat Pump (Daikin/Nibe/Bosch/Vaillant/Panasonic) `sgready_heatpump`

REST / HTTP

**Manufacturer / Vendor:** SG-Ready (Daikin, Nibe, Bosch/Nefit, Vaillant, Panasonic, Mitsubishi, Stiebel Eltron, Wolf)

**Communication Interface:** Local HTTP REST / Relay Controller (Default Port: 80, REST Endpoint: /api/sgready)

**Supported Hardware Models:** Standard SG-Ready (Smart Grid Ready) Heat Pumps via dual potential-free relay contacts or local smart module

🔍 **TELEMETRY READ OUT**
Operating State (Operating State 1: Lock / State 2: Normal / State 3: Boost / State 4: Fast Heat), Power Consumption (W).

🎛️ **CONTROLLED SETPOINTS**
SG-Ready Operating Mode (State 2 Normal vs. State 3 Thermal Boost).

⚡ **HANDLED & AUTOMATION ROLE**
Automated heat pump thermal storage: commands State 3 (increased thermal setpoint) during solar surplus or negative dynamic price hours to preheat domestic water and buffers; returns to State 2 during grid peaks.

### 5.5 Shelly Plus 1 / Pro 1 (Dry Contact Relay) `shelly_plus_1`

REST / HTTP

**Manufacturer / Vendor:** Shelly

**Communication Interface:** Local HTTP RPC (Default Port: 80, Endpoint: /rpc/Switch.Set)

**Supported Hardware Models:** Shelly Plus 1, Shelly Pro 1 (Gen 2 / Gen 3 potential-free dry contact relays)

🔍 **TELEMETRY READ OUT**
Switch State (On / Off), Device Temperature (°C), Input State.

🎛️ **CONTROLLED SETPOINTS**
Dry Contact Relay Output State (Turn ON / Turn OFF).

⚡ **HANDLED & AUTOMATION ROLE**
Potential-free contact closure to trigger SG-Ready heat pump inputs, EV charger external inhibit terminals, or domestic circulation pumps.

### 5.6 Shelly Plus 2PM / Pro 2PM (Dual Relay & Meter) `shelly_plus_2pm` REST / HTTP

**Manufacturer / Vendor:** Shelly

**Communication Interface:** Local HTTP RPC (Default Port: 80, Endpoints: /rpc/Switch.Set, /rpc/Switch.GetStatus)

**Supported Hardware Models:** Shelly Plus 2PM, Shelly Pro 2PM (Dual-channel relay with independent power metering)

| 🔍 TELEMETRY READ OUT | 🎛️ CONTROLLED SETPOINTS | ⚡ HANDLED & AUTOMATION ROLE |
|---|---|---|
| Channel 1 & Channel 2 Active Power (W), Cumulative Energy per channel (kWh), Supply Voltage (V). | Dual Relay Output States (Channel 0 & Channel 1 Independent On / Off). | Independent dual-circuit automation: controls and measures multiple resistive heating stages or controls dual-coil SG-Ready heat pump configurations (State 1 through State 4). |

### 5.7 Daikin Altherma 3 & HomeHub (Smart Grid Modbus) `daikin_altherma` MODBUS

**Manufacturer / Vendor:** Daikin

**Communication Interface:** Modbus TCP (Default Port: 502, Unit ID: 1, Input `21`–`50` & Holding `2`, `53`)

**Supported Hardware Models:** Daikin Altherma 3 R / H / M & Daikin HomeHub (`EKRHH`) Heat Pump Systems

| 🔍 TELEMETRY READ OUT | 🎛️ CONTROLLED SETPOINTS | ⚡ HANDLED & AUTOMATION ROLE |
|---|---|---|
| Instantaneous Heat Pump Electrical Power (W), DHW Tank Temperature (°C), Leaving Water Temperature (°C), Smart Grid Status. | Smart Grid Operating Mode (`53`: Free Running Normal vs Recommended On PV Surplus Boost) & DHW Temperature Setpoint (`2`). | Native Modbus TCP thermal storage automation for Daikin Altherma 3 heat pumps: converts excess PV and negative-price electricity into domestic hot water and space heating. |

### 5.8 my-PV AC•THOR / AC•THOR 9s / AC ELWA 2 Solar Diverter `mypv_acthor` MODBUS

**Manufacturer / Vendor:** my-PV

**Communication Interface:** Modbus TCP (Default Port: 502, Unit ID: 1, Holding Registers `1000`–`1003`)

**Supported Hardware Models:** my-PV AC•THOR (3 kW), AC•THOR 9s (9 kW), AC ELWA 2, AC ELWA-E Linear Photovoltaic Heaters

| 🔍 TELEMETRY READ OUT | 🎛️ CONTROLLED SETPOINTS | ⚡ HANDLED & AUTOMATION ROLE |
|---|---|---|
| Actual Immersion Heating Power (W), Domestic Hot Water Tank Temperature (`0.1 °C`), Operating Status. | Stepless Linear Heating Power Setpoint (`1000`: `0–9000 W`), Relay Boost State, and Max Tank Temperature Setpoint (`1002`). | Proportional zero-export thermal storage: diverts exact watt-for-watt PV surplus into hot water buffers before any grid export or curtailment occurs. |

### 5.9 NIBE S-Series Heat Pump (S1155 / S1255 / S2125 / VVM) `nibe_heatpump` MODBUS

**Manufacturer / Vendor:** NIBE

**Communication Interface:** Modbus TCP (Default Port: 502, Unit ID: 1, Input `8`, `2166`, `2285` & Holding `697`, `707`)

**Supported Hardware Models:** NIBE S1155, S1255, S1156, S1256, S2125, SMO S40, VVM S320 Inverter Heat Pumps

| 🔍 TELEMETRY READ OUT | 🎛️ CONTROLLED SETPOINTS | ⚡ HANDLED & AUTOMATION ROLE |
|---|---|---|
| Instantaneous Compressor Electrical Power (`2166` W), Cumulative Consumed Electrical Energy (`2285` kWh), DHW Top Temperature (`8` °C). | SG-Ready / Smart Price Luxury DHW Boost (`697` / `47`) and Comfort Temperature Setpoint (`707` `0.1 °C`). | Direct Wi-Fi/Ethernet Modbus TCP integration for NIBE S-Series heat pumps: boosts thermal storage during solar surplus and cheap EPEX hours. |

### 5.10 Stiebel Eltron / Tecalor ISG Web & Plus (WPM) `stiebel_eltron_isg` MODBUS

**Manufacturer / Vendor:** Stiebel Eltron

**Communication Interface:** Modbus TCP (Default Port: 502, Unit ID: 1, Input `521`, `2500`, `3506` & Holding `1509`, `4000`–`4002`)

**Supported Hardware Models:** Stiebel Eltron & Tecalor WPM3, WPM3i, WPMsystem Heat Pumps via ISG web / ISG plus Gateway

| 🔍 TELEMETRY READ OUT | 🎛️ CONTROLLED SETPOINTS | ⚡ HANDLED & AUTOMATION ROLE |
|---|---|---|
| DHW Cylinder Temperature (`521` °C), Compressor & DHW Pump Status (`2500`), Cumulative Electrical Energy (kWh). | SG-Ready Energy Management Enable (`4000=1`), SG-Ready State 2 / State 3 PV Boost (`4001`–`4002`), and Comfort DHW Setpoint (`1509`). | Standardized Modbus TCP SG-Ready boost and comfort temperature control for Stiebel Eltron and Tecalor heat pumps. |

### 5.11 Tasmota Smart Plug / DIN Power Relay `tasmota_relay` REST / HTTP

**Manufacturer / Vendor:** Tasmota

**Communication Interface:** Local HTTP Command API (`/cm?cmnd=Power%20On|Off` & `/cm?cmnd=Status%208`)

**Supported Hardware Models:** Tasmota Wi-Fi / Ethernet Smart Plugs, Sonoff POW, Nous A1T, and Multi-Channel DIN Contactors

| 🔍 TELEMETRY READ OUT | 🎛️ CONTROLLED SETPOINTS | ⚡ HANDLED & AUTOMATION ROLE |
|---|---|---|
| Active Load Power (W), Cumulative Energy (kWh), Supply Voltage (V), Load Current (A). | Relay Channel Output State (`Power{1..8} On` / `Off` with hysteresis protection). | 100% local cloud-free switching and power metering for resistive water heaters, pool pumps, e-bike chargers, and auxiliary relays. |

### 5.12 Askoma ASKOHEAT+ Modbus TCP PV Immersion Heater `askoheat_diverter` MODBUS

**Manufacturer / Vendor:** Askoma

**Communication Interface:** Modbus TCP (Default Port: 502, Unit ID: 1, Holding Registers `100`, `200`–`202`, `300`–`304`)

**Supported Hardware Models:** Askoma ASKOHEAT+ (AHIR-BI-PLUS 1.75 kW – 19.25 kW) 7-Step PV Flange & Screw-In Immersion Heaters

🔍 **TELEMETRY READ OUT**

Actual Immersion Heating Power (`300` W), Boiler Water Temperature (`302` `0.1 °C`), Active Heater Stage Bitmask (`301`), Cumulative Thermal Energy (`304` `0.1 kWh`).

🎛️ **CONTROLLED SETPOINTS**

Stepless Feed-In / Load Power Setpoint (`200`–`201` W), 7-Step Heater Stage Control (`100` Stages 0–7), Legionella Boost, and Maximum Water Temperature Setpoint (`202` `0.1 °C`).

⚡ **HANDLED & AUTOMATION ROLE**

Proportional 7-step and stepless solar thermal diversion for Askoma ASKOHEAT+ immersion heaters: converts excess solar generation and negative-price electricity into domestic hot water buffer storage.

# 5. Communication Protocols & Field Wiring Reference

Connecting physical energy devices to the GEMS Raspberry Pi requires standard industrial interfaces. Below is the reference wiring and protocol architecture:

## 1. Modbus TCP & Modbus RTU (RS485)

### Modbus TCP (Ethernet / Wi-Fi)

Standard industrial port 502 (or 5020/8899). Connects directly via local LAN. Supports simultaneous pollers and sub-second update latency. Zero wiring required when devices reside on the same local subnet.

### Modbus RTU RS485 & Persistent USB IDs

Requires shielded twisted-pair cable (A+, B−, GND) in a daisy-chain topology with 120Ω end-of-line termination and unique Slave IDs. GEMS shares multi-drop RS485 buses safely and exposes persistent `/dev/serial/by-id/*` symlinks so USB adapters never swap across reboots.

## 2. Local OCPP 1.6-J Server (EVSE)

GEMS operates a high-performance, embedded local OCPP 1.6-J Central System on both dedicated port `8887` and reverse-proxy-friendly port `8080`. No external cloud subscription or third-party backend is needed for standalone installations:

**LOCAL OCPP 1.6-J SERVER CONNECTION ENDPOINTS**

`ws://<GEMS-IP>:8887/<ChargePointID>` or `ws://<GEMS-IP>:8080/api/ocpp/<ChargePointID>`

When configured, the EV charger sends real-time MeterValues (W, Wh, Amps, Volts) and receives dynamic `SetChargingProfile` setpoints calculated every second by the GEMS control loop.

## 3. P1 Port DSMR Smart Meters (Fluvius, Dutch Smart Meter)

Direct connection from the digital meter's RJ12 P1 port into the GEMS Raspberry Pi via USB-to-RJ12 cable:

| PIN | SIGNAL NAME | DESCRIPTION | CONNECTION ON RASPBERRY PI USB ADAPTER |
|---|---|---|---|
| **Pin 1** | +5V Power | Meter Power Supply (+5V DC, max 250mA on DSMR 5.0) | Connected to USB VCC (Optional, powers external dongle) |
| **Pin 2** | Data Request | Active-High Request Line | Tied to +5V (pull-up) to continuously enable telemetry output |
| **Pin 3** | Data GND | Ground Reference | Connected to USB GND (Ground) |
| **Pin 4** | NC | Not Connected | Unused |
| **Pin 5** | Data Line (TxD) | Inverted Serial Telemetry (115200 baud, 8N1) | Connected to USB RxD (via hardware transistor or FTDI inverted RX) |
| **Pin 6** | Power GND | Ground Reference for +5V | Connected to USB GND |

GEMS Supported Hardware & Devices Manual • End of Technical Reference

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